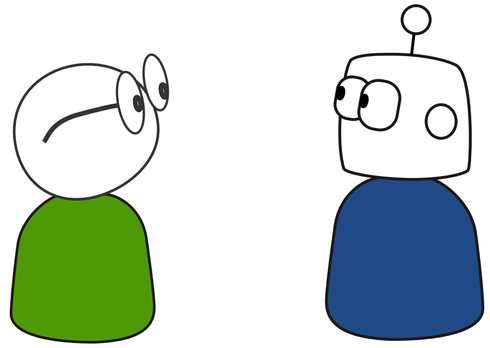
Learning to Build Common Ground in Multi-turn Interaction

Thesis Proposal



Saujas Vaduguru | ***Committee:*** *Daniel Fried, Graham Neubig, Yonatan Bisk, Yoav Artzi*

Collaborative agents

BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents

Jason Wei* Zhiqing Sun* Spencer Papay
Scott McKinney Jeffrey Han Isa Fulford Hyung Won Chung
Alex Tachard Passos William Fedus Amelia Glaese

OpenAI

RESEARCHQA: Evaluating Scholarly Question Answering at Scale Across 75 Fields with Survey-Mined Questions and Rubrics

Li S. Yifei*, Allen Chang*, Chaitanya Malaviya, Mark Yatskar
University of Pennsylvania
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Information gathering and research

SWE-BENCH: CAN LANGUAGE MODELS RESOLVE REAL-WORLD GITHUB ISSUES?

Carlos E. Jimenez*[1,2] John Yang*[1,2] Alexander Wettig[1,2]
Shunyu Yao[1,2] Kexin Pei[3] Ofir Press[1,2] Karthik Narasimhan[1,2]
[1]Princeton University [2]Princeton Language and Intelligence [3]University of Chicago

SWT-Bench: Testing and Validating Real-World Bug-Fixes with Code Agents

Niels Mündler[1], Mark Niklas Müller[1,2], Jingxuan He[1], Martin Vechev[1]
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mark@logicstar.ai [2]

Software engineering

WorkArena: How Capable are Web Agents at Solving Common Knowledge Work Tasks?

Alexandre Drouin*[1 2] Maxime Gasse*[1 2 3] Massimo Caccia[† 1] Issam H. Laradji[1]
Manuel Del Verme[2 4] Tom Marty[1 2 5] Léo Boisvert[1 3] Megh Thakkar[1 2 5] Quentin Cappart[2 3]
David Vazquez[1] Nicolas Chapados[1 2 3] Alexandre Lacoste[† 1]

TheAgentCompany: Benchmarking LLM Agents on Consequential Real World Tasks

Frank F. Xu[1*] Yufan Song[2*] Boxuan Li[2*] Yuxuan Tang[2] Kritanjali Jain[1]
Mengxue Bao[2] Zora Z. Wang[1] Xuhui Zhou[1] Zhitong Guo[1] Murong Cao[2]
Mingyang Yang[2] Hao Yang Lu[2] Amaad Martin[1] Zhe Su[1] Leander Melroy Maben[1]
Raj Mehta[1] Wayne Chi[1] Lawrence Jang[1] Yiqing Xie[1] Shuyan Zhou[3] Graham Neubig[1]
[1]Carnegie Mellon University [2]Independent [3]Duke University
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Administrative knowledge work

Joint actions in collaboration

"Two people have a coordination problem whenever they have ***common interests, or goals****, and each person's actions depend on the actions of the other. To reach their goals, they have to coordinate their individual actions in a* ***joint action****."*

"People cannot take joint actions without assuming certain pieces of ***common ground****."*

—Herbert Clark, *Using Language*

Language models in collaborative agents

Language models in collaborative agents

MULTITASK PROMPTED TRAINING ENABLES
ZERO-SHOT TASK GENERALIZATION

Victor Sanh* A
Hugging Face B

Lintang Sutawika Za
BigScience KF

Arun Raja Manan
I²R, Singapore SAP

Shanya Sharma Eliza
Walmart Labs BigSc

Debajyoti Datta J
University of Virginia A

Sheng Shen Zheng-X
UC Berkeley Brown U

Scaling Instruction-Finetuned Language Models

Hyung Won Chung* **Le Hou*** **Shayne Longpre*** **Barret Zoph†** **Yi Tay†**
William Fedus† **Yunxuan Li** **Xuezhi Wang** **Mostafa Dehghani** **Siddhartha Brahma**
Albert Webson **Shixiang Shane Gu** **Zhuyun Dai** **Mirac Suzgun** **Xinyun Chen**
Aakanksha Chowdhery **Alex Castro-Ros** **Marie Pellat** **Kevin Robinson**
Dasha Valter **Sharan Narang** **Gaurav Mishra** **Adams Yu** **Vincent Zhao**
Yanping Huang **Andrew Dai** **Hongkun Yu** **Slav Petrov** **Ed H. Chi**
Jeff Dean **Jacob Devlin** **Adam Roberts** **Denny Zhou** **Quoc V. Le**

SUPER-NATURALINSTRUCTIONS: Generalization via Declarative Instructions on 1600+ NLP Tasks

♢Yizhong Wang[2] **♢Swaroop Mishra[3]** **♣Pegah Alipoormolabashi[4]** **♣Yeganeh Kordi[5]**
Amirreza Mirzaei[4] **Anjana Arunkumar[3]** **Arjun Ashok[6]** **Arut Selvan Dhanasekaran[3]**
Atharva Naik[7] **David Stap[8]** **Eshaan Pathak[9]** **Giannis Karamanolakis[10]** **Haizhi Gary Lai[11]**
Ishan Purohit[12] **Ishani Mondal[13]** **Jacob Anderson[3]** **Kirby Kuznia[3]** **Krima Doshi[3]** **Maitreya Patel[3]**
Kuntal Kumar Pal[3] **Mehrad Moradshahi[14]** **Mihir Parmar[3]** **Mirali Purohit[15]** **Neeraj Varshney[3]**
Phani Rohitha Kaza[3] **Pulkit Verma[3]** **Ravsehaj Singh Puri[3]** **Rushang Karia[3]** **Shailaja Keyur Sampat[3]**
Savan Doshi[3] **Siddhartha Mishra[16]** **Sujan Reddy[17]** **Sumanta Patro[18]** **Tanay Dixit[19]** **Xudong Shen[20]**
Chitta Baral[3] **Yejin Choi[1,2]** **Noah A. Smith[1,2]** **Hannaneh Hajishirzi[1,2]** **Daniel Khashabi[21]**

[1]Allen Institute for AI [2]Univ. of Washington [3]Arizona State Univ. [4]Sharif Univ. of Tech. [5]Tehran Polytechnic [6]PSG College of Tech. [7]IIT Kharagpur
[8]Univ. of Amsterdam [9]UC Berkeley [10]Columbia Univ. [11]Factored AI [12]Govt. Polytechnic Rajkot [13]Microsoft Research [14]Stanford Univ. [15]Zycus Infotech
[16]Univ. of Massachusetts Amherst [17]National Inst. of Tech. Karnataka [18]TCS Research [19]IIT Madras [20]National Univ. of Singapore [21]Johns Hopkins Univ.

Language models in collaborative agents

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REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

Shunyu Yao*,[1], Jeffrey Zhao[2], Dian Yu[2], Nan Du[2], Izhak Shafran[2], Karthik Narasimhan[1], Yuan Cao[2]

SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering

.com

The BrowserGym Ecosystem for Web Agent Research

John Y.

Thibault Le Sellier De Chezelles*[123] **Maxime Gasse*[123]** **Alexandre Lacoste*[1]**

Core contributors:
Alexandre Drouin[12] **Massimo Caccia[1]**
Léo Boisvert[123] **Megh Thakkar[12]** **Tom Marty[27]** **Rim Assouel[27]** **Sahar Omidi Shayegan[125]**

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Affiliated advisors:
Siva Reddy[125] **Quentin Cappart[23]** **Graham Neubig[4]** **Ruslan Salakhutdinov[4]** **Nicolas Chapados[123]**

Lead:
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Agents don’t yet interact well with people

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Code with Me or for Me? How Increasing AI Automation Transforms Developer Workflows

Valerie Chen[1,2], Ameet Talwalkar[1], Robert Brennan[2], and Graham Neubig[1,2]

Desiderata 2: Agents should balance the level of proactivity. Many participants observed, or even complained, about how OpenHands would take more actions than necessary: one participant wrote in a message to the agent, “*could the code have been simplified, I did not expect 10 files to be created with more than 1000 lines each.*” In the post-study feedback, one participant noted that “*even though OpenHands agents performs all tasks, it does not seem to*

ImpossibleBench: Measuring LLMs’ Propensity of Exploiting Test Cases

Ziqian Zhong[1] **Aditi Raghunathan**[1] **Nicholas Carlini**[2]

[1] Carnegie Mellon University [2] Anthropic

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LLMs Get Lost In Multi-Turn Conversation

Philippe Laban[*◇] **Hiroaki Hayashi**[*♣] **Yingbo Zhou**[♣] **Jennifer Neville**[◇]
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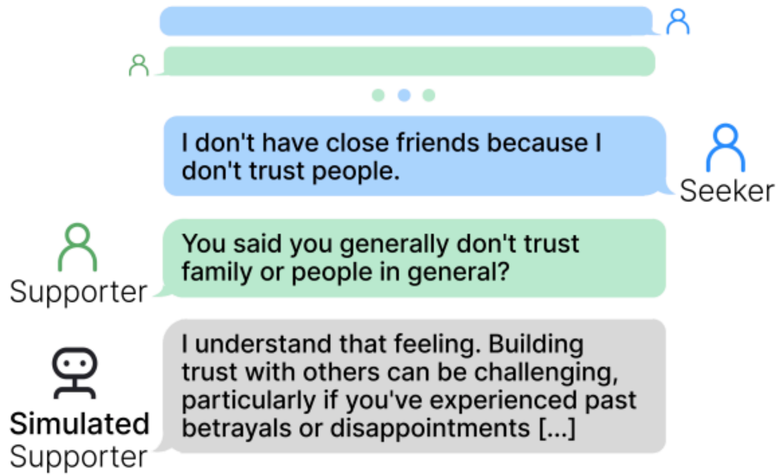
Grounding Gaps in Language Model Generations

Omar Shaikh[⋆] **Kristina Gligorić**[⋆] **Ashna Khetan** **Matthias Gerstgrasser**
Diyi Yang **Dan Jurafsky**
Stanford University



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◇Microsoft Research ♣Salesforce Research
{plaban,jenneville}@microsoft.com
{hiroakihayashi,yingbo.zhou}@salesforce.com

Collaborative agents and common ground

- We can view these as failures to reason about common ground
  - Cooperation is mutually assumed, so a user not asking for complex solutions likely means they don’t want them
  - It is natural for intent to be revealed across turns, and each turn is meant to be interpreted in the shared context of previous ones
  - When something is unclear, users expect requests for clarification or help rather than presumption

Communicating in games

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- Games provide a framework to model goal-oriented actors

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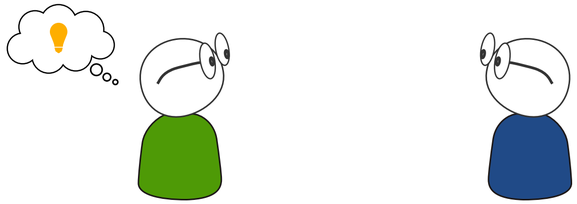
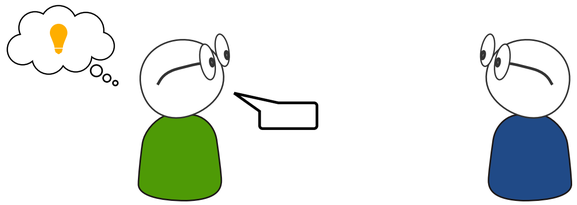
- Cooperative games



Communicating in games

- Games provide a framework to model goal-oriented actors
- Cooperative games
  - Players communicate to coordinate their actions



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- Games provide a framework to model goal-oriented actors
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  - Players communicate to coordinate their actions
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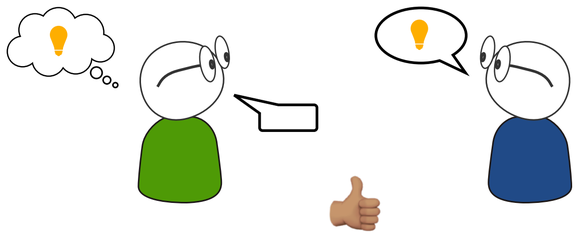

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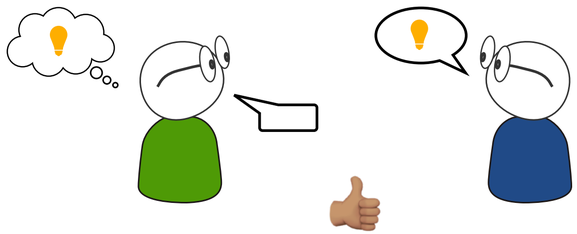

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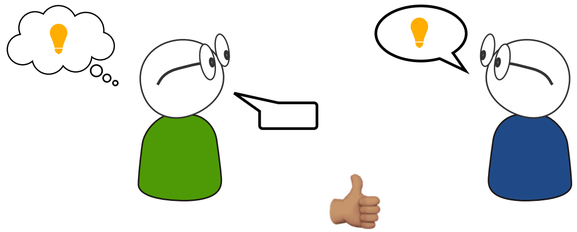
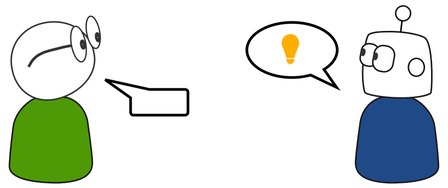
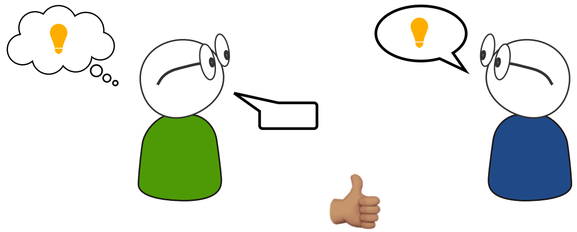
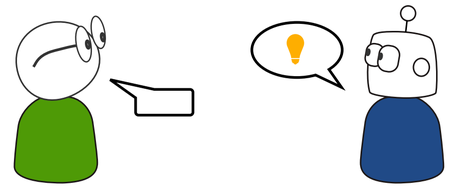


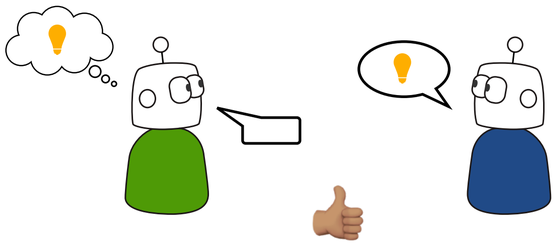


Communicating in games

- Games provide a framework to model goal-oriented actors
- Cooperative games
  - Players communicate to coordinate their actions
  - All players are rewarded if the goal is met
- Model players as acting to maximize their chance of winning
  - Evaluate agents with human partners
  - Train agents in simulated interactions with other agents







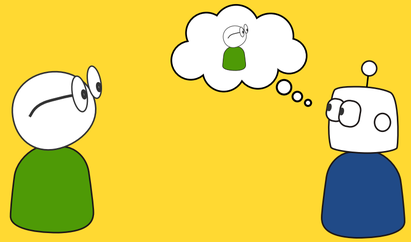
Objectives

- Improve and evaluate how agents reason about common ground in multi-turn interactions
  - Frame different human-agent interaction scenarios as communication games
  - Train agents in simulated games to improve their grounded interaction ability
  - Evaluate models' ability to reason about common ground through game performance

Thesis overview

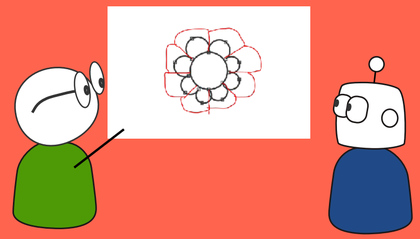
Part I: Reasoning about user intent

Make agents reason about how the user is acting to achieve a shared goal



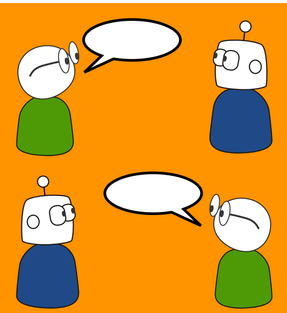
Part II: Reasoning about shared observation

Evaluate agents with instructions grounded in the shared environment and interaction so far



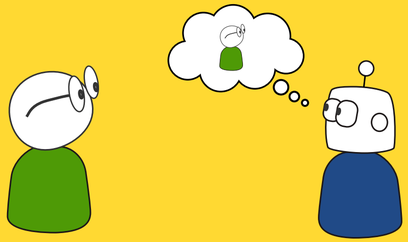
Part III: Reasoning about common ground over time

Build agents that adapt over the course of multiple interactions with a user



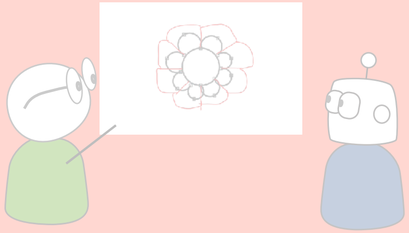
Part I: Reasoning about user intent

*[**Vaduguru, Fried, Pu, ICLR 2024**; proposed work]*



Part II: Reasoning about shared observation

[McCarthy, Vaduguru*,* et al.*, Findings of EMNLP, 2025]*



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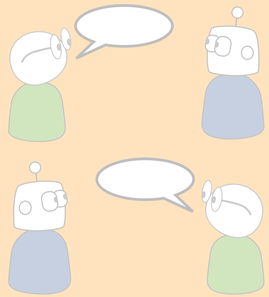
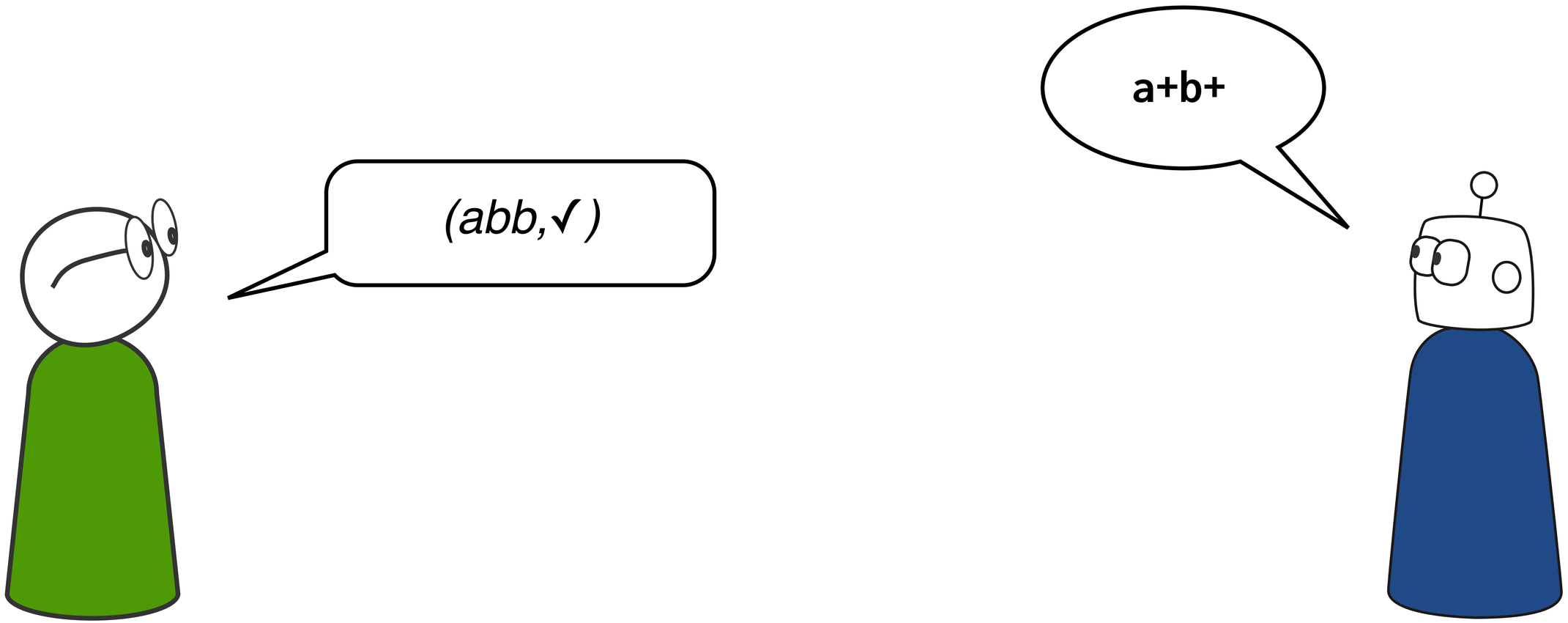
[Vaduguru, Hua, Artzi, Fried, ACL 2026; proposed work]



Programming-by-example



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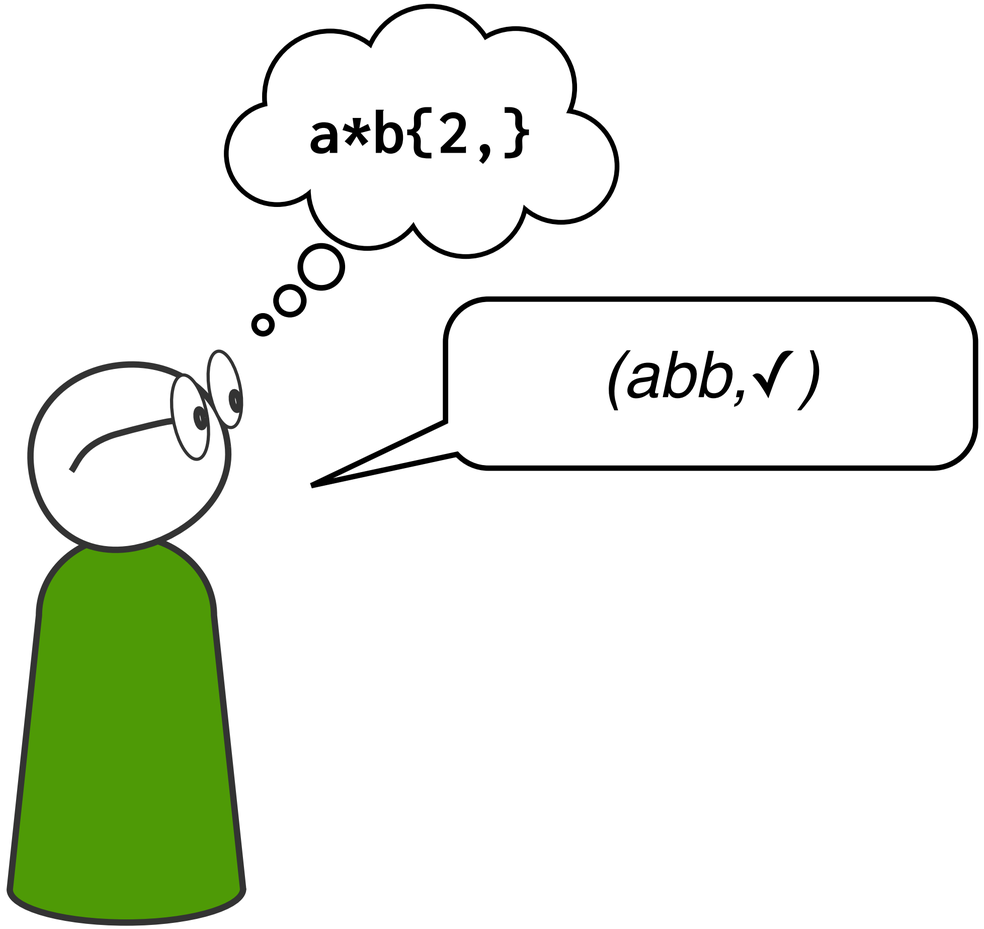

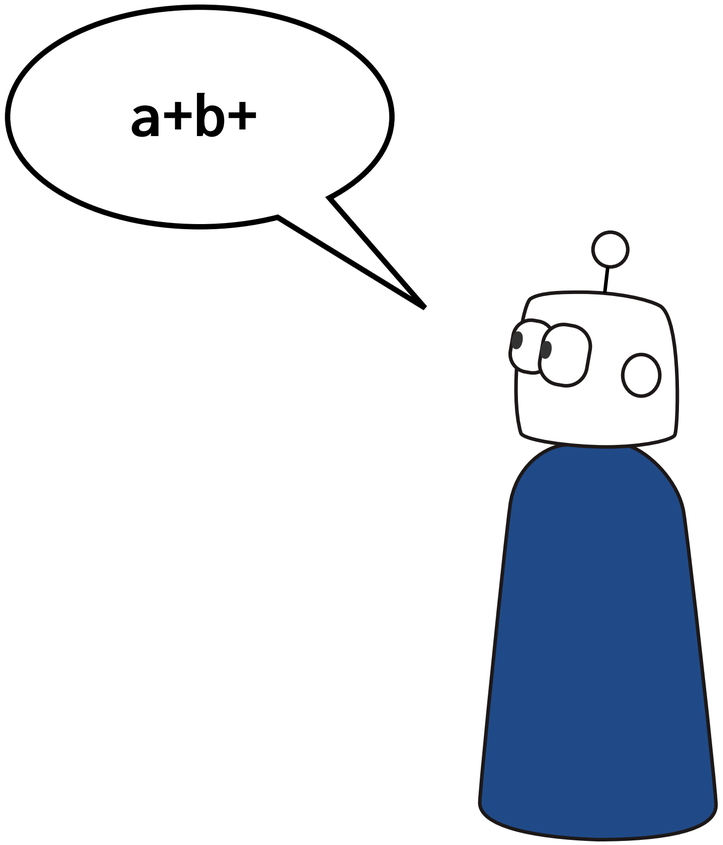

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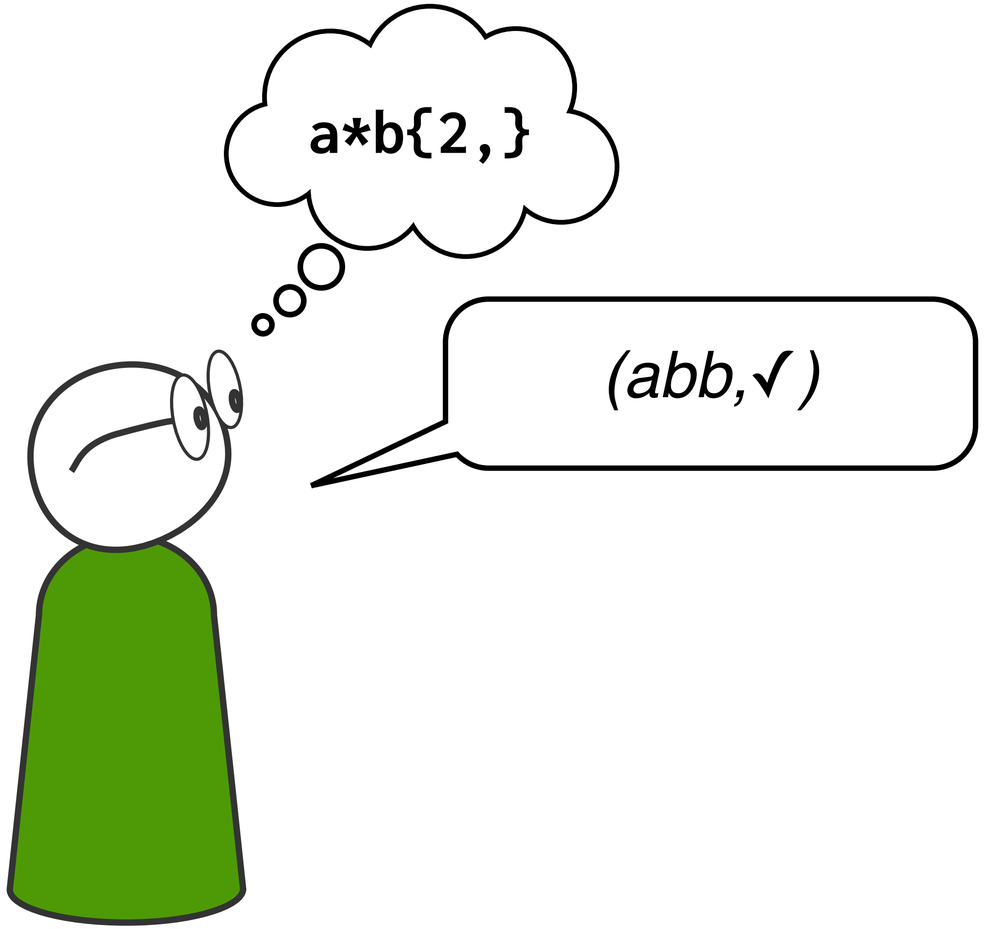
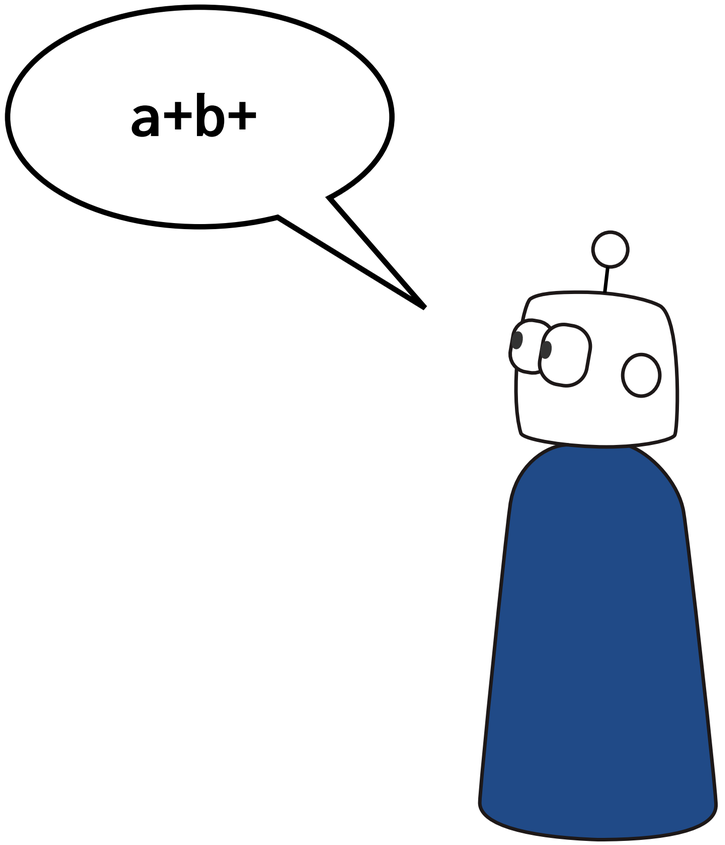
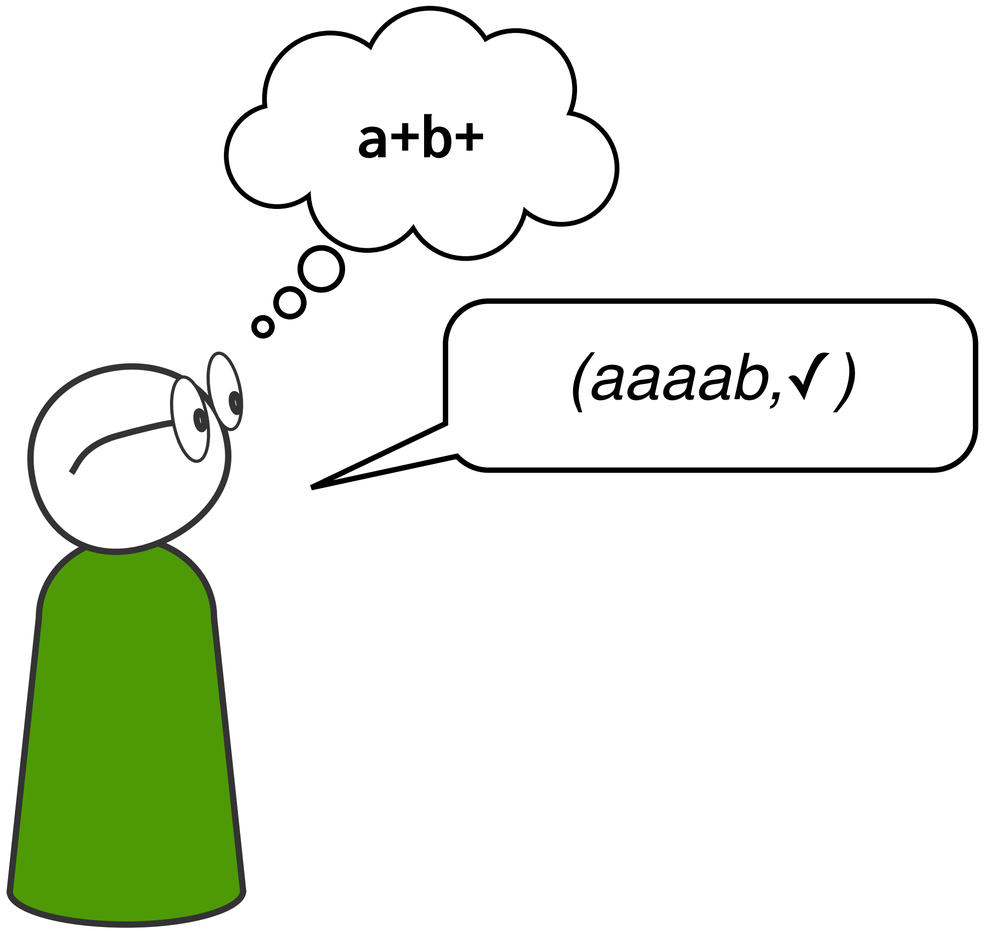


Speaker

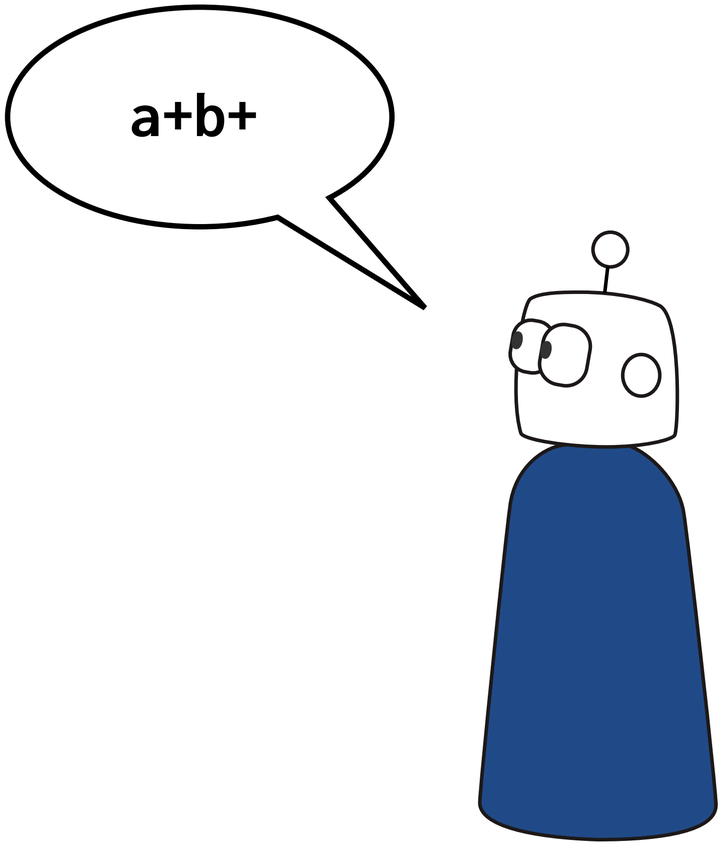


Listener

Pragmatic examples



Speaker



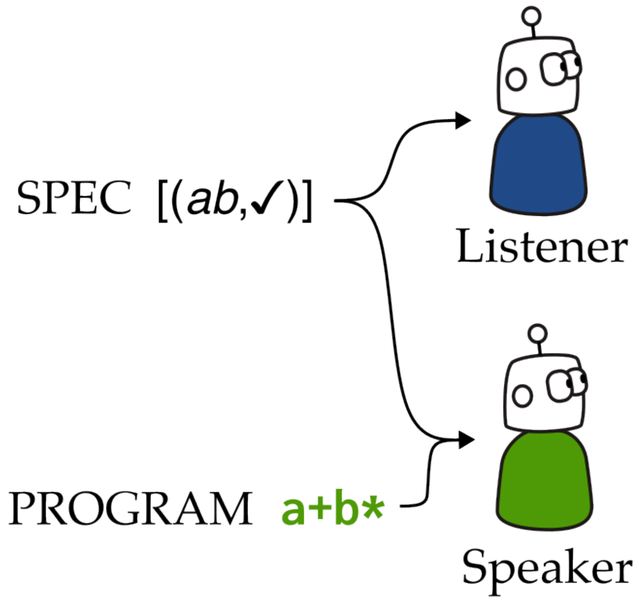
Listener

Pragmatic program synthesis

- Users choose examples pragmatically
- Reasoning about the pragmatic choice reduces ambiguity of examples
  - Reach the target program with fewer examples
- Train a synthesizer to interpret pragmatic examples chosen by a speaker
  - Model a speaker that chooses pragmatic examples

[Pu, et al., 2020]

Generating pragmatic examples



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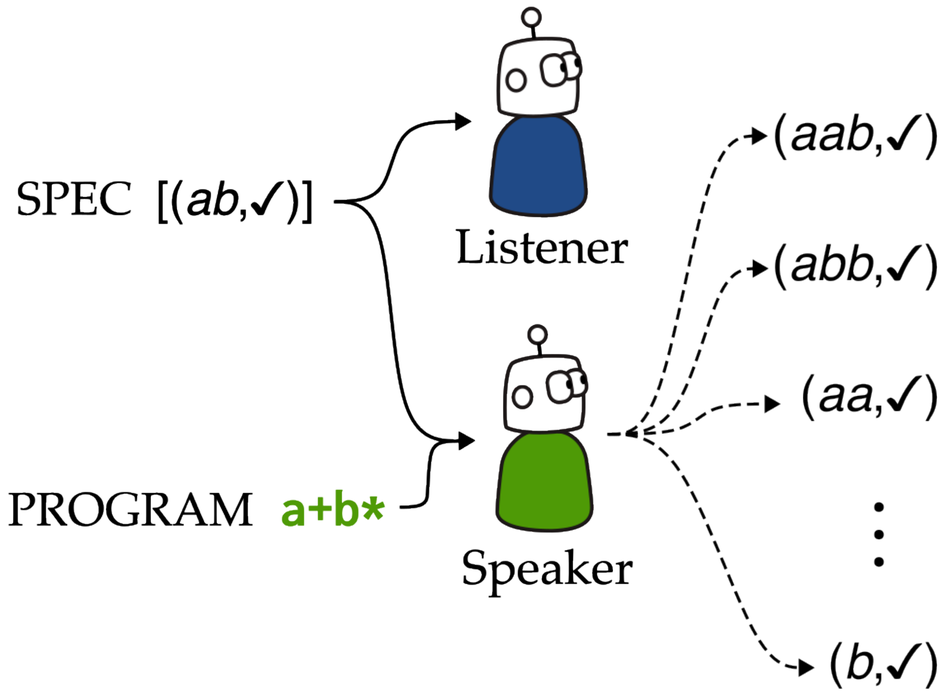

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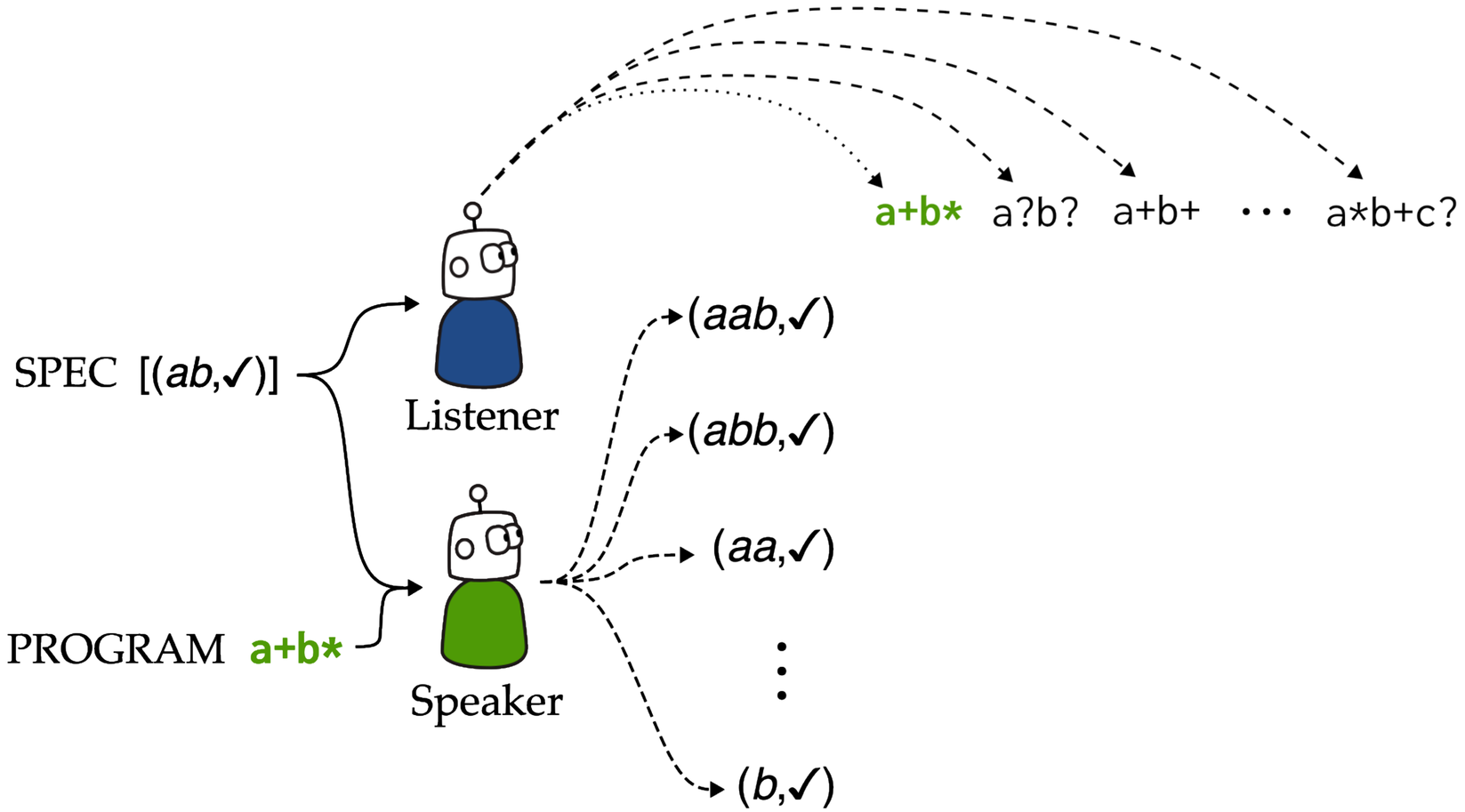

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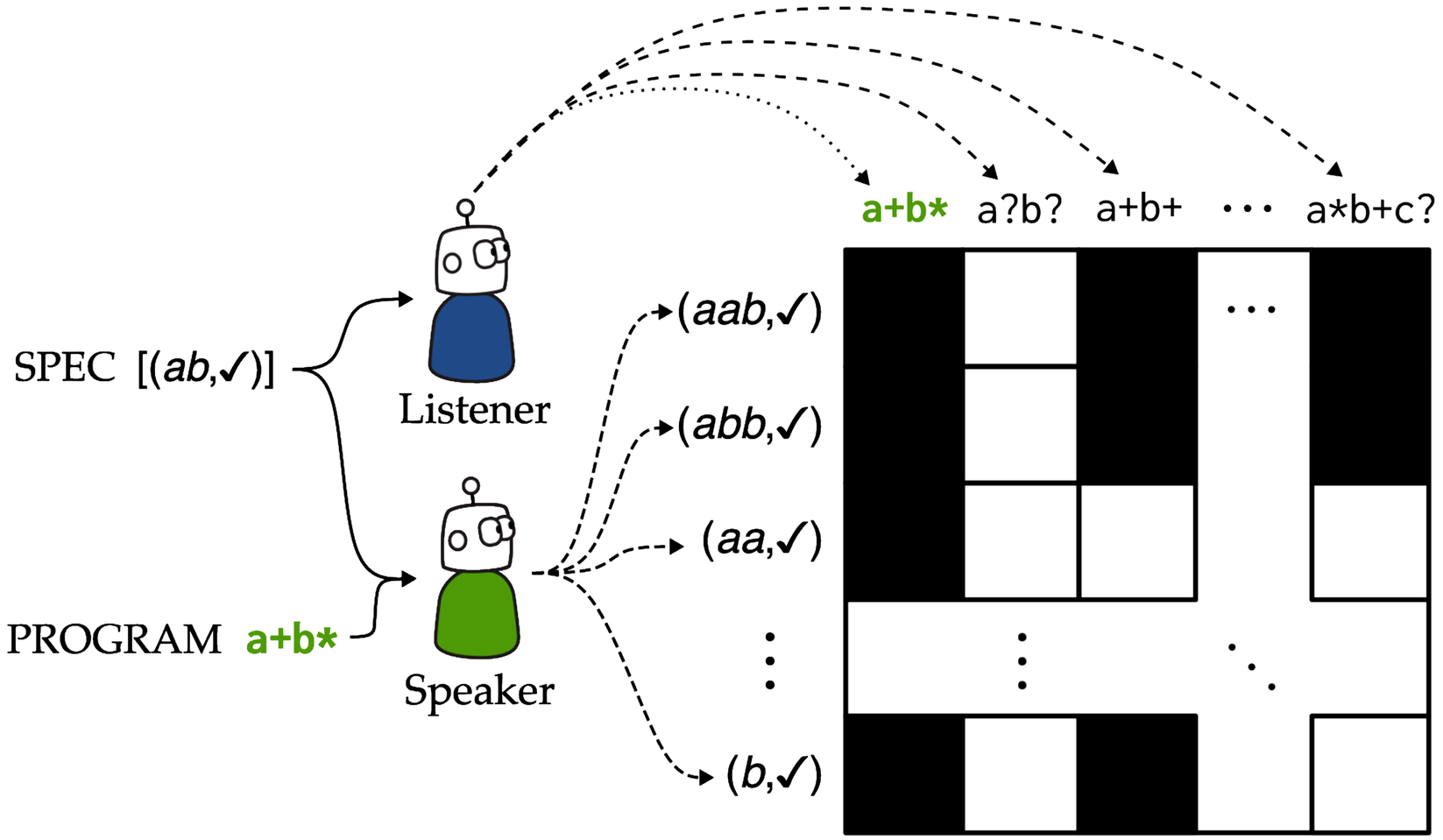

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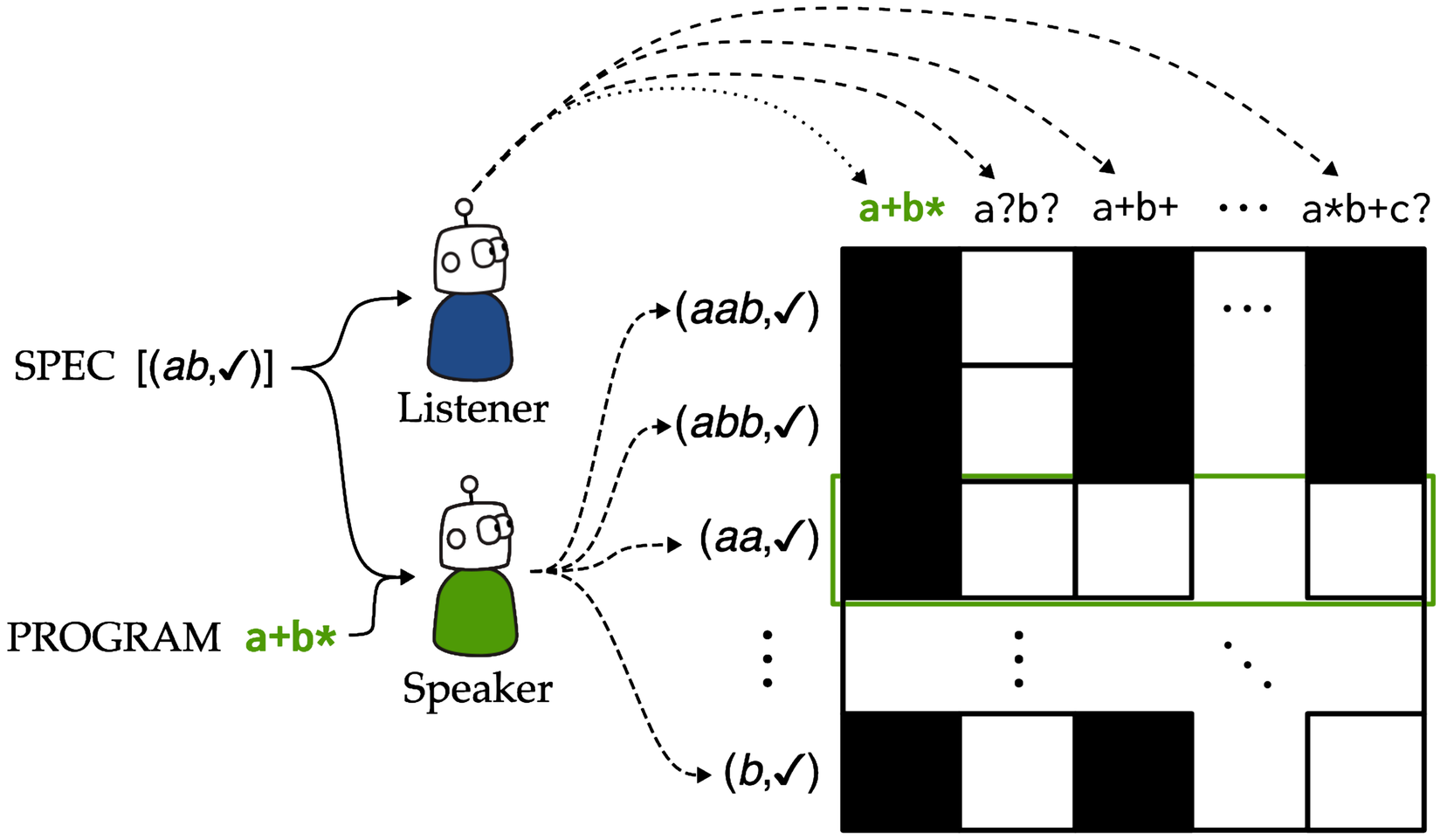

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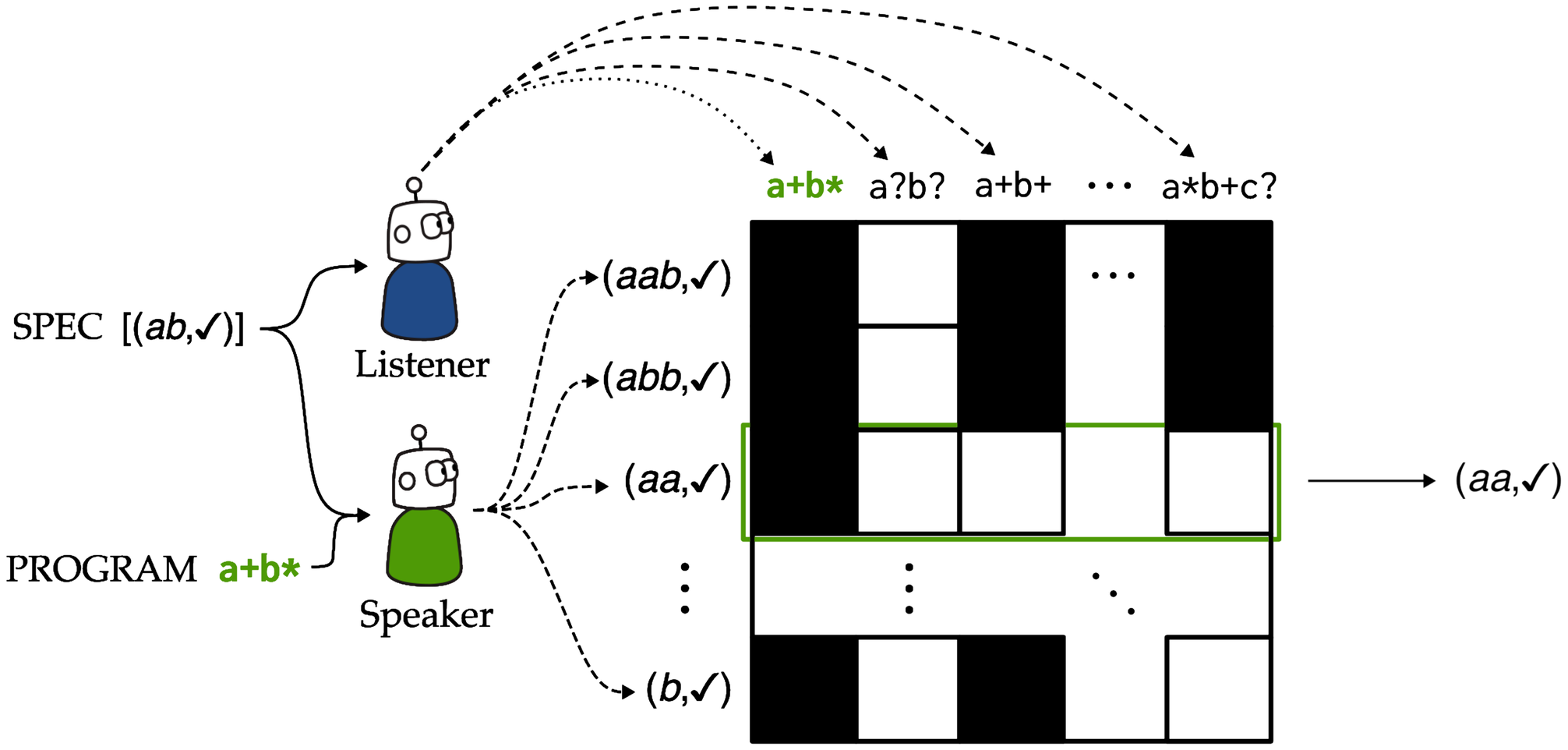

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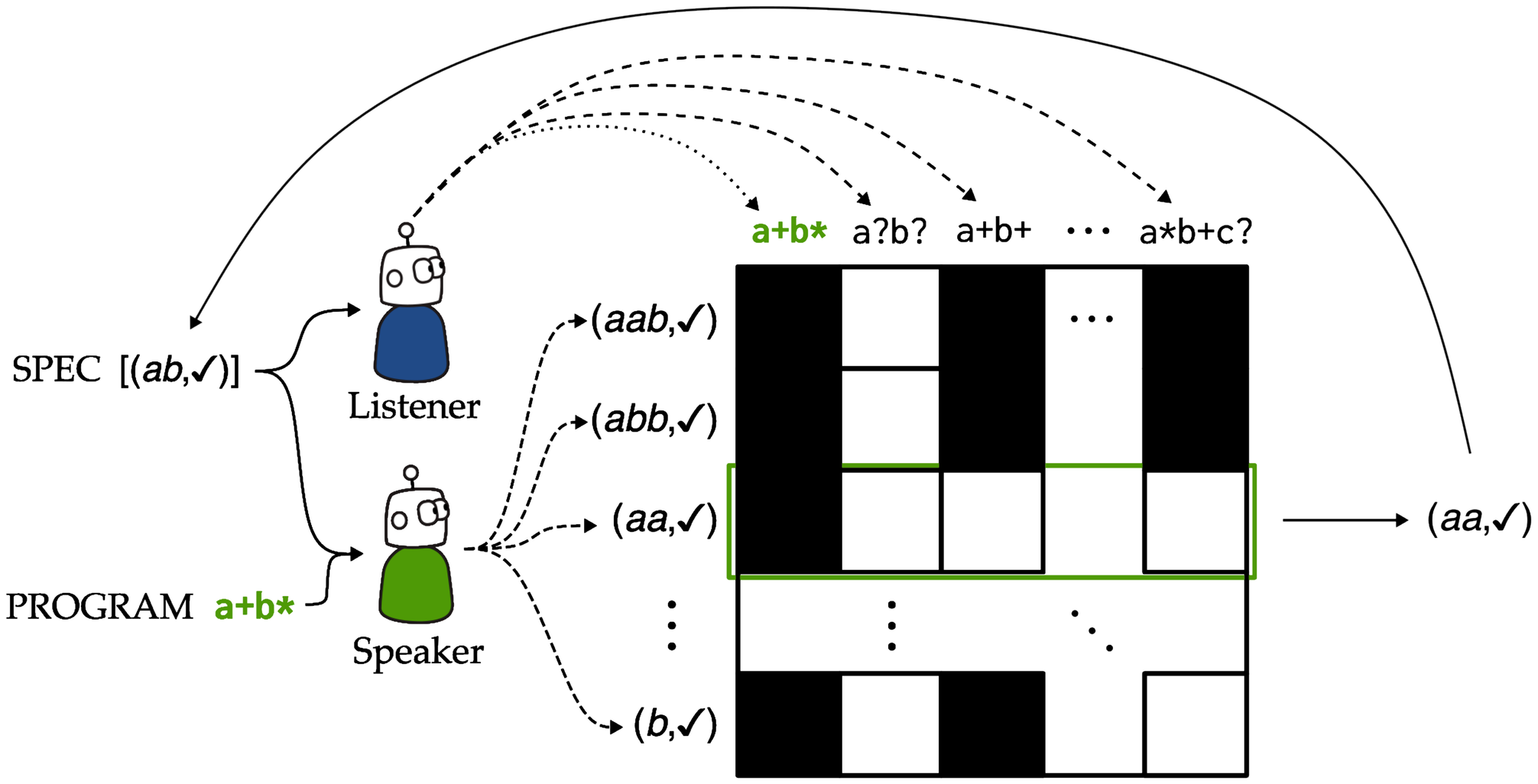

PraX: Training a pragmatic synthesizer

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- Start with base listener and speaker

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- Start with base listener and speaker
- For a set of target programs, generate sequences of examples

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- Update speaker and listener on generated data

PraX: Training a pragmatic synthesizer

- Start with base listener and speaker
- For a set of target programs, generate sequences of examples
- Update speaker and listener on generated data
- Repeat with updated speaker and listener

Evaluation

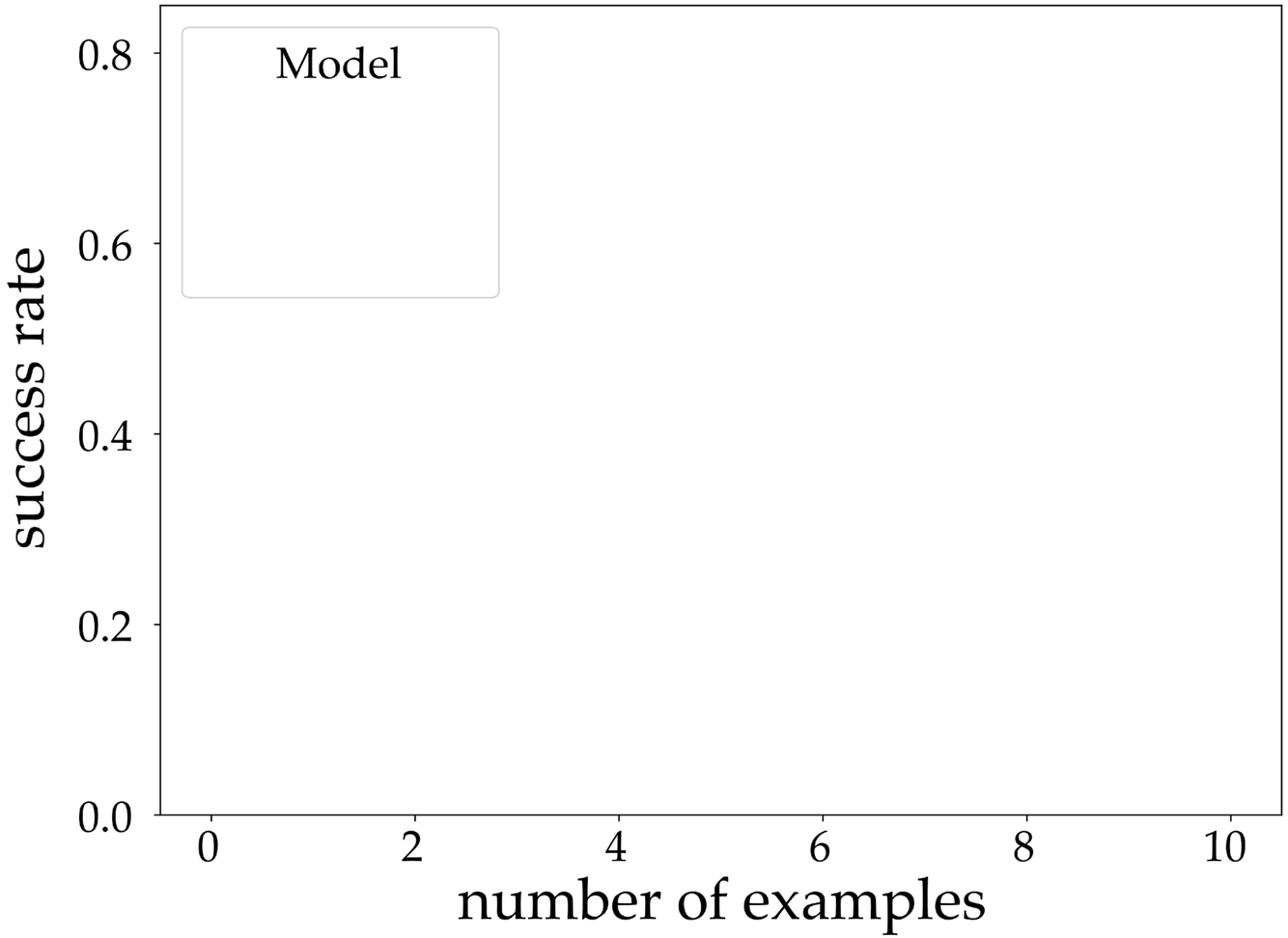
Evaluation

- See how well the listener does in interactions with people
  - Give a person a target program, and ask them to provide examples to illustrate this program to the synthesizer
  - Stop when the target program is recovered
  - Measure success rate

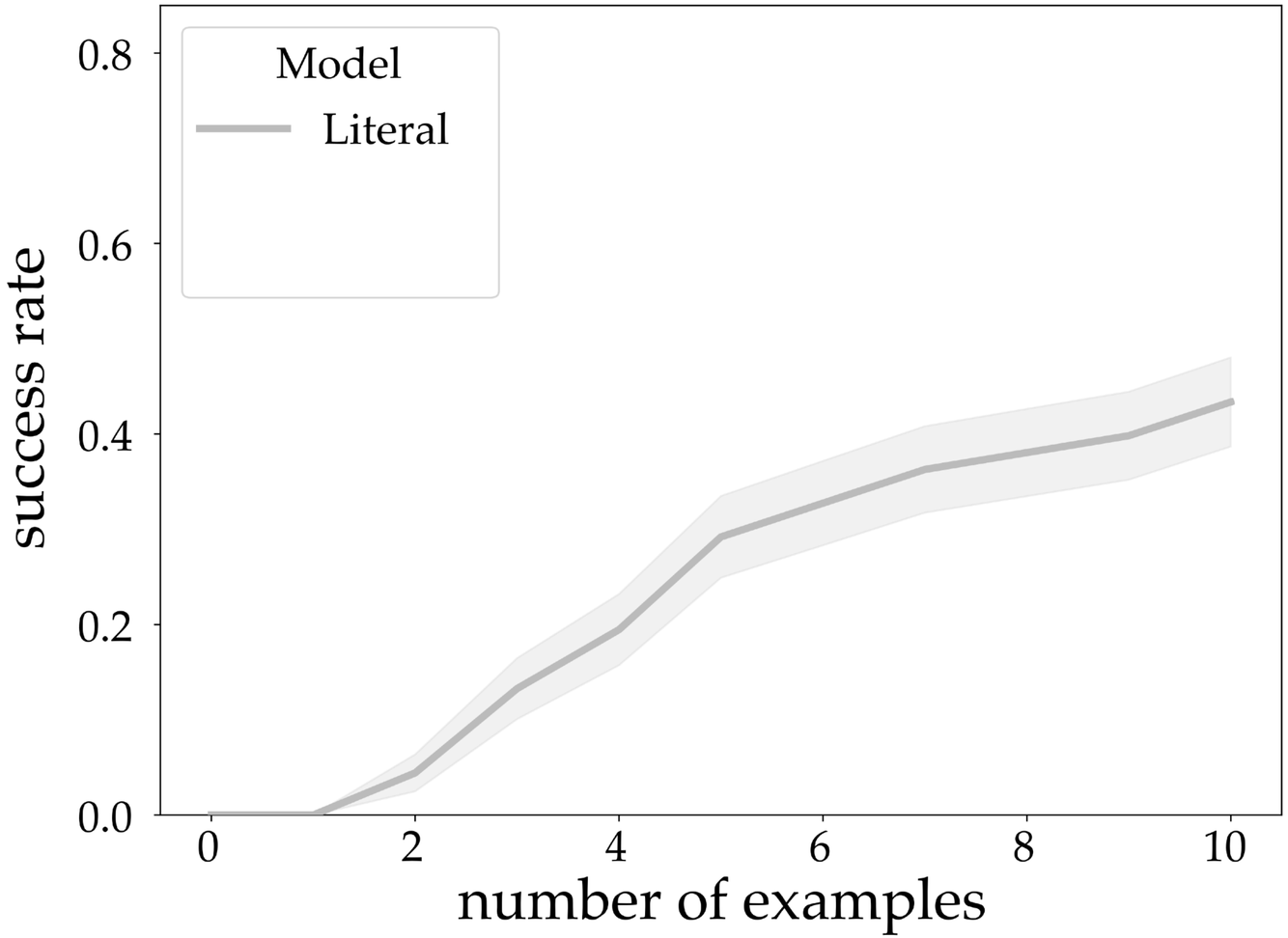
Evaluation

- See how well the listener does in interactions with people
  - Give a person a target program, and ask them to provide examples to illustrate this program to the synthesizer
  - Stop when the target program is recovered
  - Measure success rate
- Baselines
  - Literal: use the base model
  - HFT: Finetune on human provided examples

Pragmatic listeners perform better



Pragmatic listeners perform better



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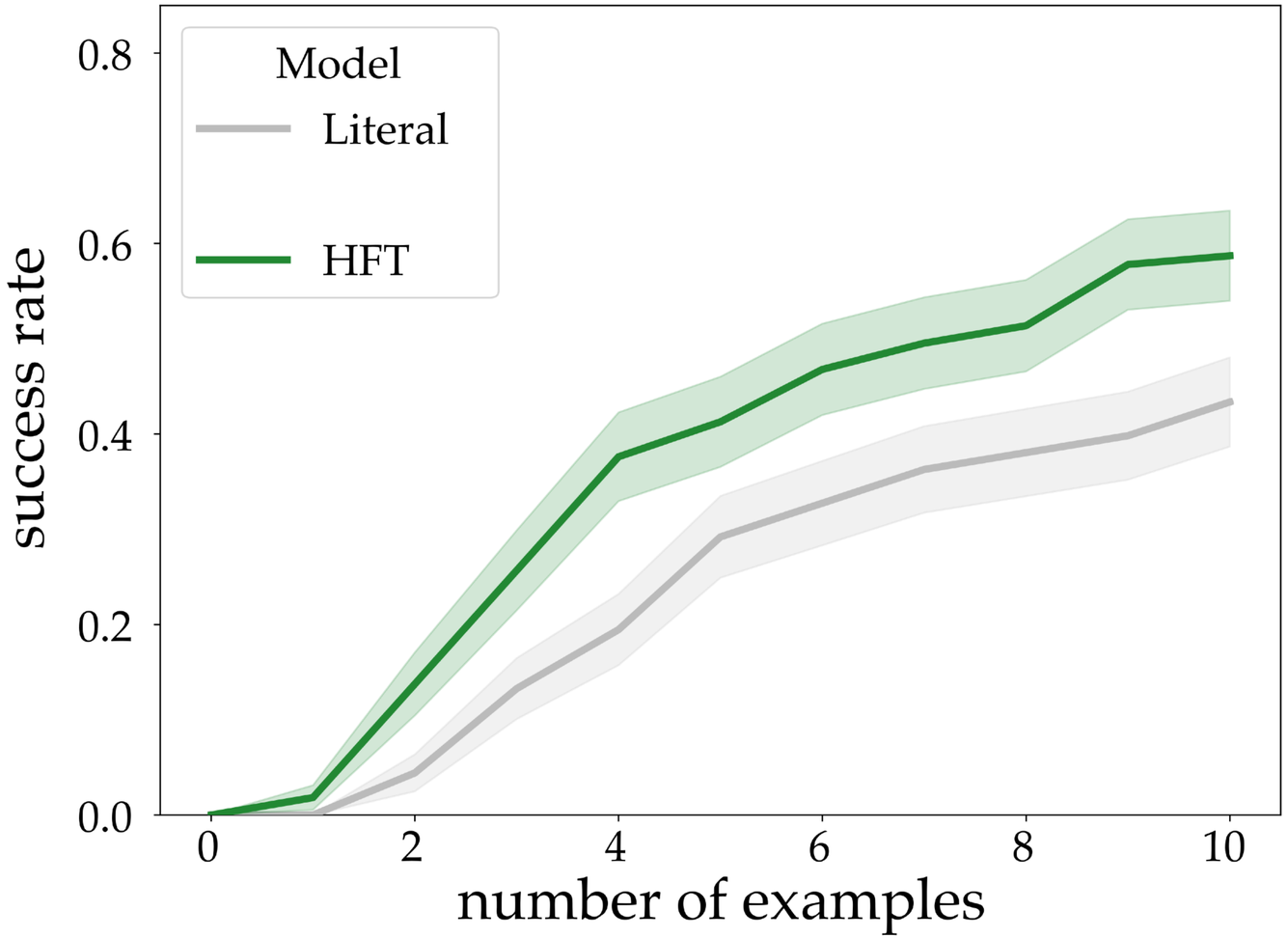

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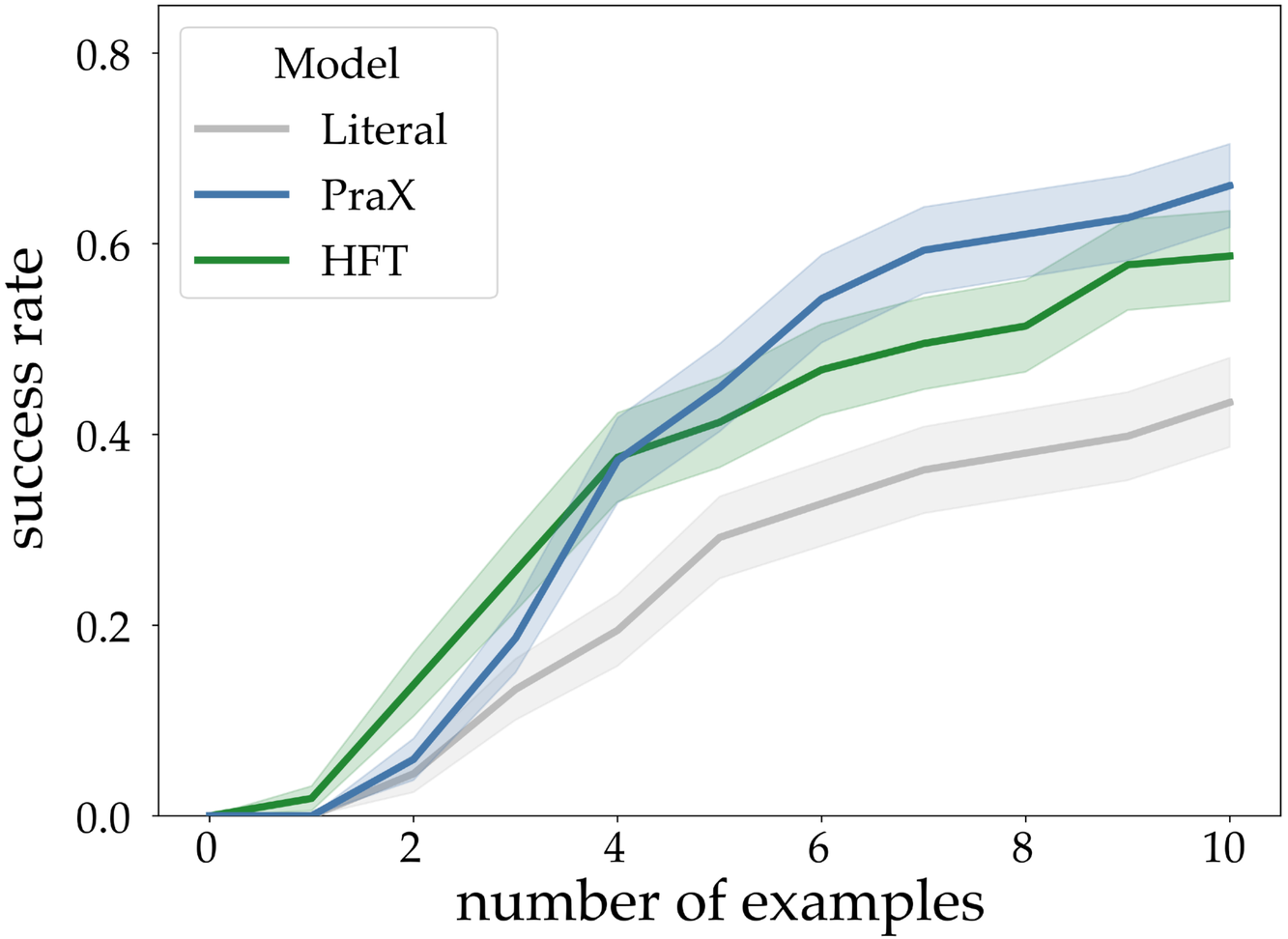


Speakers learn to be pragmatic

Speakers learn to be pragmatic

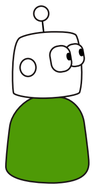
Regular expression

```
1{3,}h{3,}1{3,}
```

Speakers learn to be pragmatic

Regular expression

`1{3,}h{3,}1{3,}`



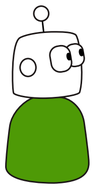
Round 0

1111hhhhh1111 ✅

Speakers learn to be pragmatic

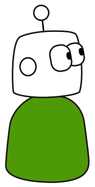
Regular expression

`1{3,}h{3,}1{3,}`



Round 0

1111hhhhh1111 ✅



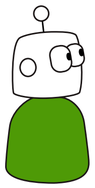
Round 8

1111111111111hhh111 ✅

Speakers learn to be pragmatic

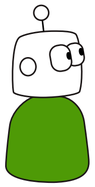
Regular expression

`1{3,}h{3,}1{3,}`



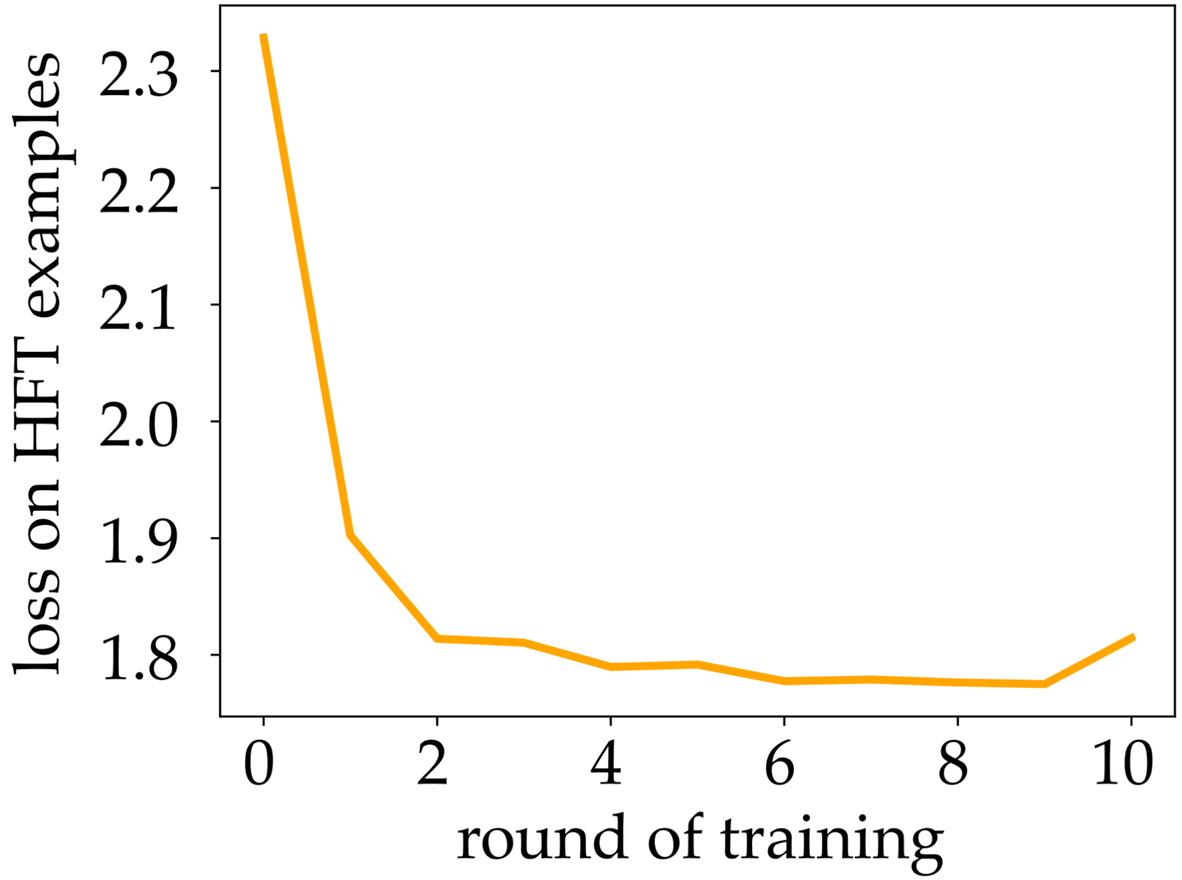
Round 0

1111hhhhh1111 ✅



Round 8

1111111111111hhh111 ✅



Takeaways

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- It is in common ground that users are being cooperative
  - This can be used to reduce ambiguity
  - Can be used to interact more efficiently

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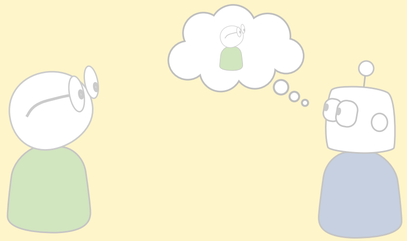
- It is in common ground that users are being cooperative
  - This can be used to reduce ambiguity
  - Can be used to interact more efficiently
- We can train agents to do this based on simulated interactions alone
  - We don’t need demonstrations of how ambiguous communication is resolved
  - We can elicit sophisticated behaviors just from training in simulation

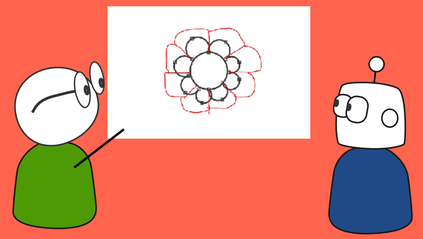
Part I: Reasoning about user intent

[Vaduguru, Fried, Pu, ICLR 2024; proposed work]



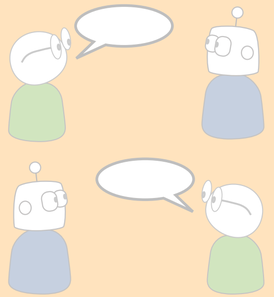
Part II: Reasoning about shared observation

[McCarthy, Vaduguru*,* et al.*, Findings of EMNLP, 2025]*

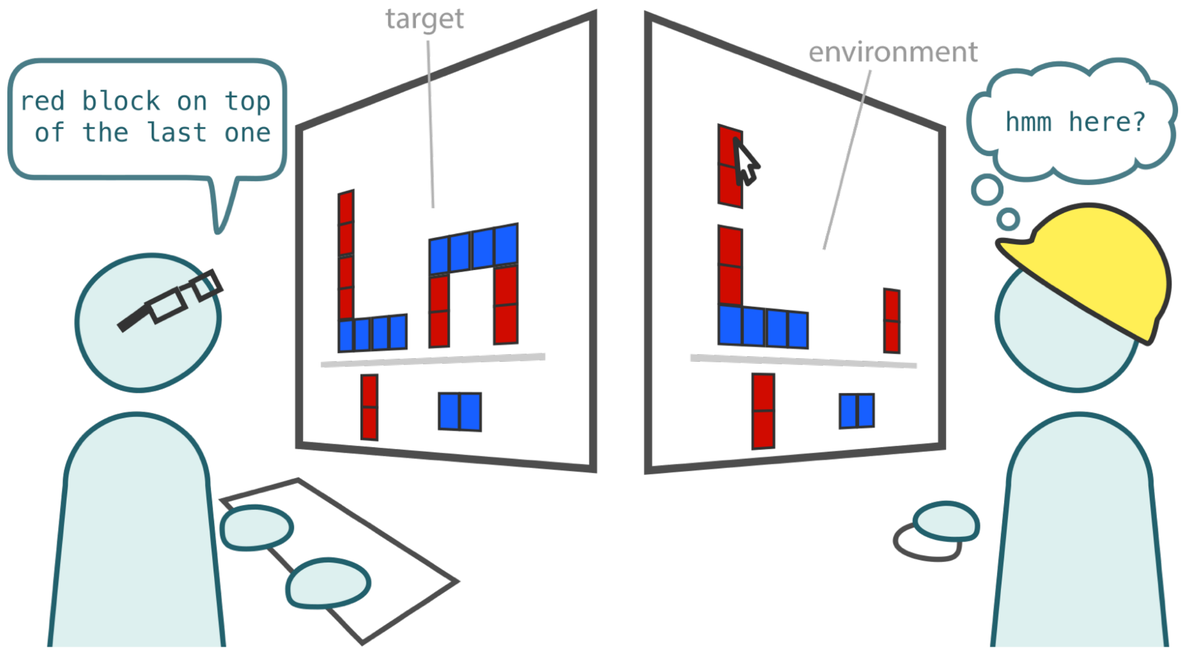


Part III: Reasoning about common ground over time

[Vaduguru, Hua, Artzi, Fried, ACL 2026; proposed work]

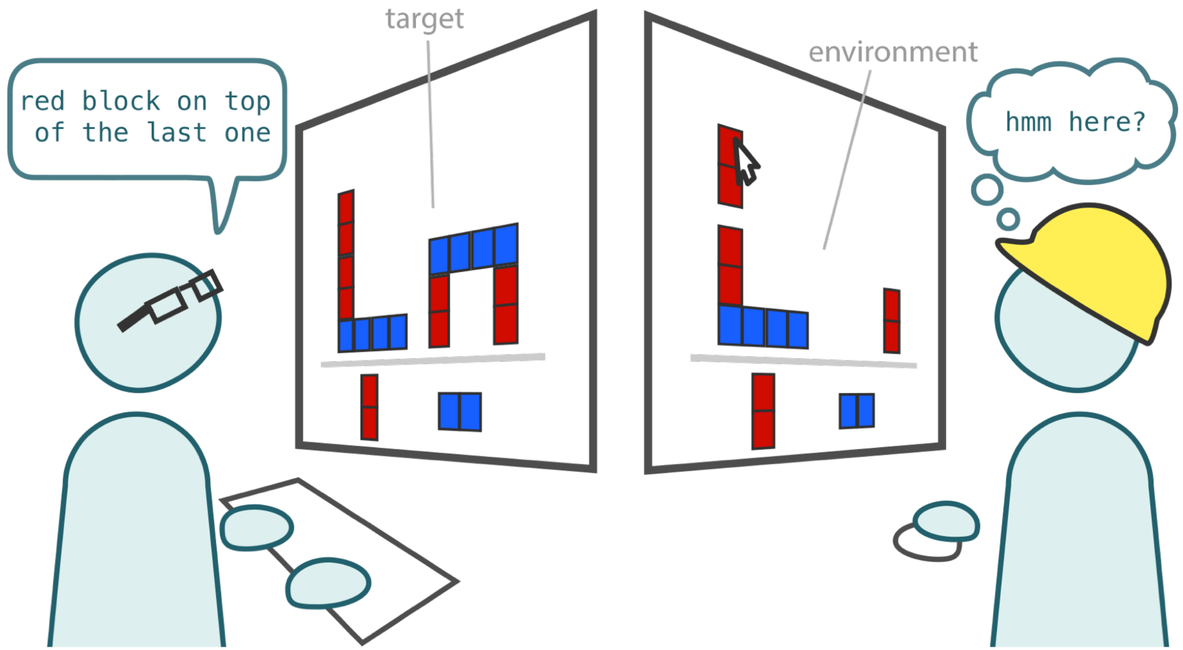


Collaborative construction



[McCarthy, Hawkins, et al., 2021]

Collaborative construction



post_process.py > process_directory

```python
def process_directory(source_root, target_root):
    # Create target root directory if it doesn't exist
    os.makedirs(target_root, exist_ok=True)

    # Get all subdirectories in source root
    subdirs = [d for d in os.listdir(source_root)
               if os.path.isdir(os.path.join(source_root, d))]

    print(f"Found {len(subdirs)} subdirectories to process")

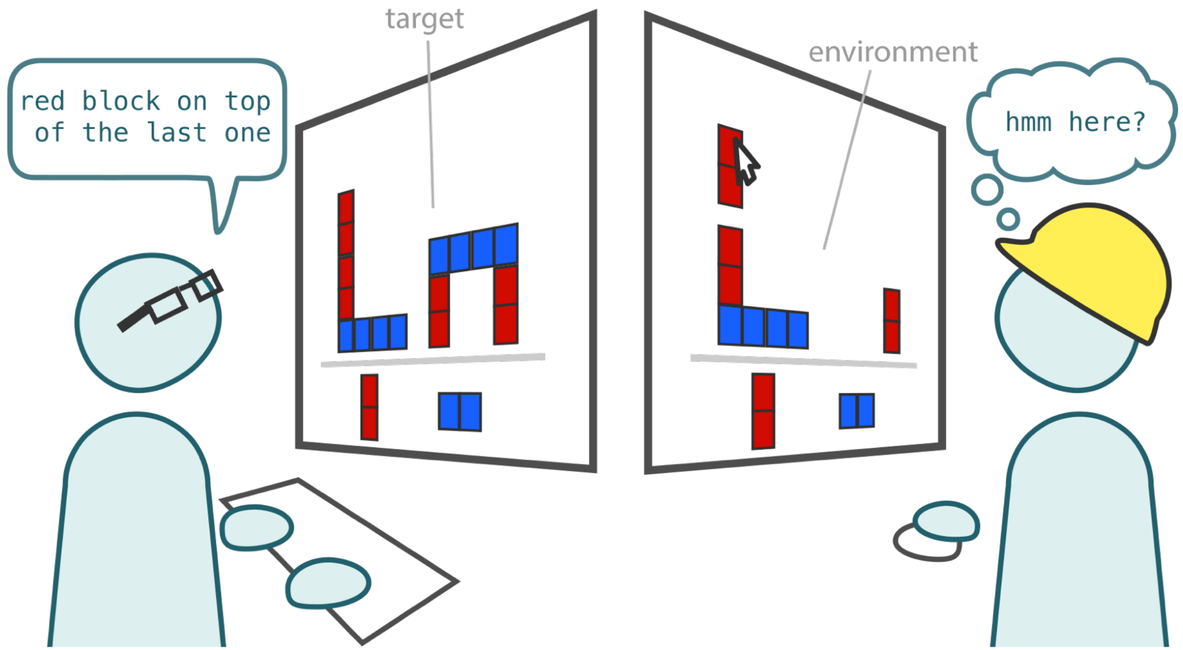
    # Create lists to store sandbox and pair IDs
    sandbox_ids = []
    pair_ids = []
```

Optimize This

[McCarthy, Hawkins, et al., 2021]

[Chi et al., 2026]

Collaborative construction



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```

Optimize This

- Farming, as a part of agriculture, involves ~~growing~~ crops cultivation and animal rearing for food and raw materials.
- Originated ~~It began~~ thousands of years ago, likely in the Fertile Crescent, leading to the Neolithic Revolution
- Transition ~~as people transitioned~~ from nomadic hunting to settled farming. resulted in significant human population increase

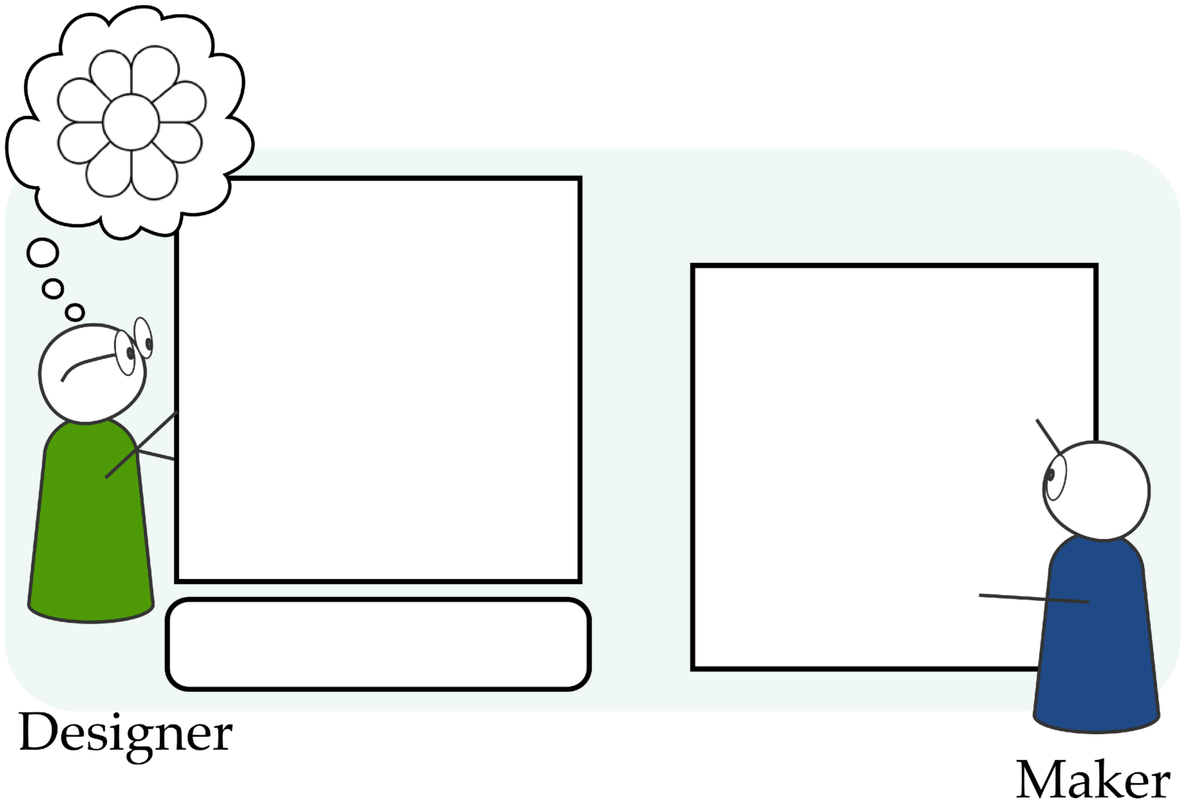
[McCarthy, Hawkins, et al., 2021]

[Chi et al., 2026]

[Gao et al., 2024]

mrCAD game

mrCAD game



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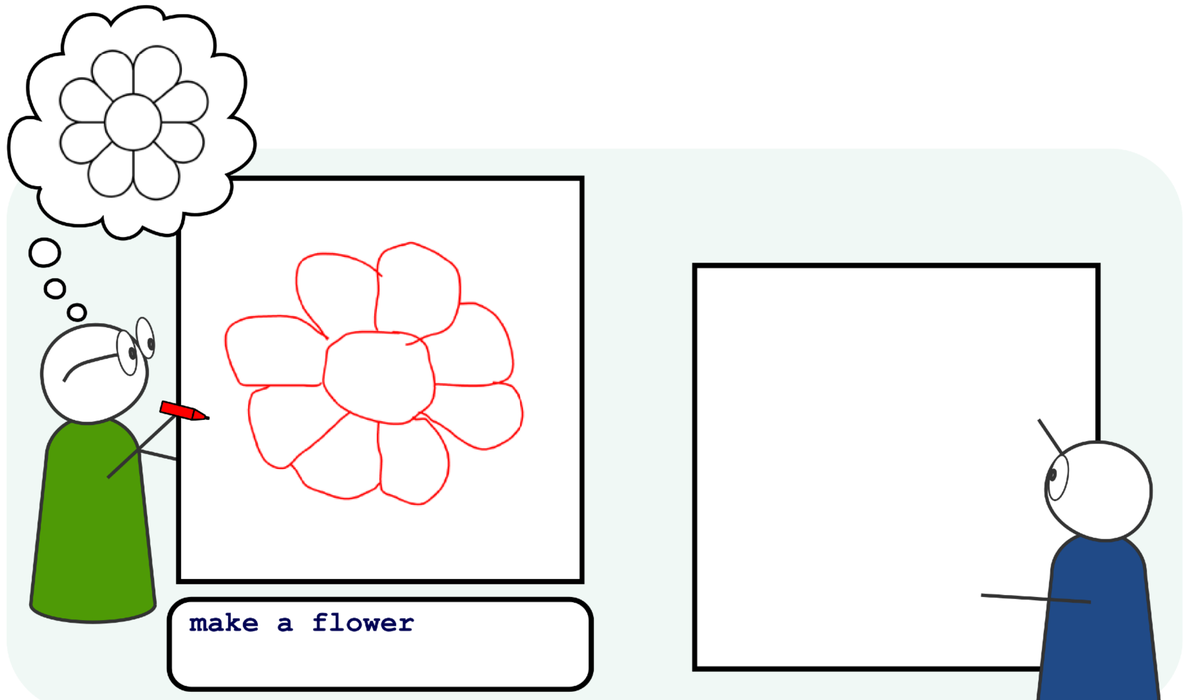

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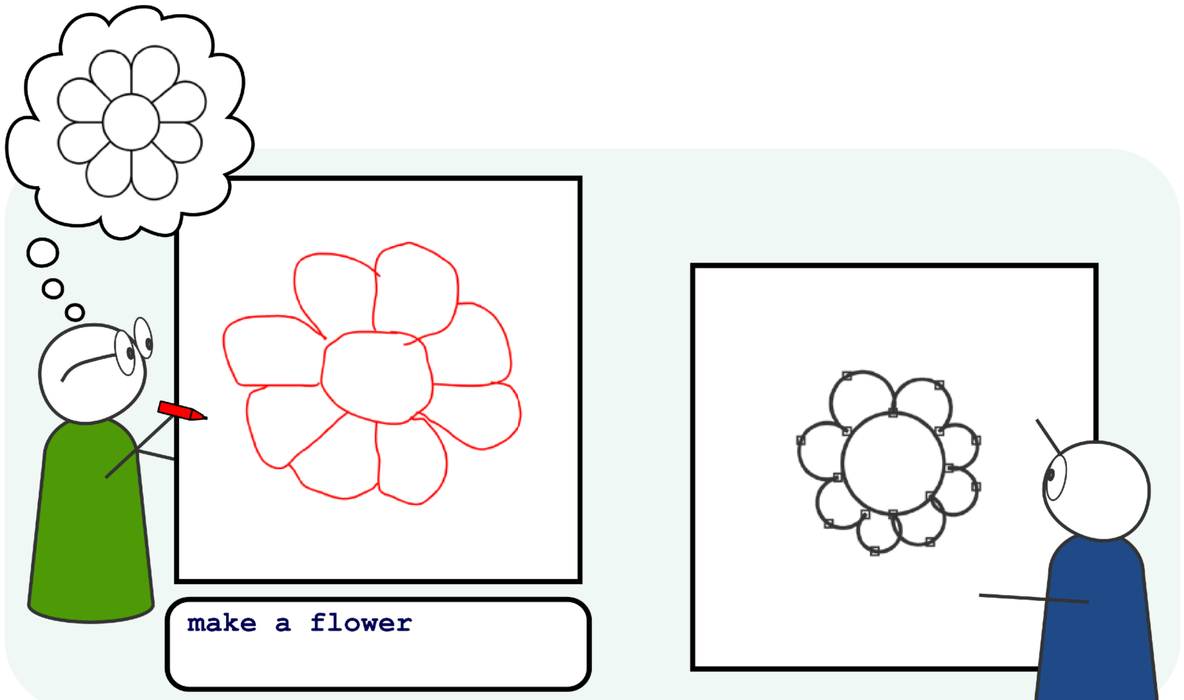
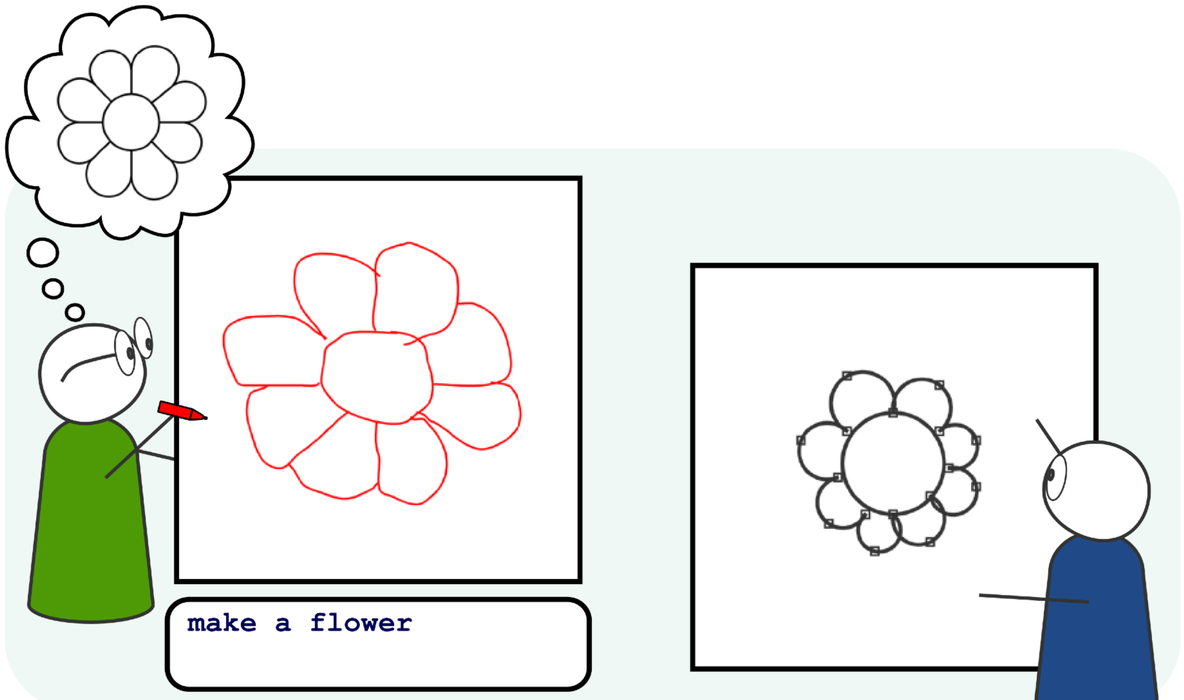


Designer

Maker

mrCAD game



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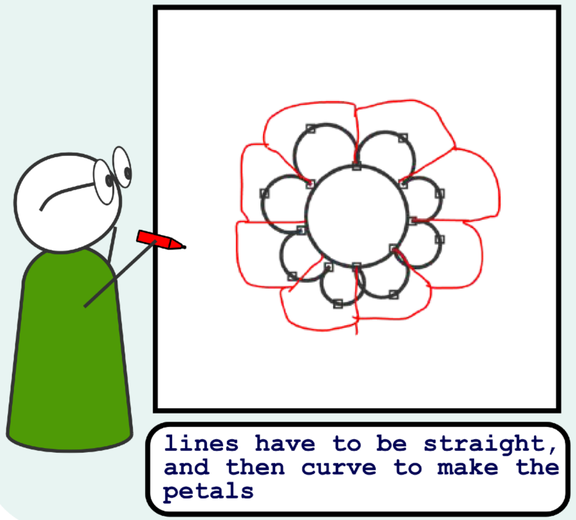
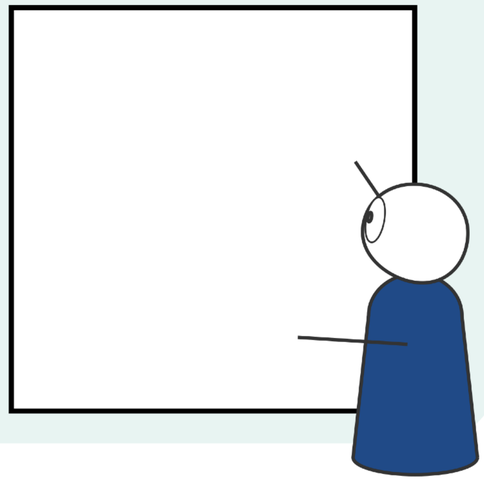
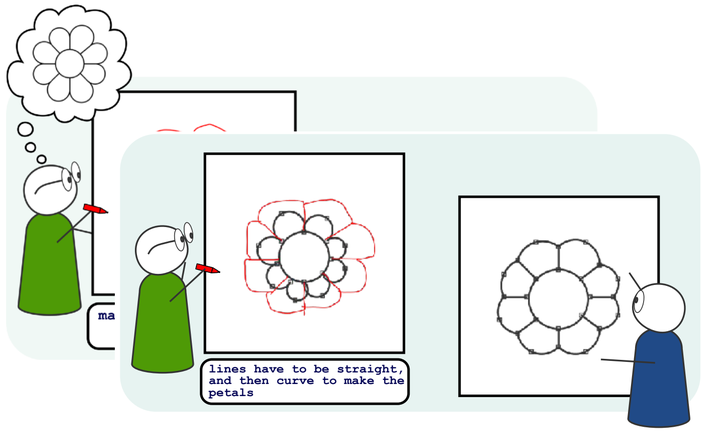




Dataset collection

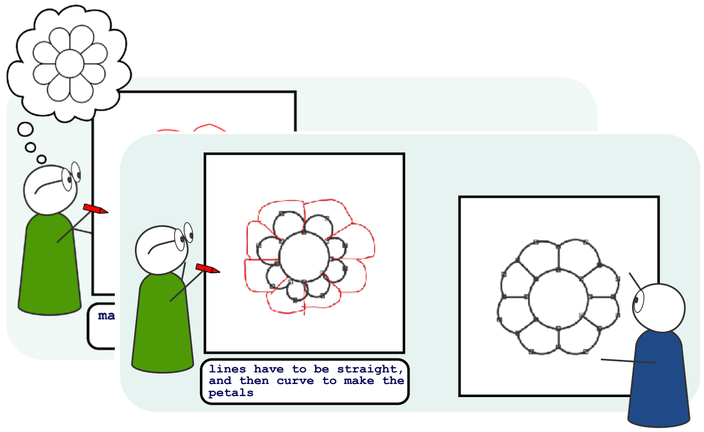
Dataset collection

- A *dyad* reconstructs a *target CAD* in a *rollout*



Dataset collection

- A *dyad* reconstructs a *target CAD* in a *rollout*
  - Participants from Prolific on a web interface



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- A *dyad* reconstructs a *target CAD* in a *rollout*
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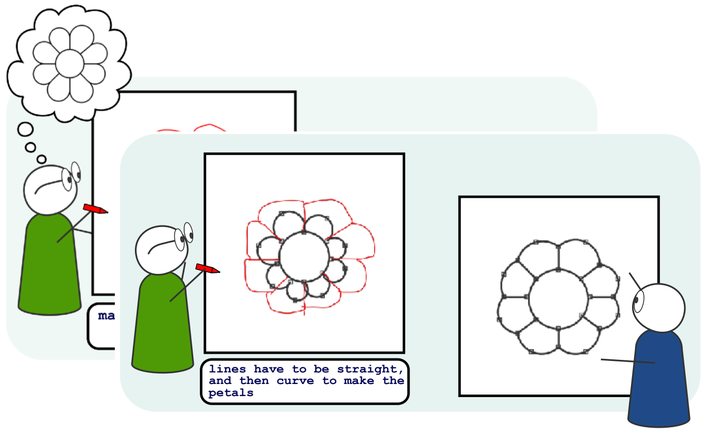

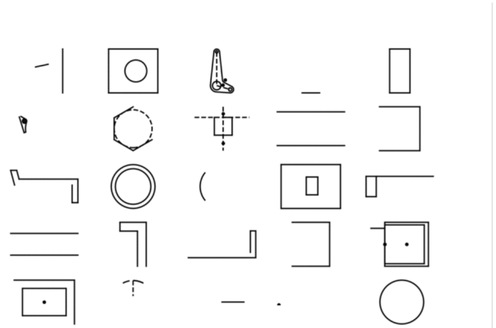

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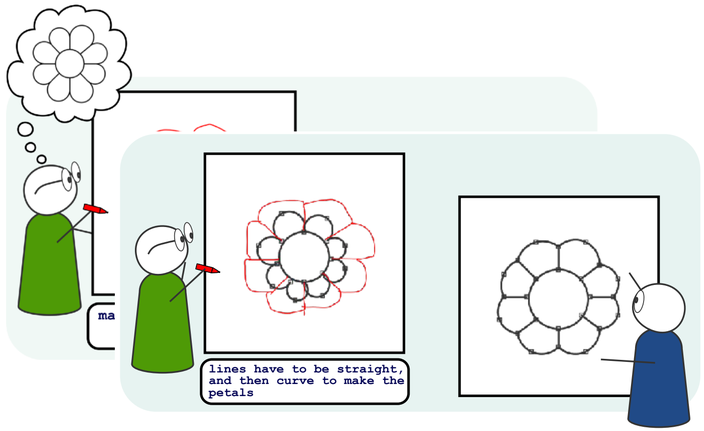

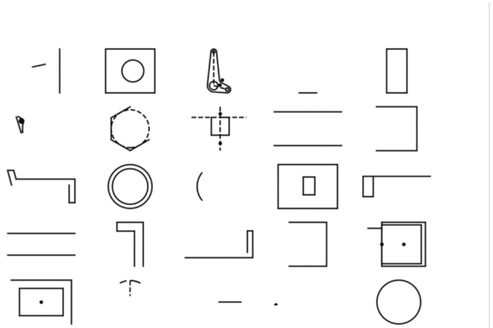

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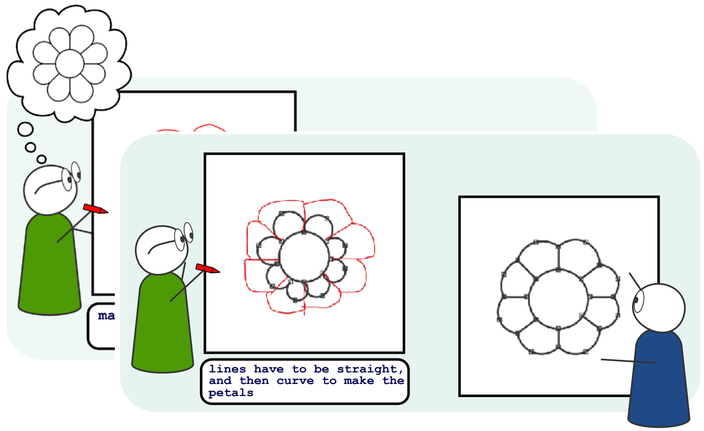
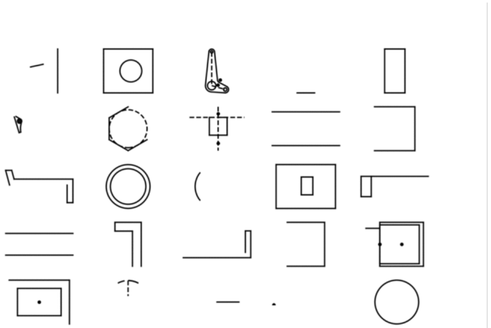
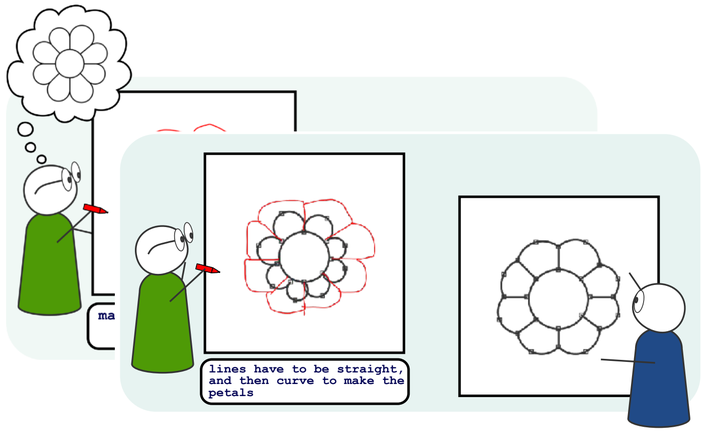
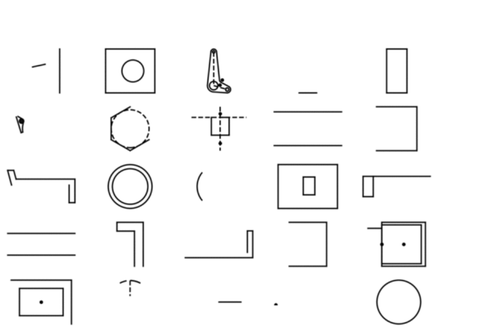


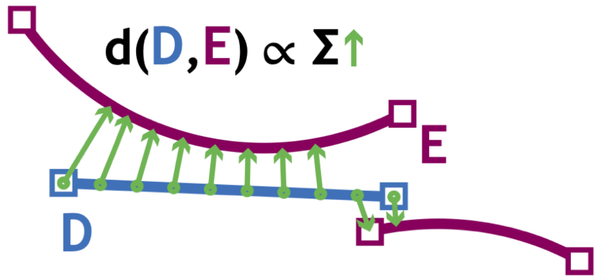


Dataset collection

- A *dyad* reconstructs a *target CAD* in a *rollout*
  - Participants from Prolific on a web interface
  - Diverse, interesting targets from SketchGraphs
- Duration of *rounds* is unconstrained
  - Total time for a game is bounded
  - Stop when close enough to target

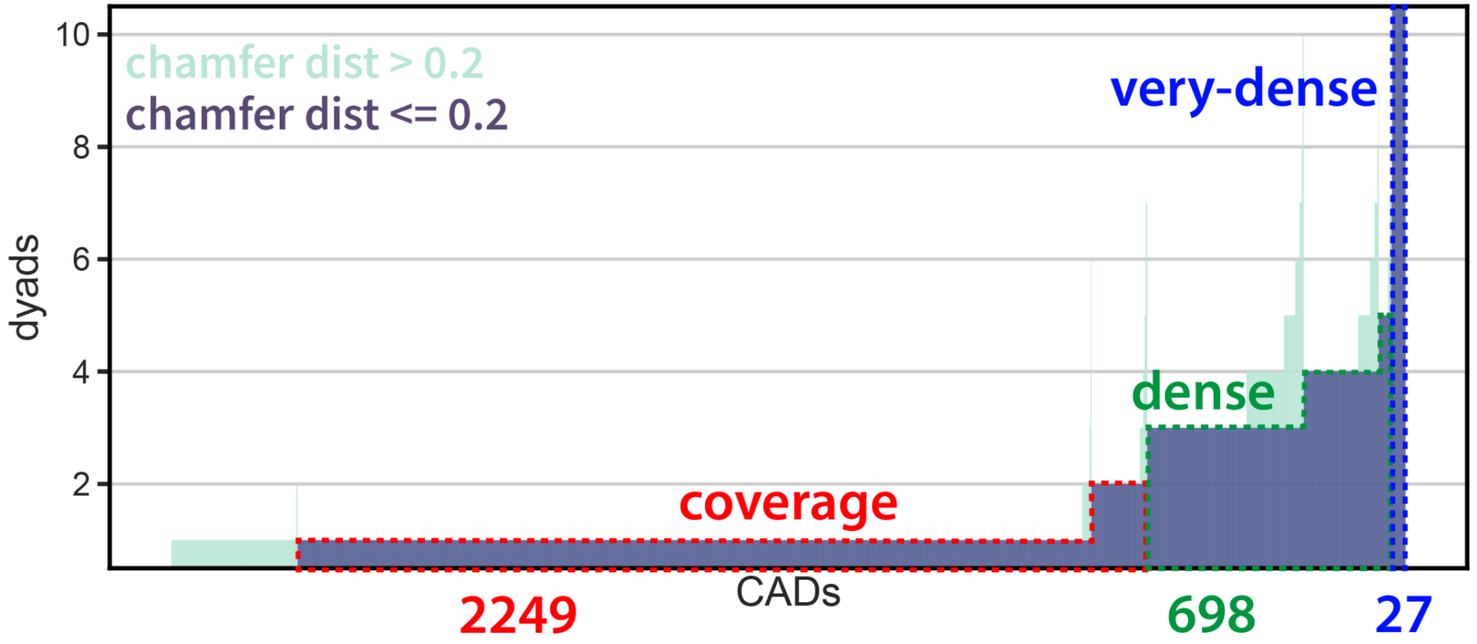




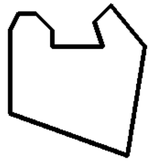


Dataset statistics

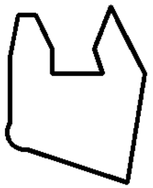
- **6,082** *rollouts*
- **15,163** *rounds* { **6,078** *generation* / **9,085** *refinement*
- **1,092** *dyads*
- **2,974** *unique CADs*



Refinement interactions



I imagine it a bit like bringing your thumb and finger together. IT's a little smaller than I've drawn there

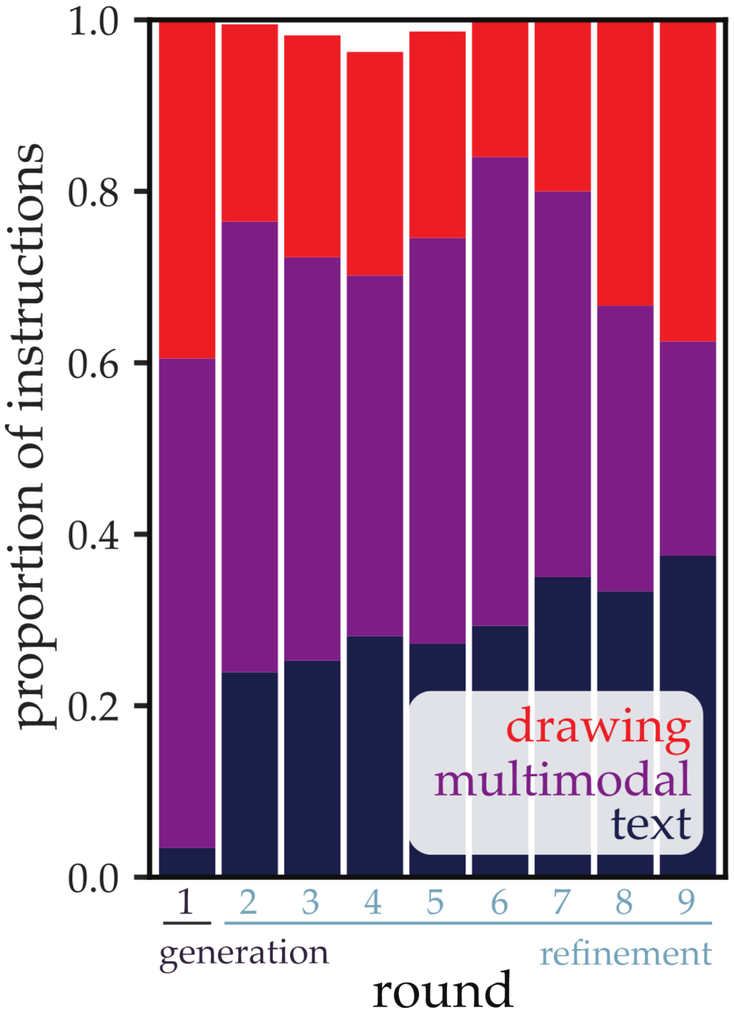


Really good start, just lengthen those top bits and curve the bottom left!

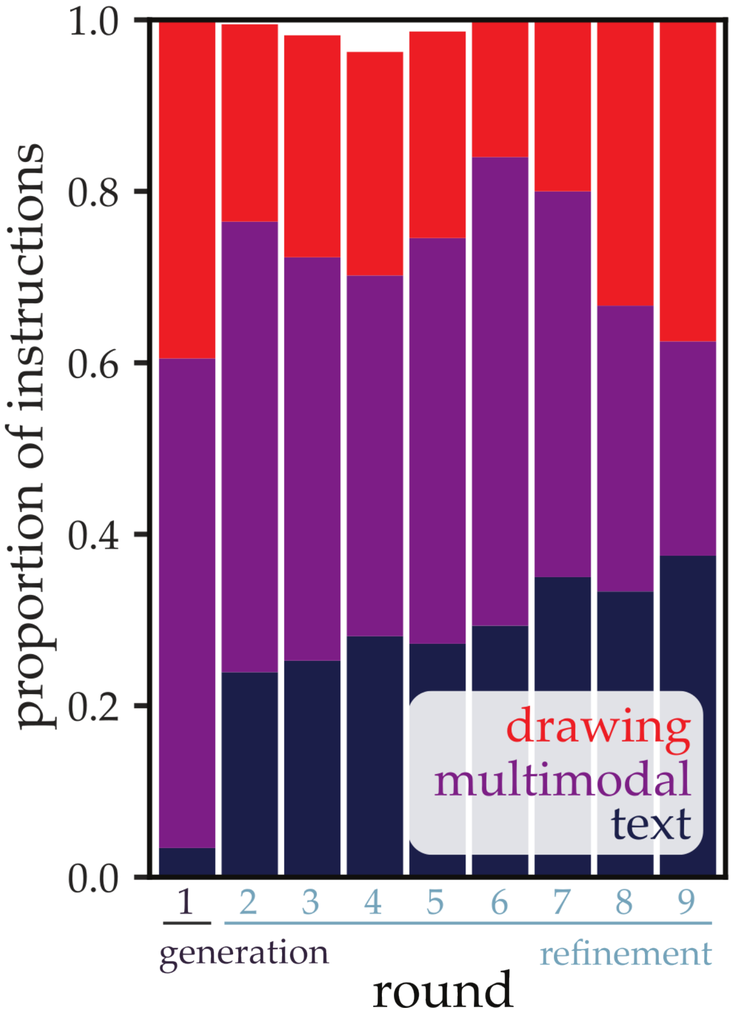
almost there! Just do curved lines for the 'fingers'

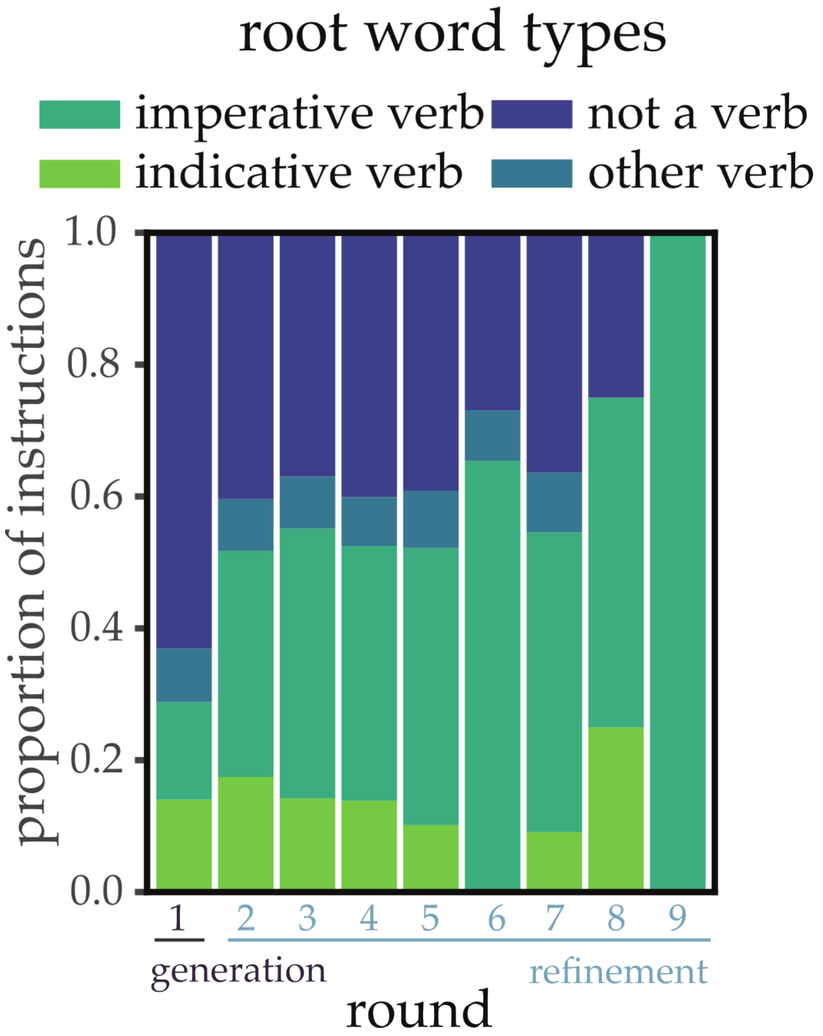
Communication about refinements

Communication about refinements



Communication about refinements





Evaluating multimodal agents

Evaluating multimodal agents

- Evaluate a multimodal agent in the *Maker* role
  - Given a sequence of steps leading up to the current state, and the instruction, produce *editing actions* that modify the design
  - Results of previous steps are from human actions

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$$1 - \frac{\text{distance to target after edit}}{\text{distance to target before edit}}$$

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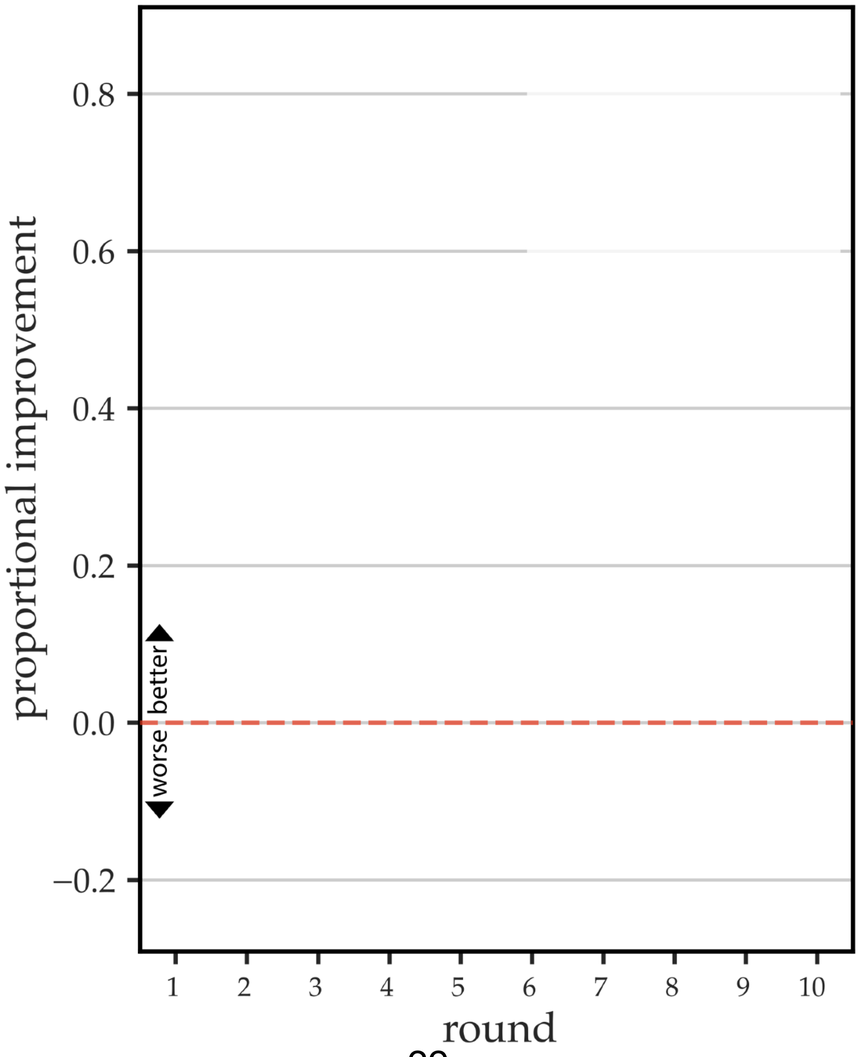
- Off-the-shelf models
  - GPT-4o, Claude-3.7-Sonnet, Qwen2.5-VL-7B-Instruct

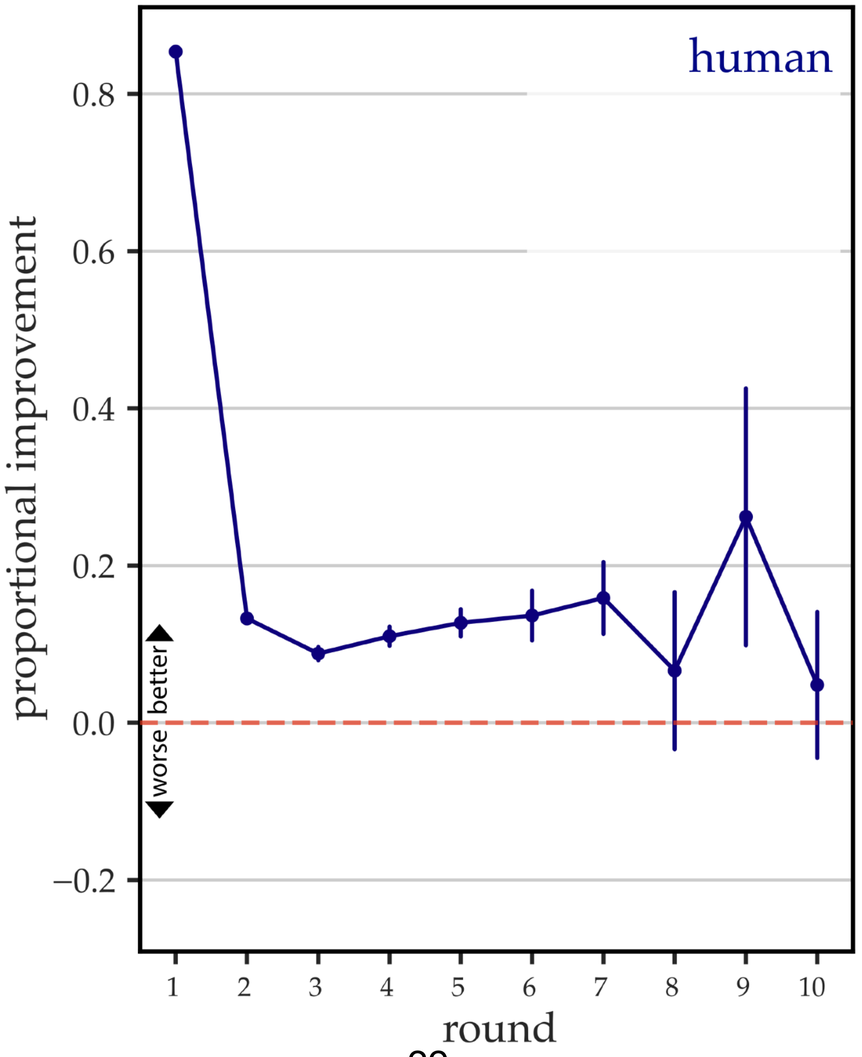
Evaluating multimodal agents

- Off-the-shelf models
  - GPT-4o, Claude-3.7-Sonnet, Qwen2.5-VL-7B-Instruct
- Finetuned model
  - Supervised finetuning on human editing actions given prior turns and instruction
  - Qwen2.5-VL-7B-Instruct

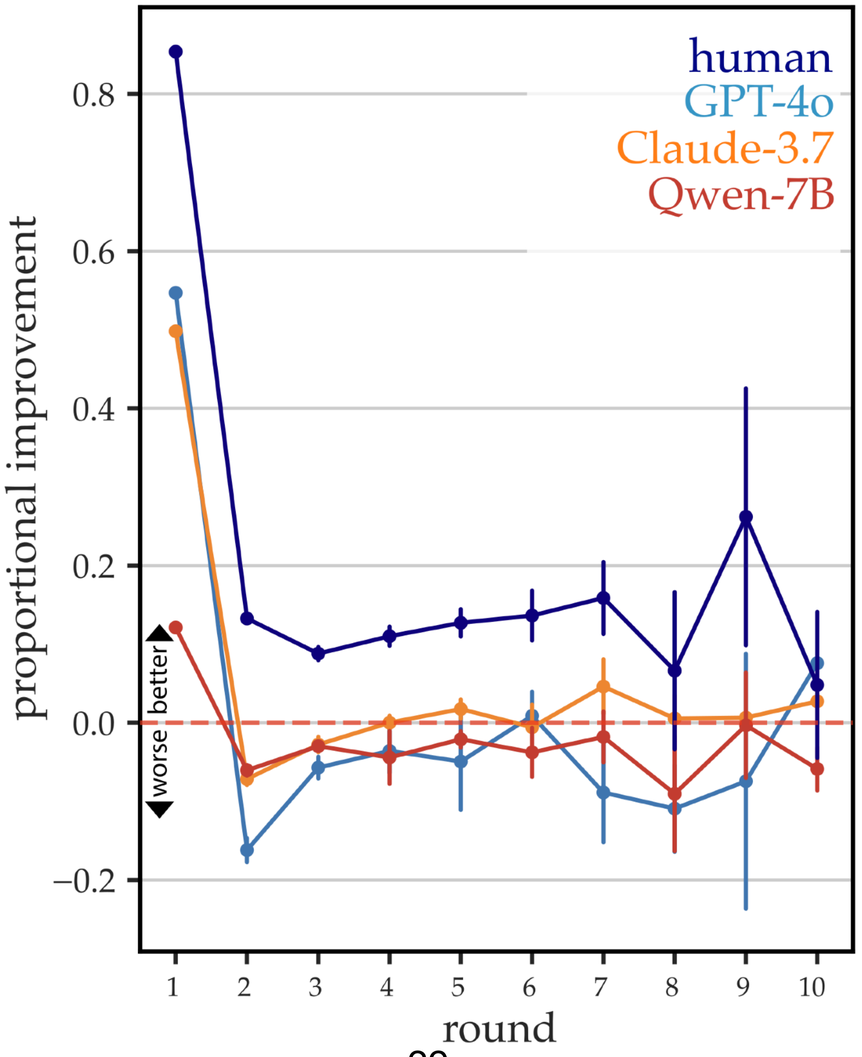
Evaluating multimodal agents



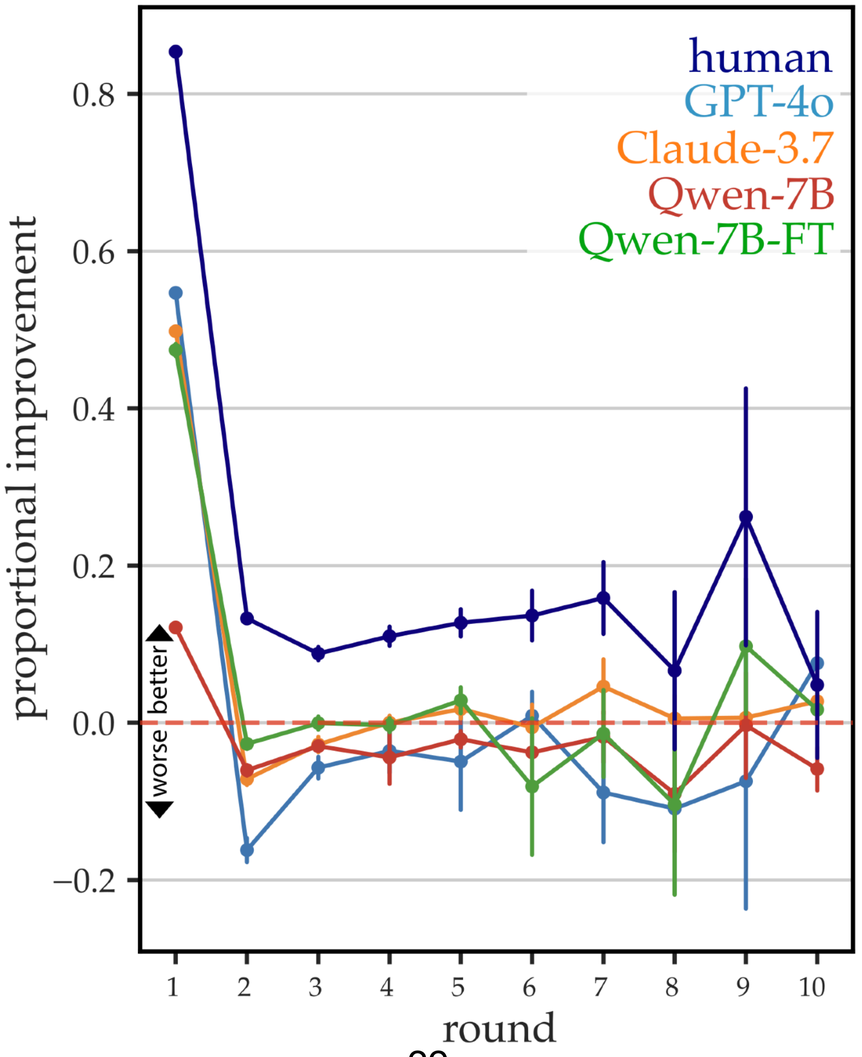
Evaluating multimodal agents



Evaluating multimodal agents



Evaluating multimodal agents

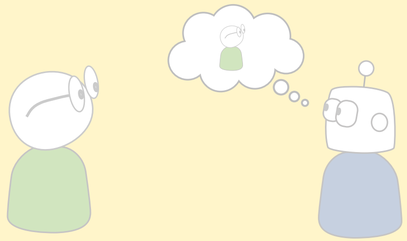


Takeaways

- People use language differently in the context of iterative refinements to something they are creating
- Refinement instructions need to be interpreted against the common ground of the current state of the object and prior turns
- Agents are better at understanding generation instructions than refinement instructions

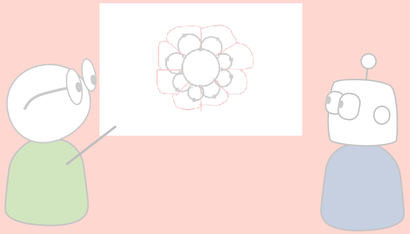
Part I: Reasoning about user intent

[Vaduguru, Fried, Pu, ICLR 2024; proposed work]



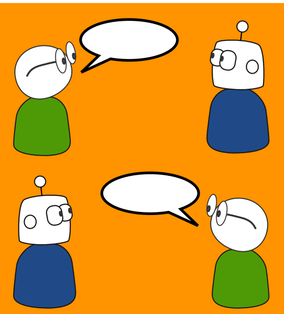
Part II: Reasoning about shared observation

[McCarthy, Vaduguru*,* et al.*, Findings of EMNLP, 2025]*



Part III: Reasoning about common ground over time

[Vaduguru, Hua, Artzi, Fried, ACL 2026; proposed work]



Ad hoc linguistic conventions

September 2025

I'm thinking of submitting a *paper on our work on convention formation* to the October ARR cycle

Yeah, sounds good!

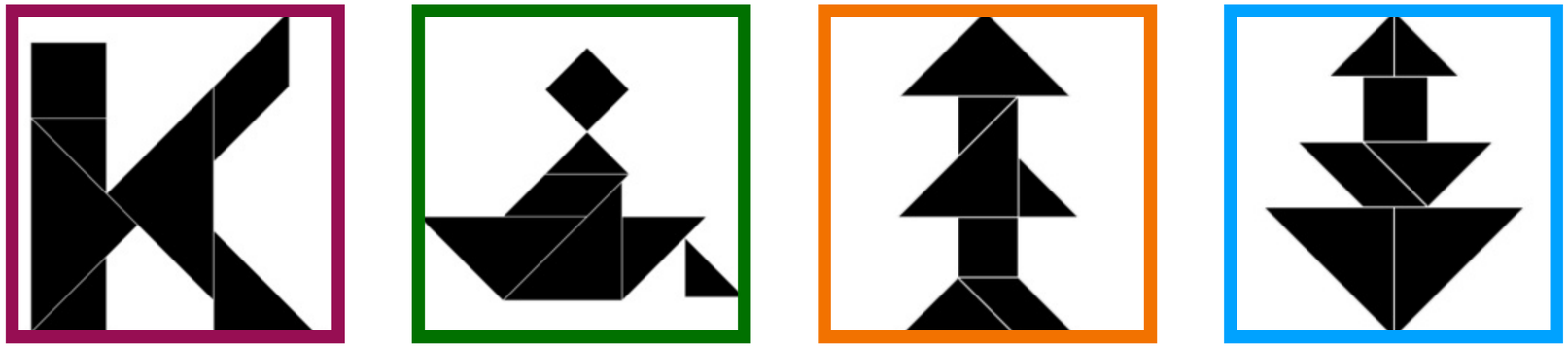
Ad hoc linguistic conventions

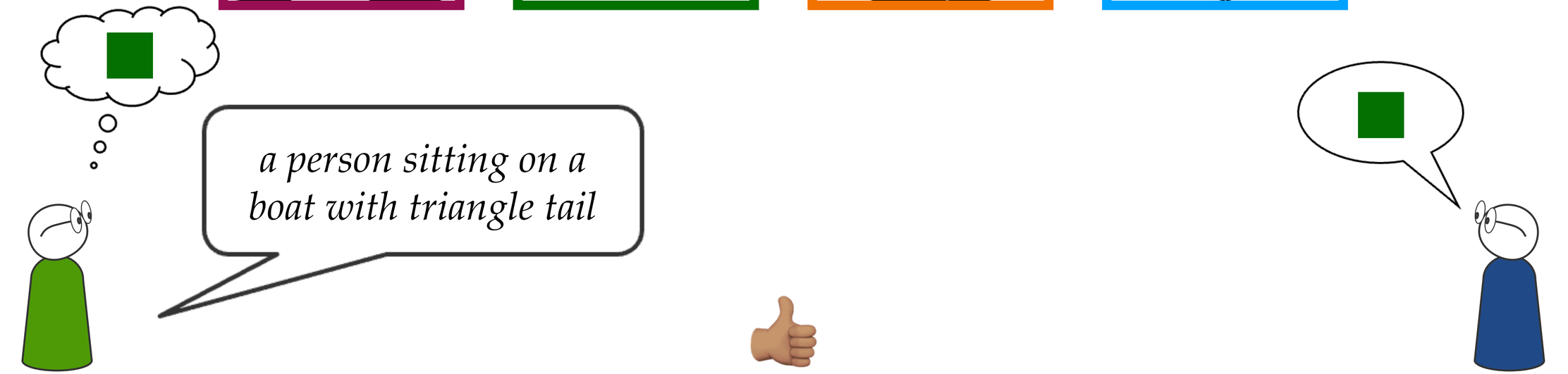
November 2025

The reviews for the *conventions paper* are out

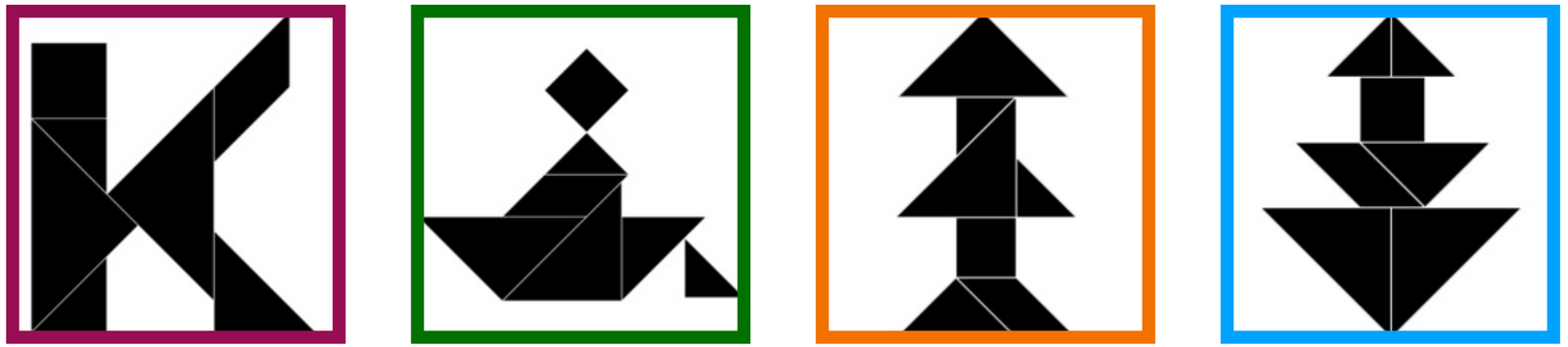
Looks promising!

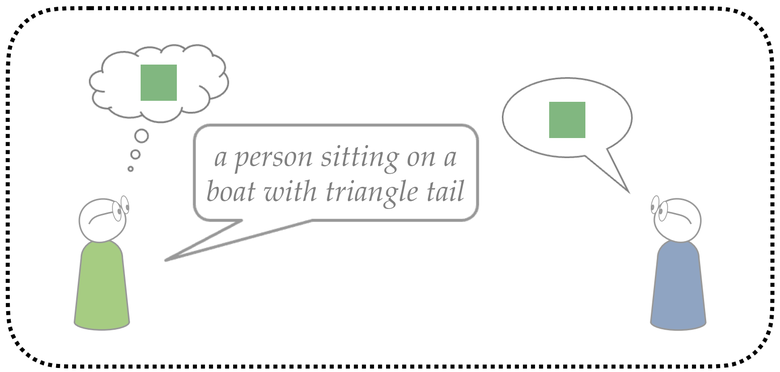
Reference games

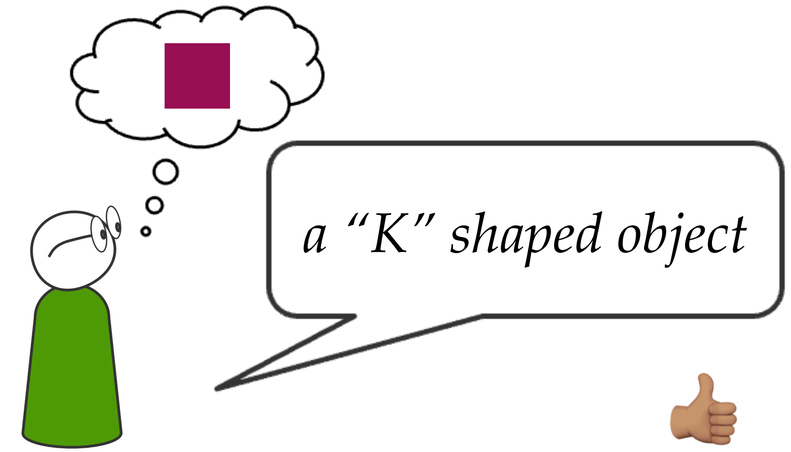




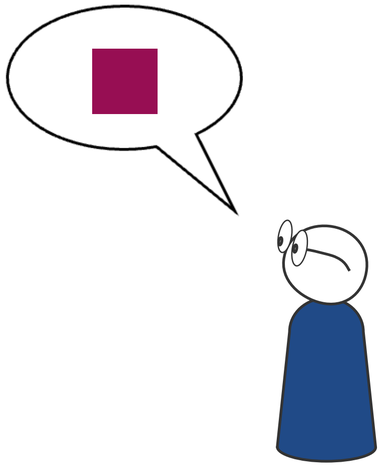
Repeated reference games







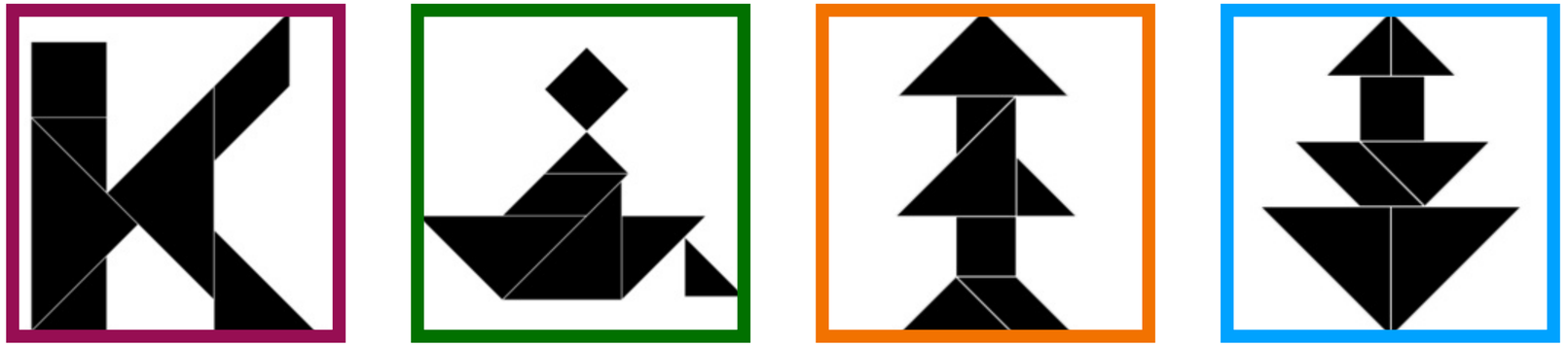
Speaker

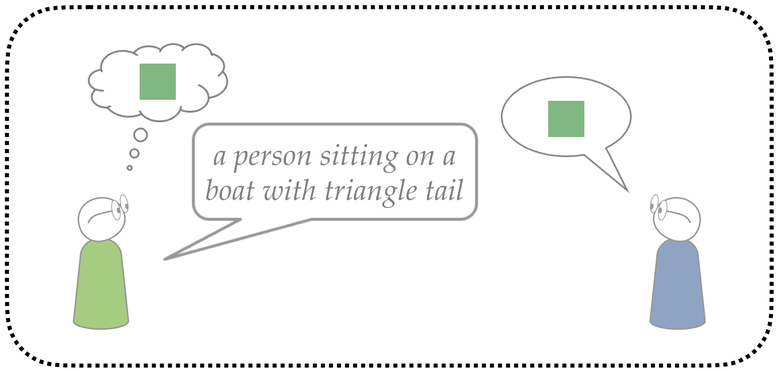


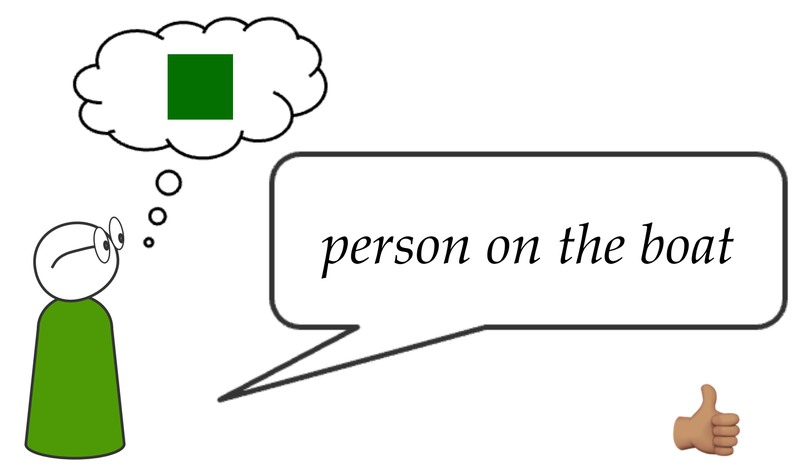
Listener

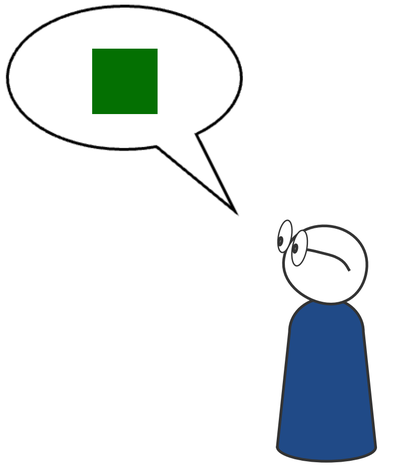
[Kraus and Weinheimer, 1966; Clark and Wilkes-Gibbs, 1986]

Ad hoc conventions in repeated reference games









[Hawkins, et al., 2020]

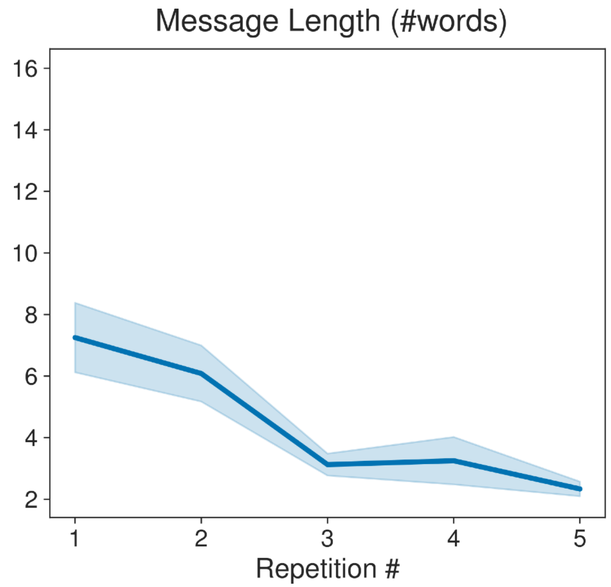
Conventions are *efficient* and *consistent*

[Hua and Artzi, 2024]

Conventions are *efficient* and *consistent*

a person sitting on a boat with triangle tail

person on the boat



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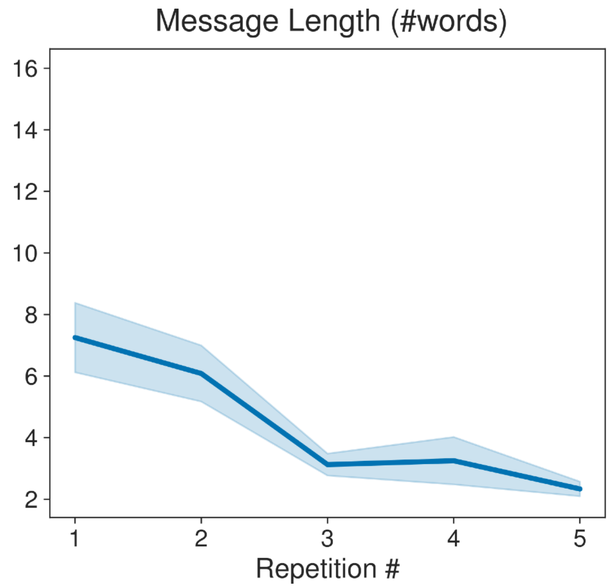

Word Novelty Distance = 1 × insertions and substitutions + 0 × alignments + 0 × deletions = 1

[Hua and Artzi, 2024]

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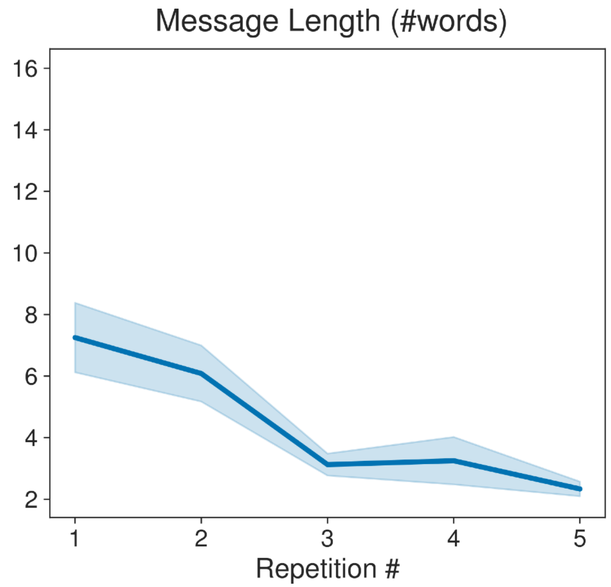
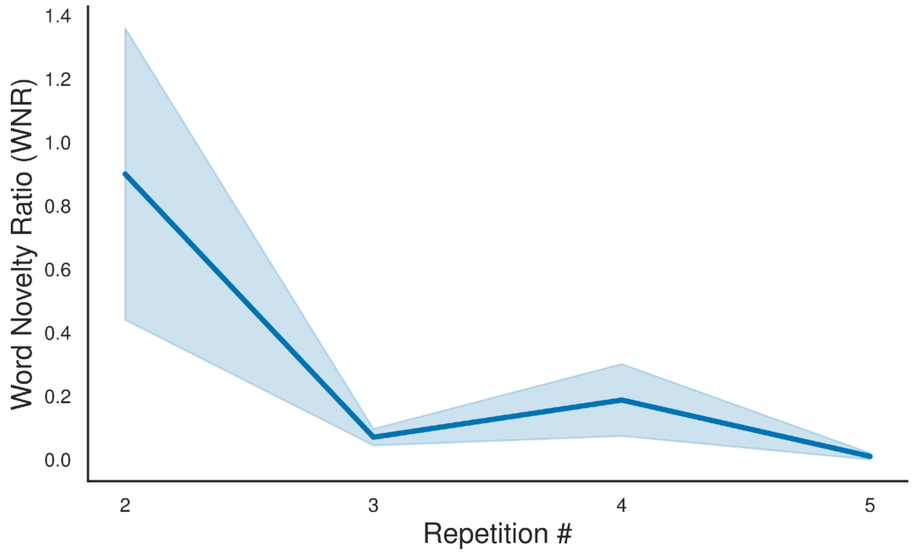




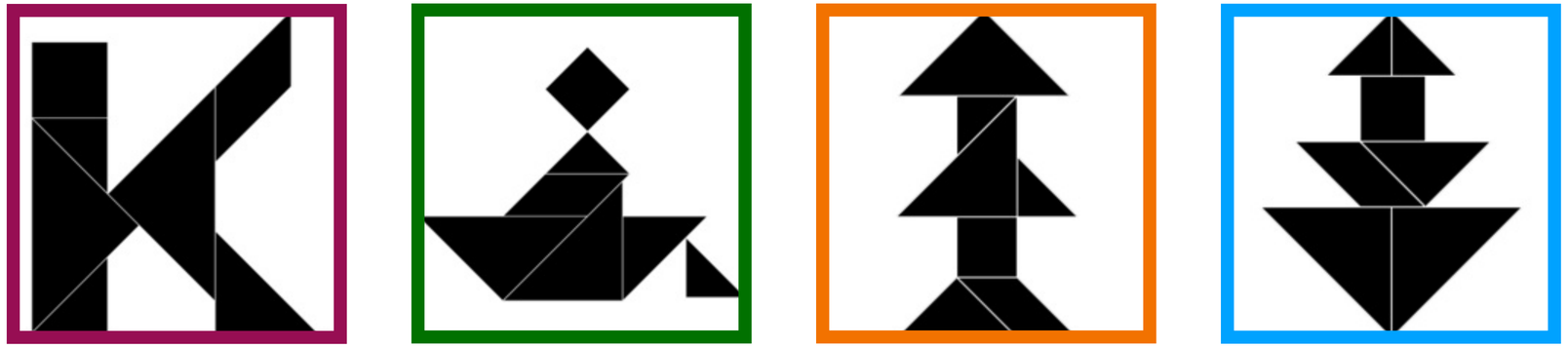
Word Novelty Distance = 1 × insertions and substitutions + 0 × alignments + 0 × deletions = 1

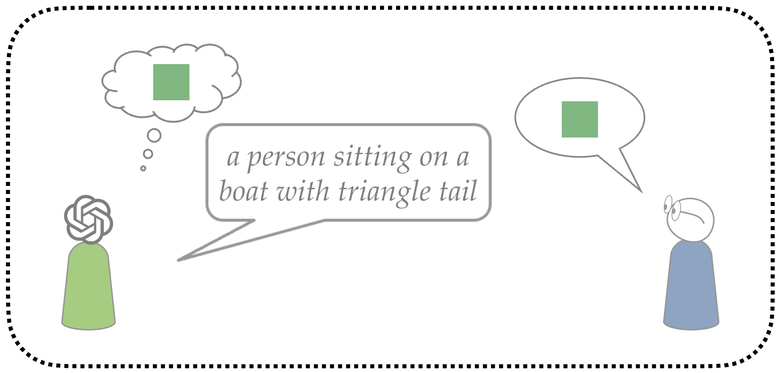
[Hua and Artzi, 2024]

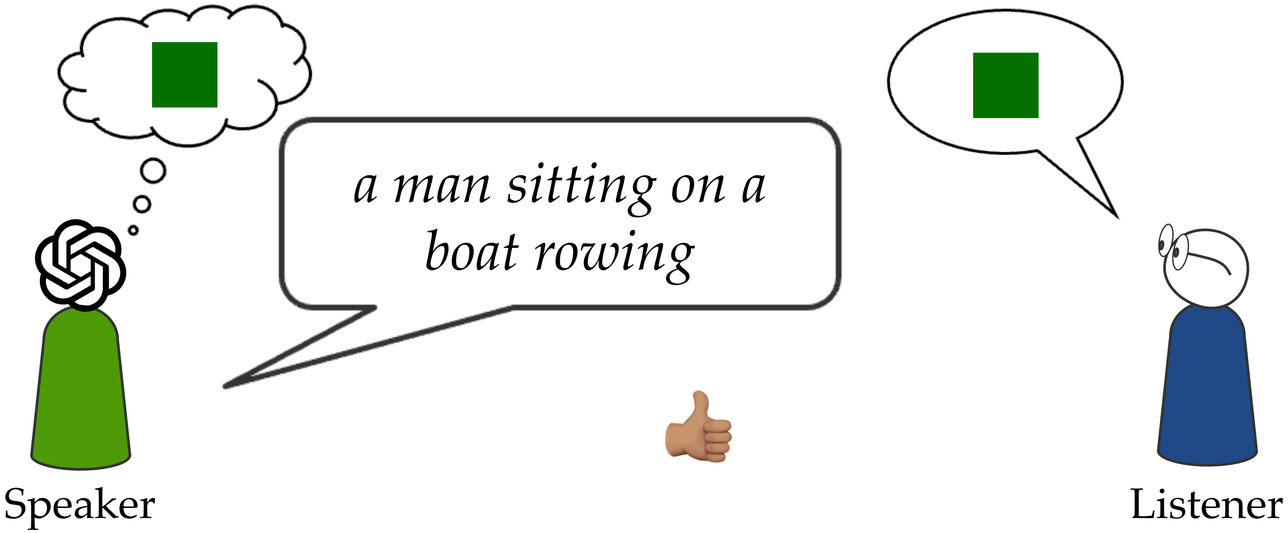
Multimodal language agents don't form conventions

Talk Less, Interact Better: Evaluating In-context Conversational Adaptation in Multimodal LLMs

Yilun Hua and Yoav Artzi
Department of Computer Science and Cornell Tech
Cornell University
{yilunhua, yoav}@cs.cornell.edu





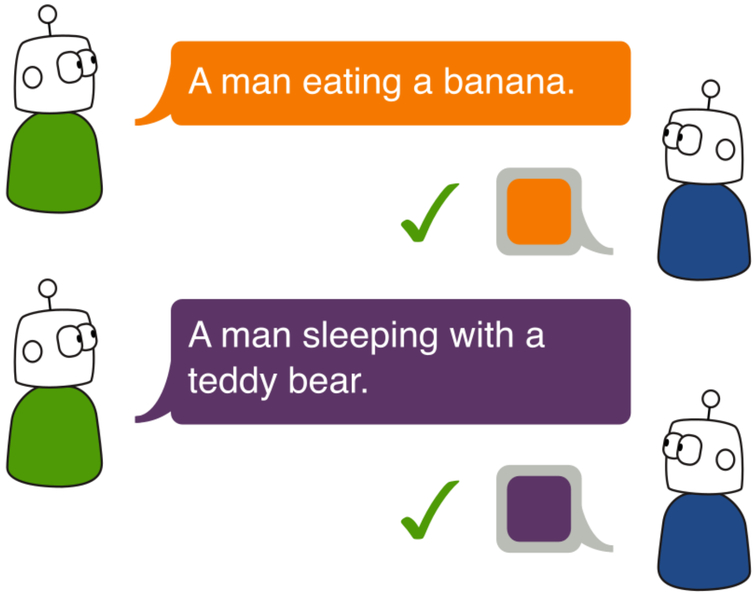


How can we get models to communicate efficiently in context?

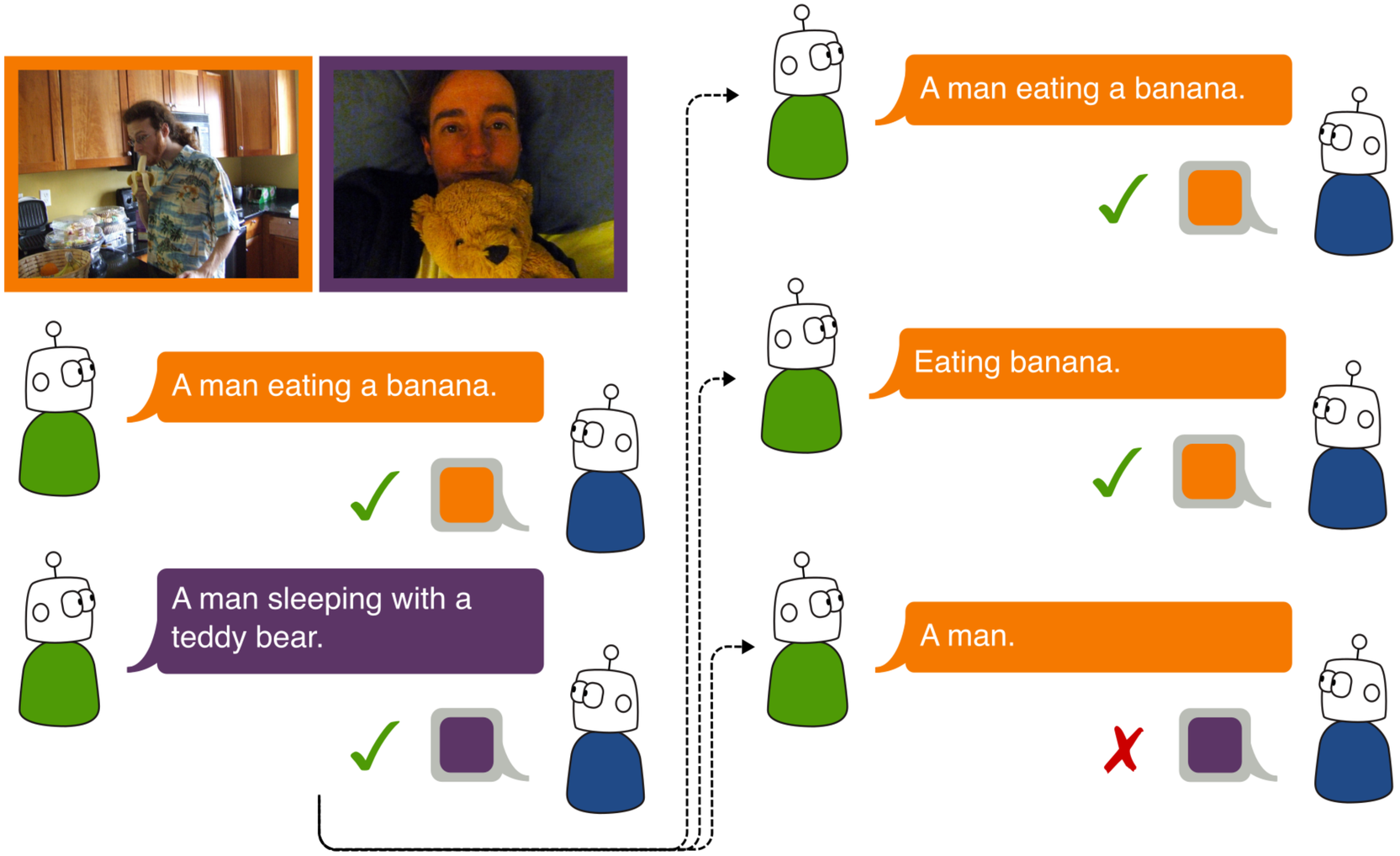
- Conventions are formed between a pair of agents over multiple rounds of interaction
- This kind of data is hard to collect
  - You need to have a model interact with the same person for a long time, explore different strategies
- An alternative is to train agents in *simulated* interactions with another agent

Simulating repeated reference games

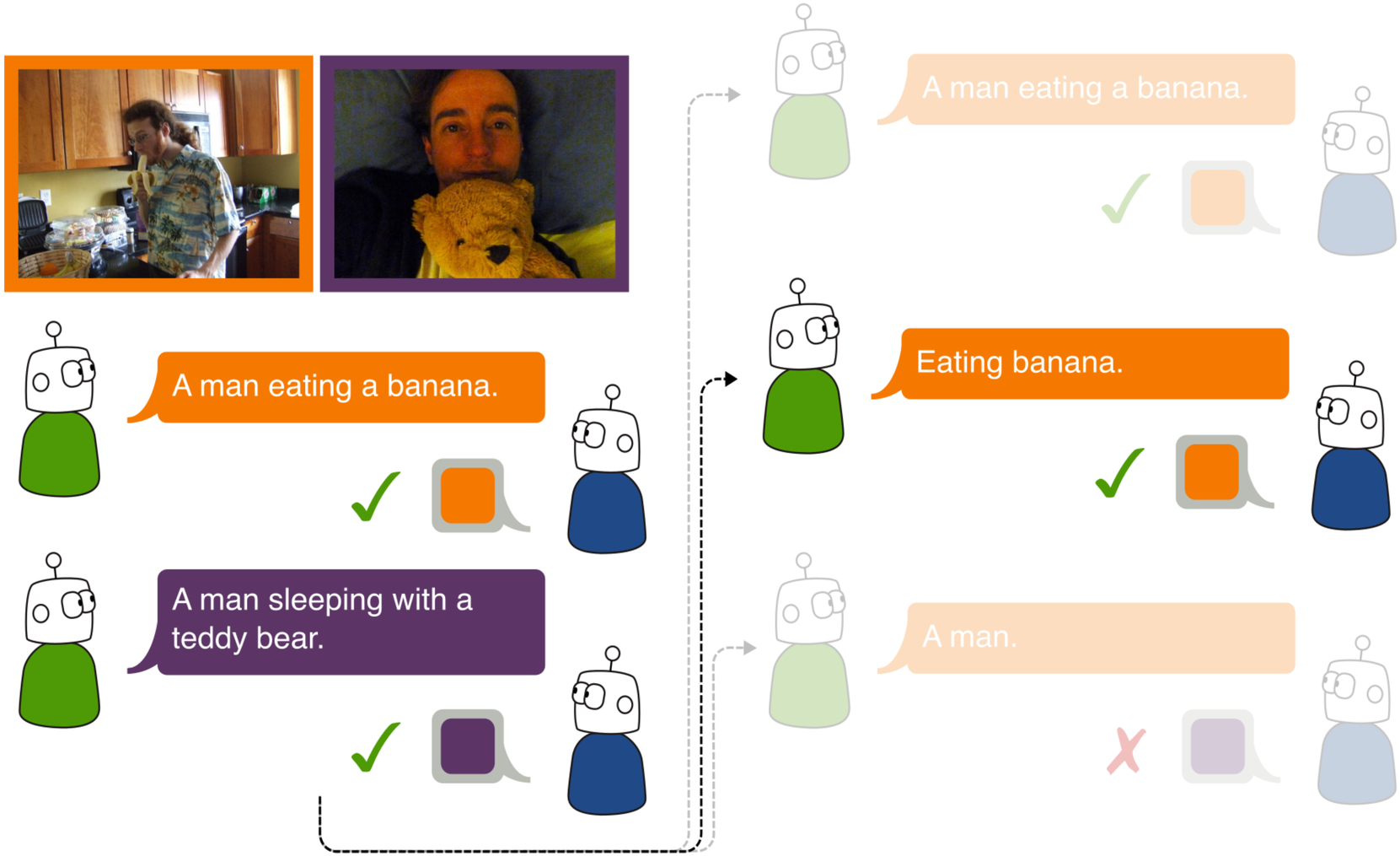
Simulating repeated reference games



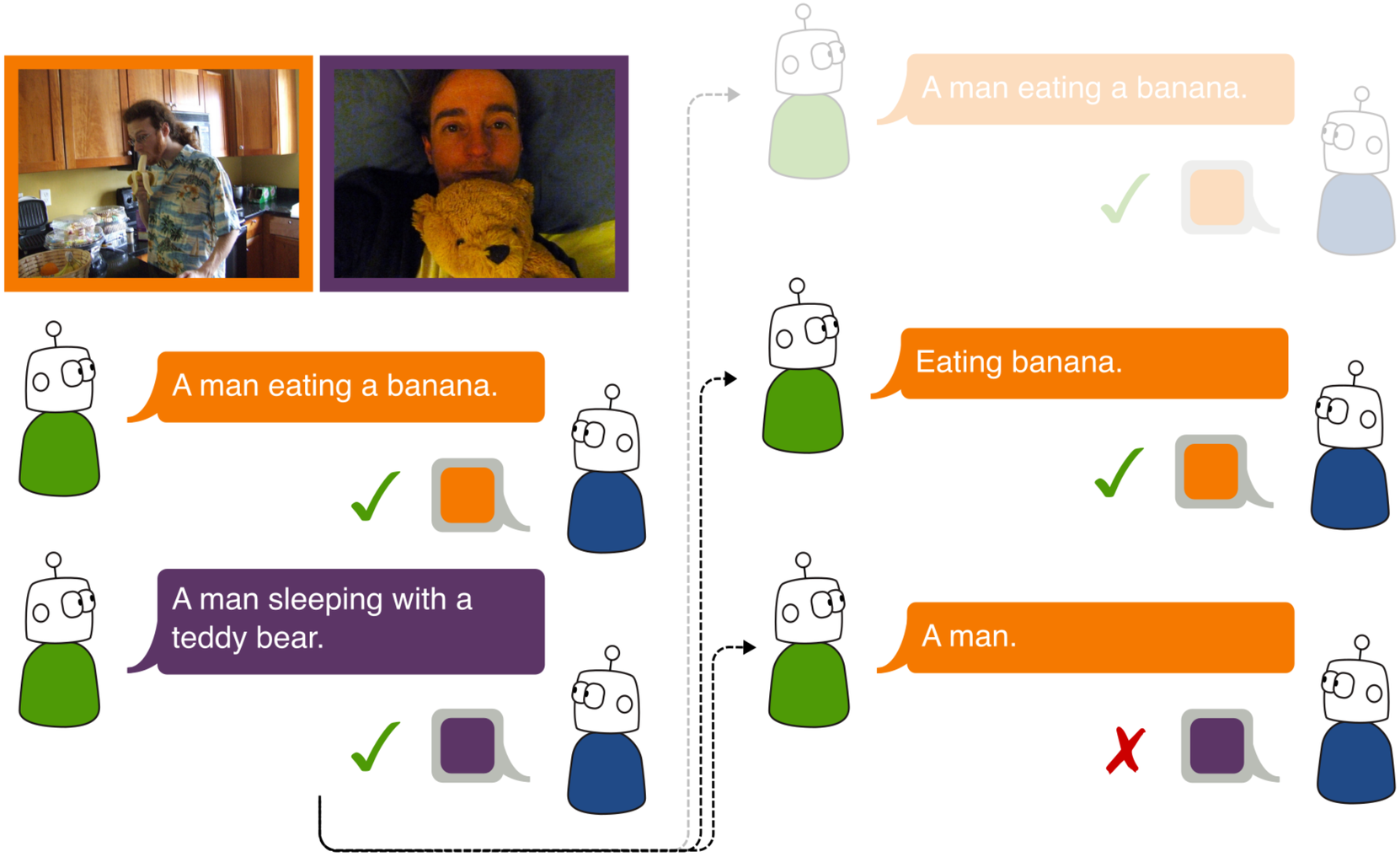
Simulating repeated reference games



Simulating repeated reference games

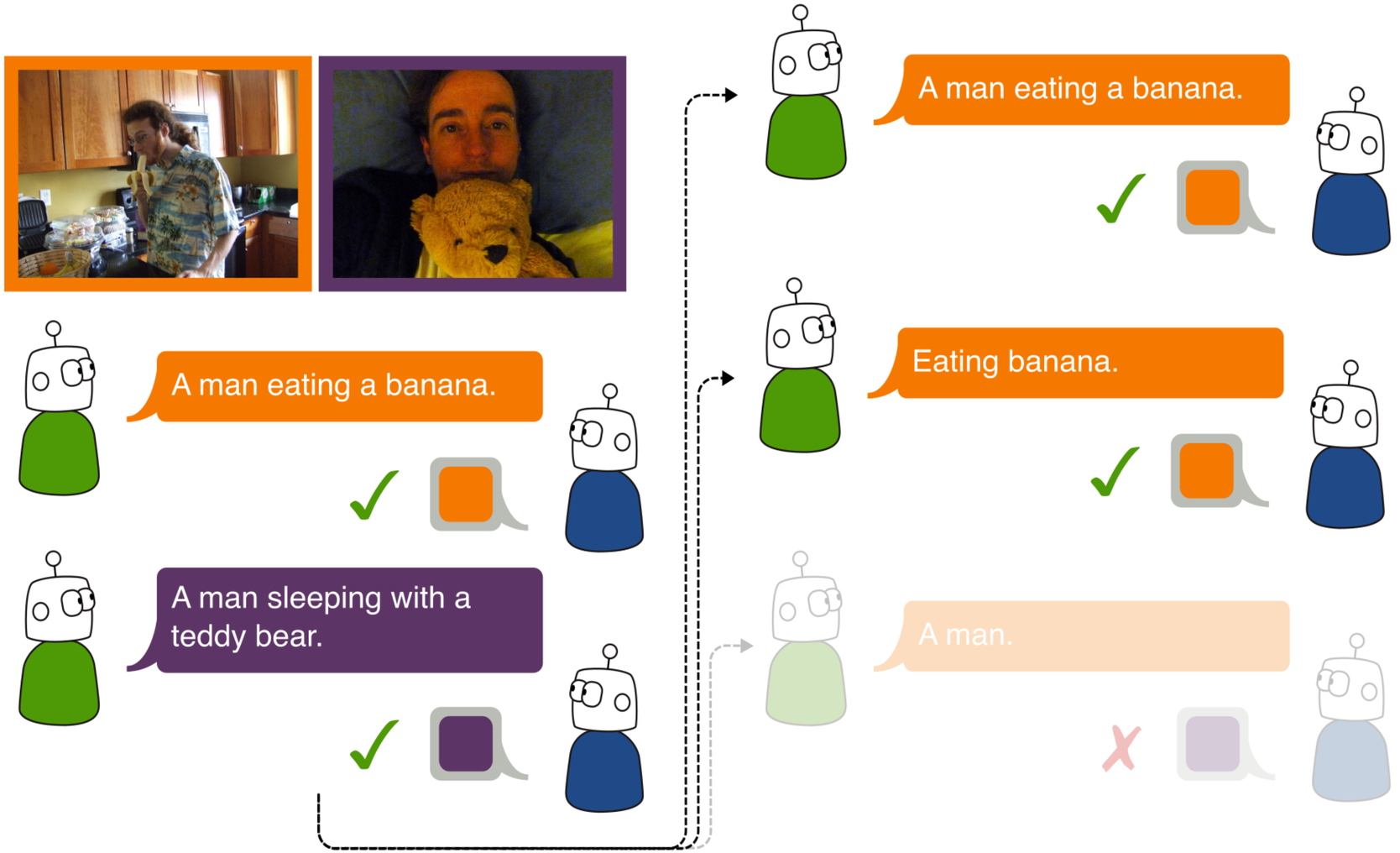


Simulating repeated reference games

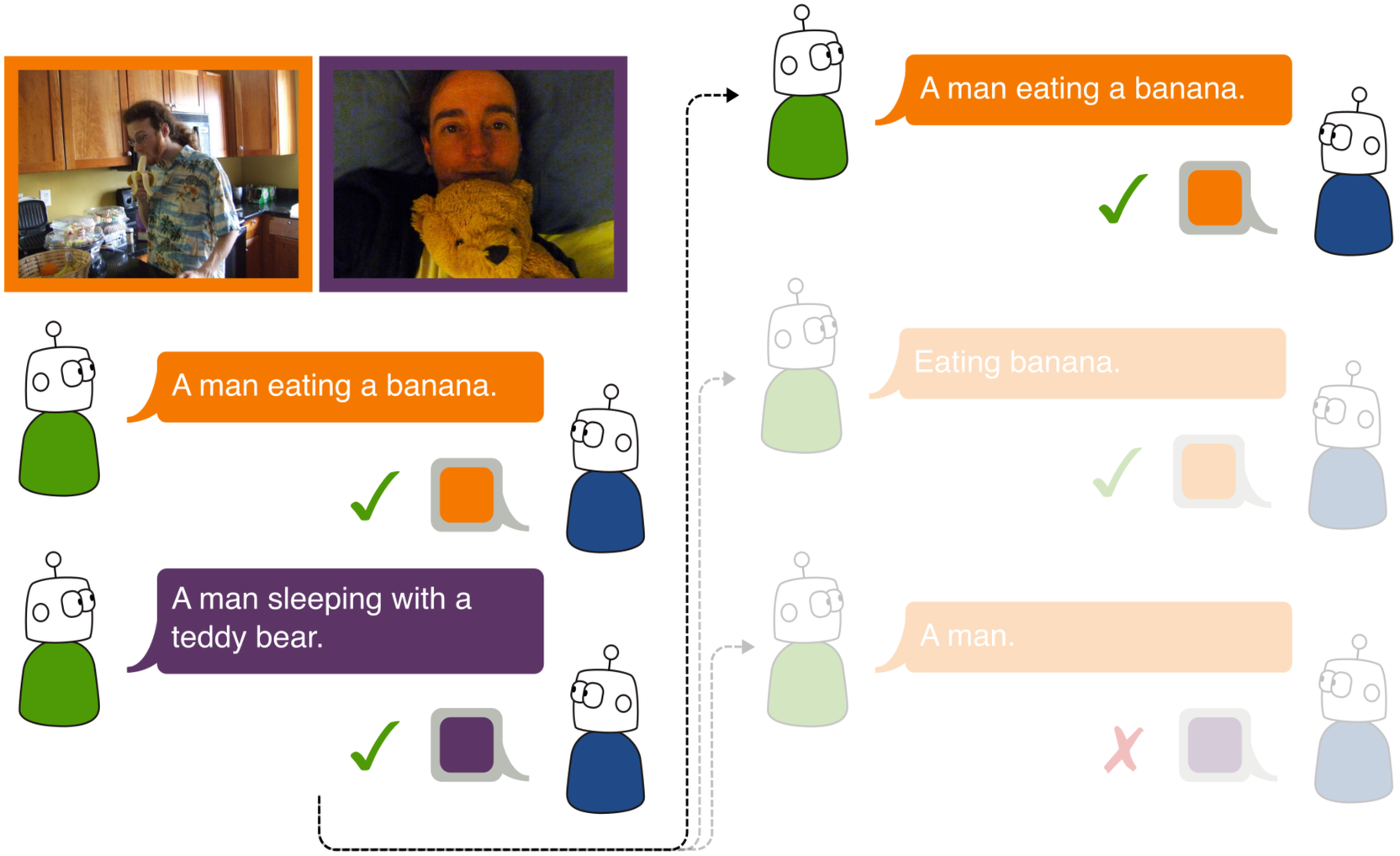


SUCCESS

Simulating repeated reference games

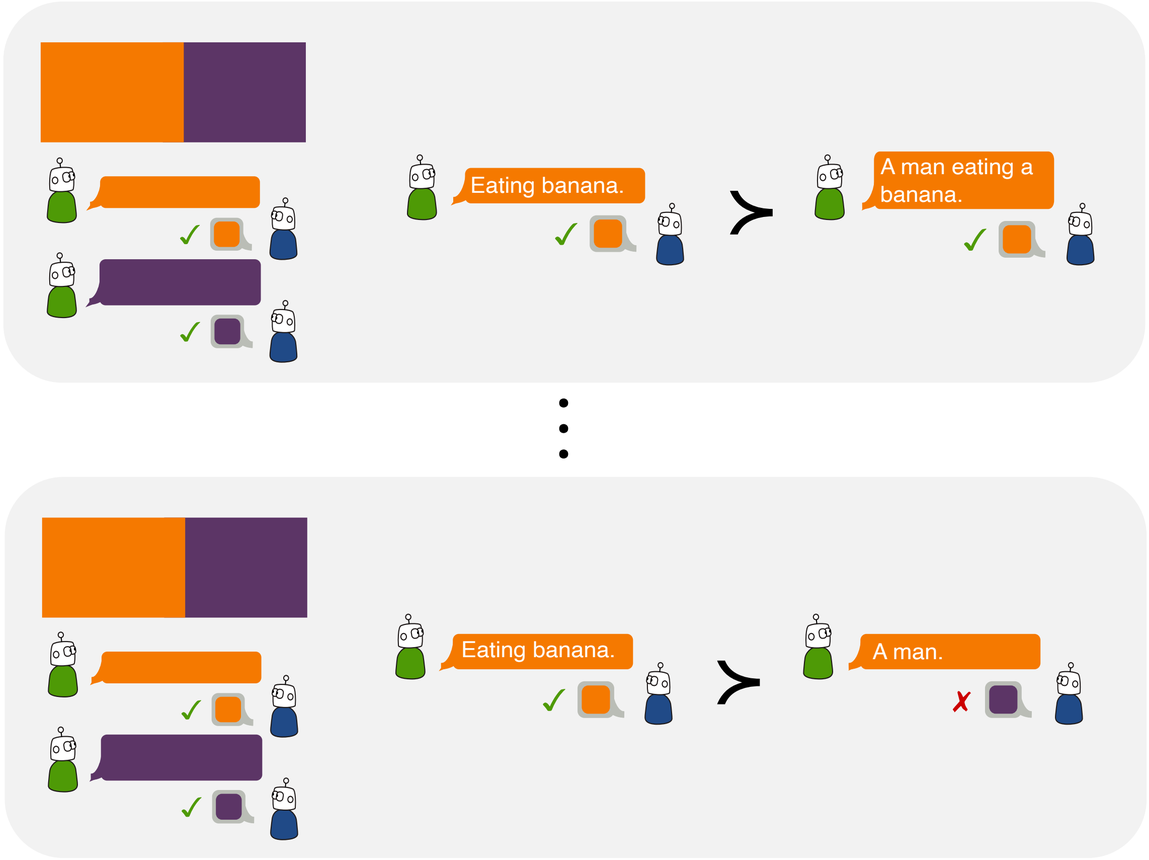


Simulating repeated reference games



Training on dataset of simulated games

Gather dataset from simulated games and train a VLM speaker agent with offline preference optimisation (IPO)



Models

- All games are simulated using a Gemma-3-12B-Base model for both the speaker and the listener
- We then update the *same* model using preference data
  - SUCCESS+COST
  - SUCCESS
  - COST

Evaluation

- Evaluate in interactions with people
  - Model speaker-human listener games
  - Each listener plays 2 games
    - One with the SUCCESS+COST model
    - Other with either SUCCESS model or instruction-tuned Gemma model
  - Record all messages, listener responses and success, and response time

Training with SUCCESS and COST elicits conventions

A baby playing with a kite.

Trial #2

Kites in the shape of octopus and shark.

Trial #4

Baby.

Trial #6

Octopus and shark.

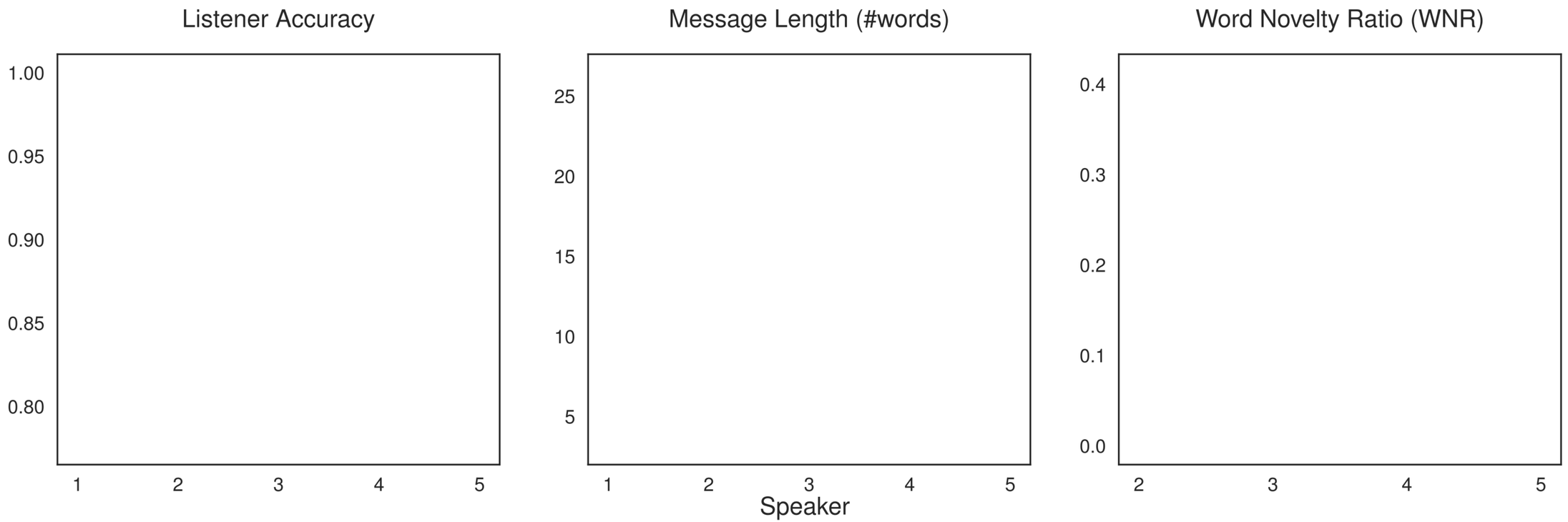
Trial #7

...

Baby.

Trial #19

Training with SUCCESS and COST elicits conventions



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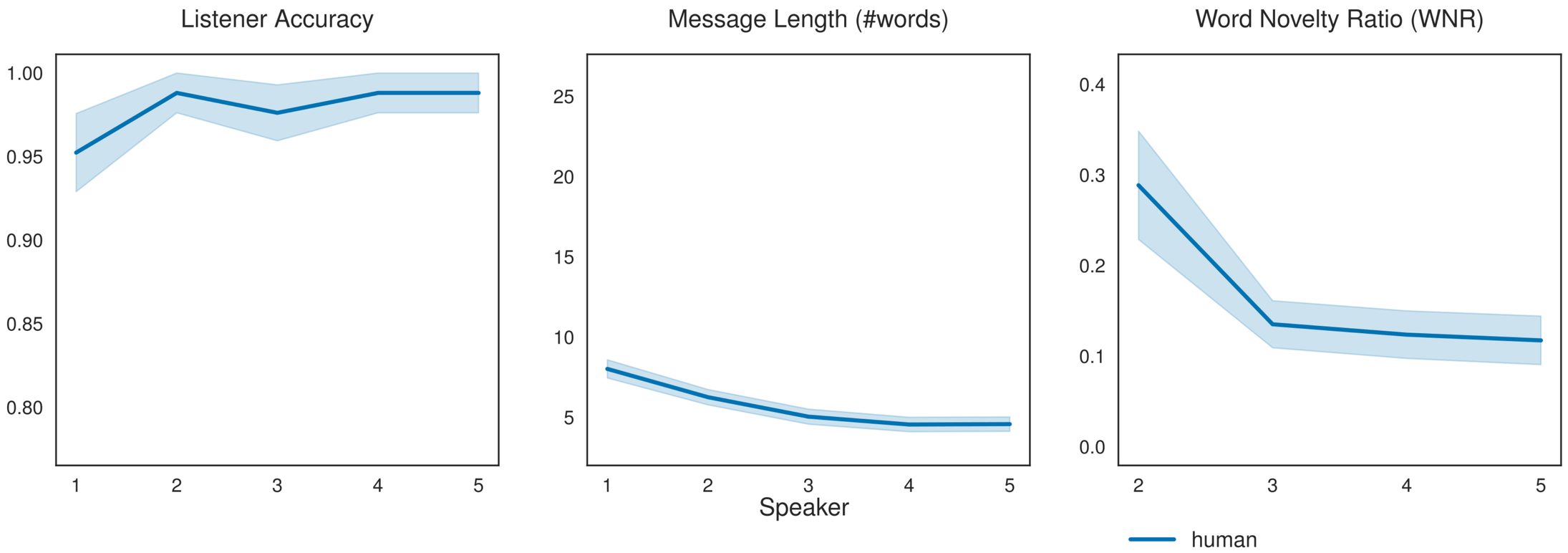

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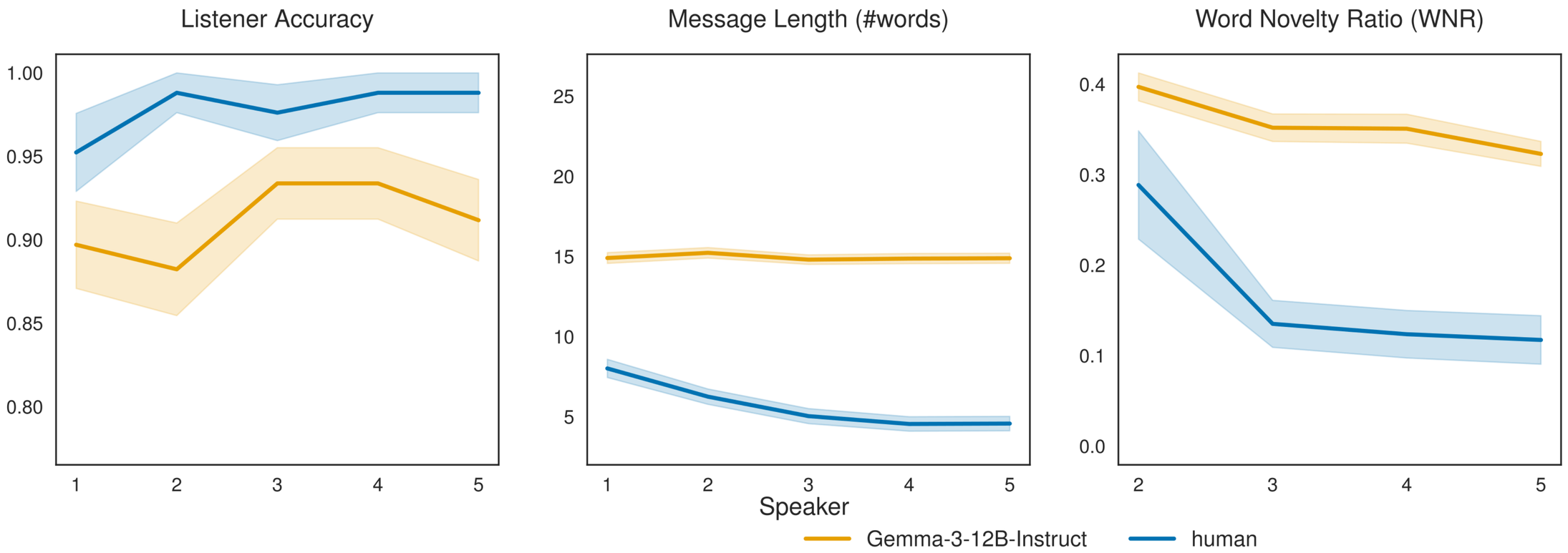

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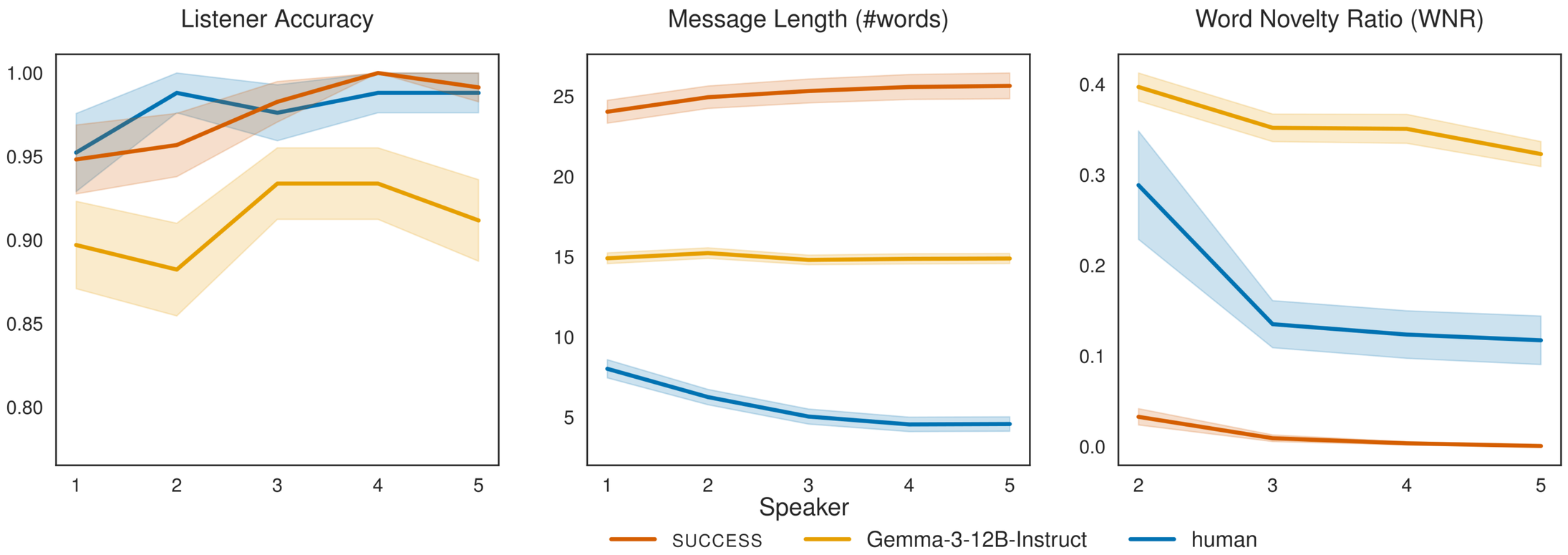

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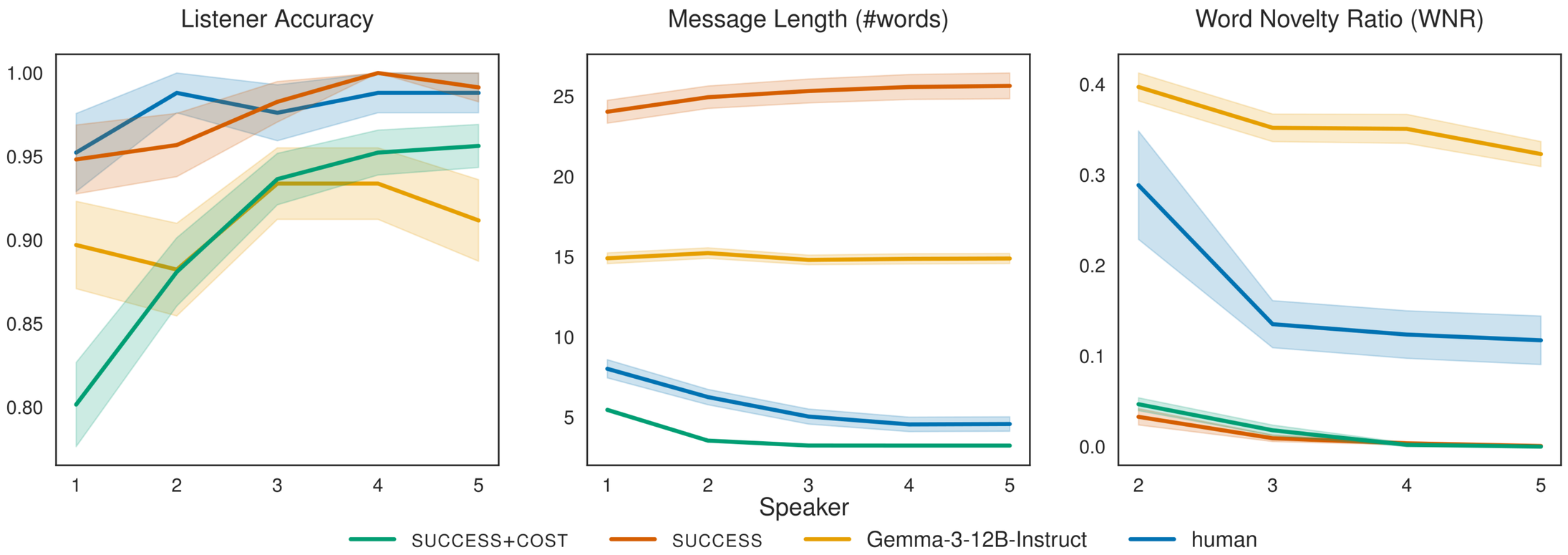
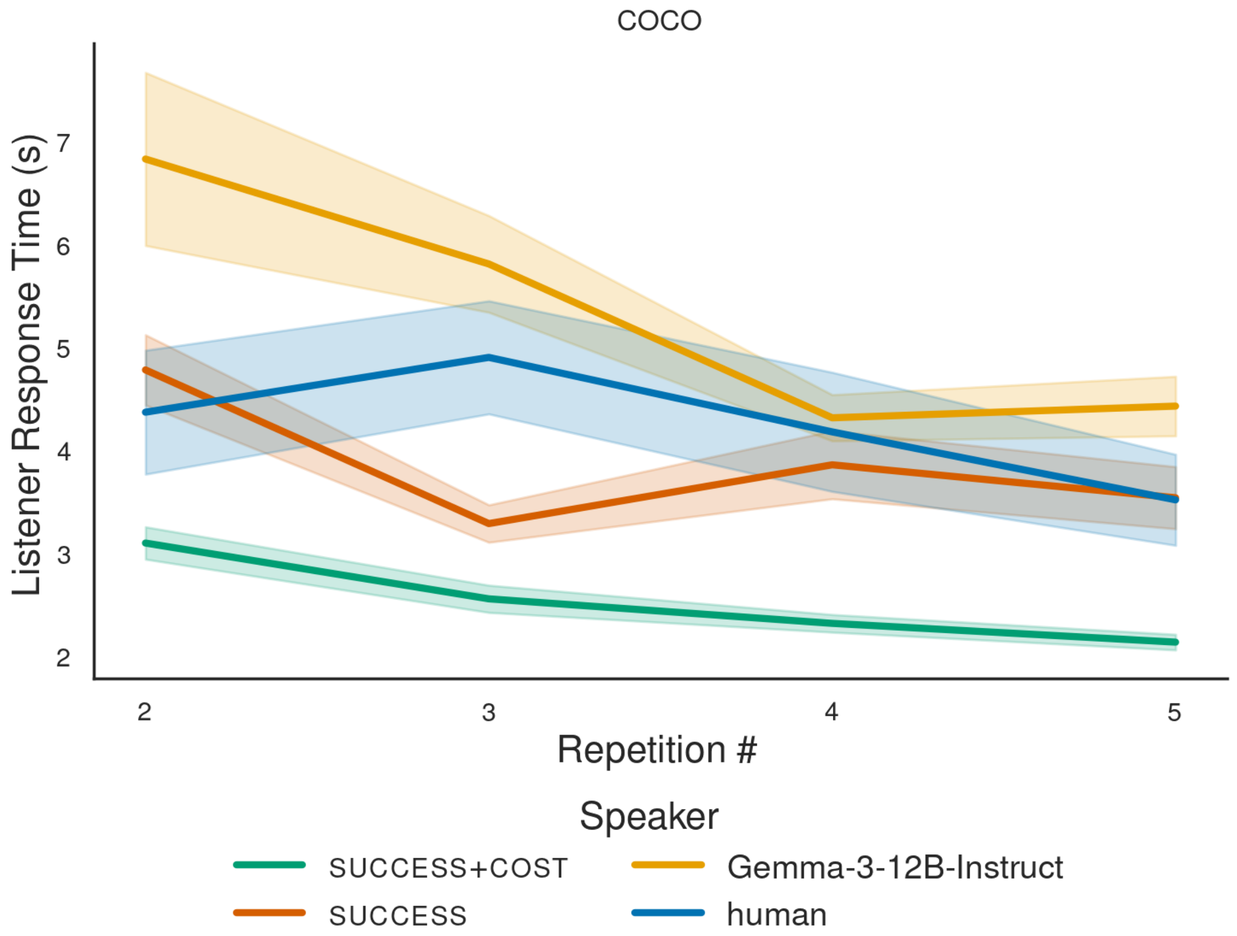
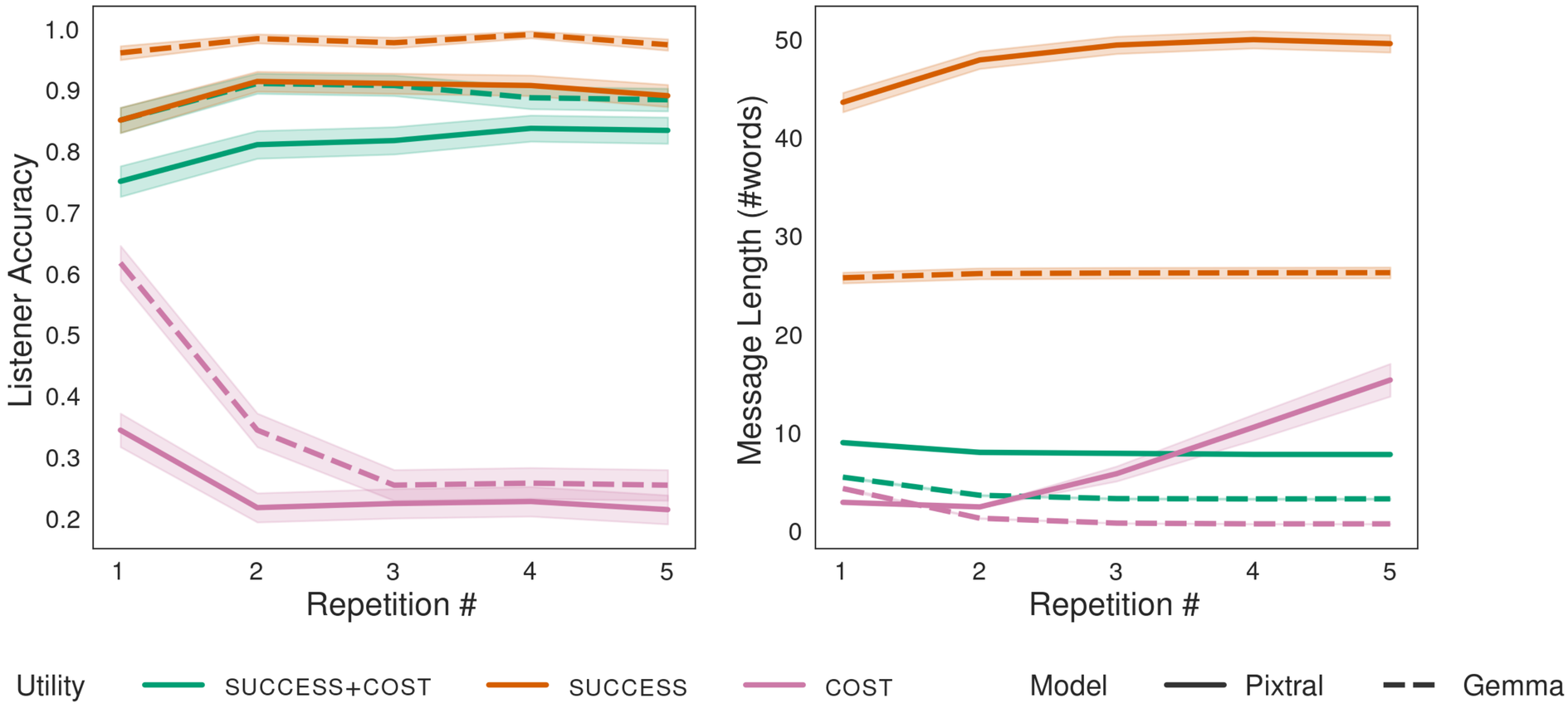


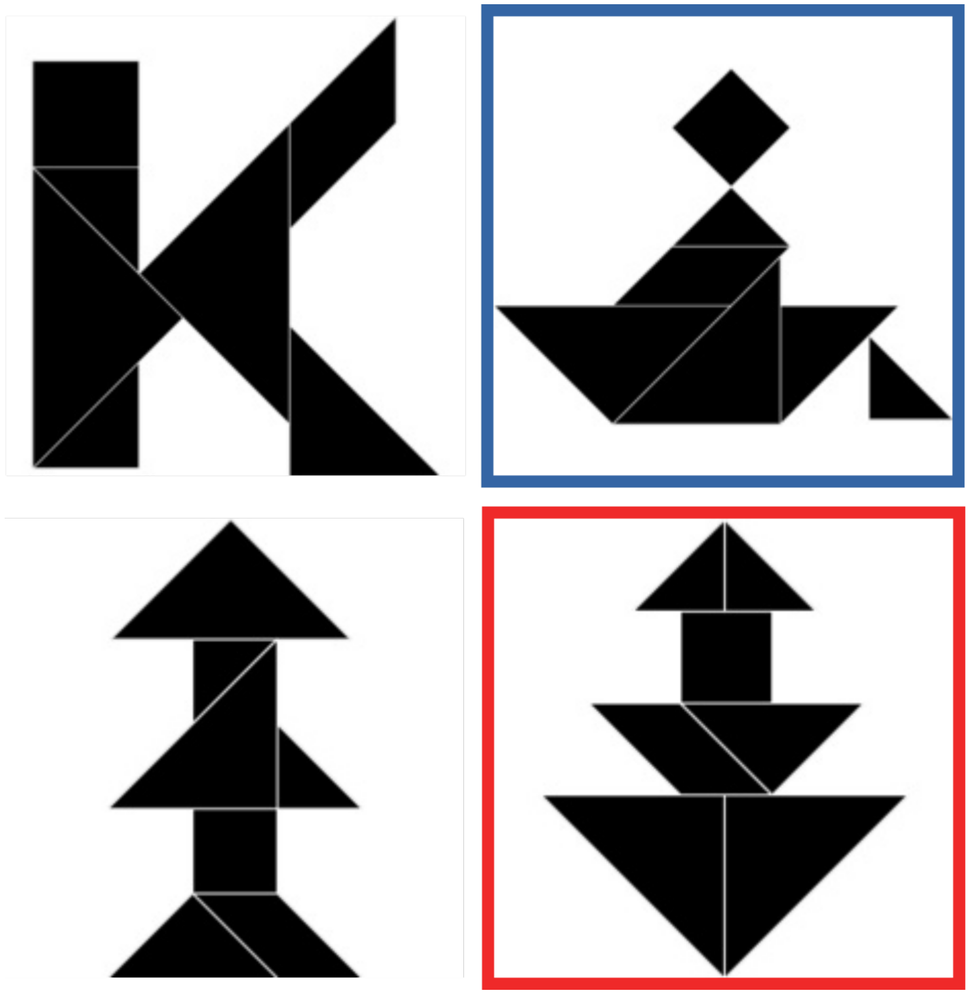
People respond faster to models that form convention



Neither SUCCESS not COST alone is sufficient



Experiments with tangram images



a Person sitting on boat with triangle tail

Trial #2

a Boat with Arrow pointing up

Trial #3

Person on boat

Trial #8

Boat Arrow up

Trial #11

Person boat

Trial #20

Takeaways

- We can train agents to form communicate efficiently with people by training only on simulated interactions
- These agents learn behaviors like convention formation even without being shown demonstrations of how people do it
- Training for success alone isn't sufficient for making agents communicate better with people, we need to reason about other factors like cost or effort

Asking for help

Asking for help

- Shared knowledge can go beyond conventions of reference, to procedural knowledge
  - How do you accomplish a certain task?

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- Seeking help allows for acquiring procedural knowledge proactively

Asking for help

- Shared knowledge can go beyond conventions of reference, to procedural knowledge
  - How do you accomplish a certain task?
- Seeking help allows for acquiring procedural knowledge proactively
- Once the knowledge is made available, it may be assumed to be known from that point onwards

Agents that learn from help they seek

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- *Proactivity:* Are agents able to identify when they don't know something and ask for help?

Agents that learn from help they seek

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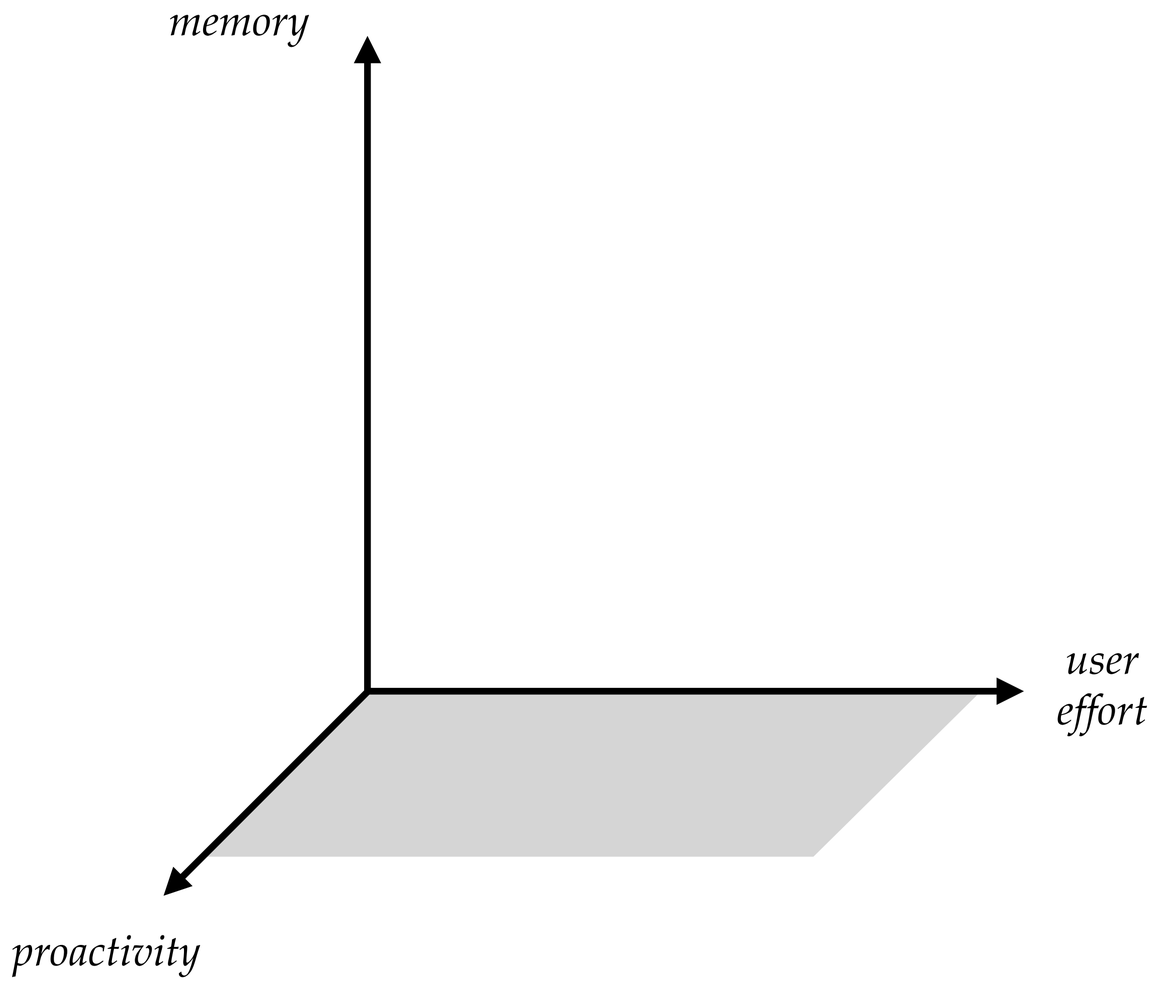
  - Natural language feedback
  - Demonstrations

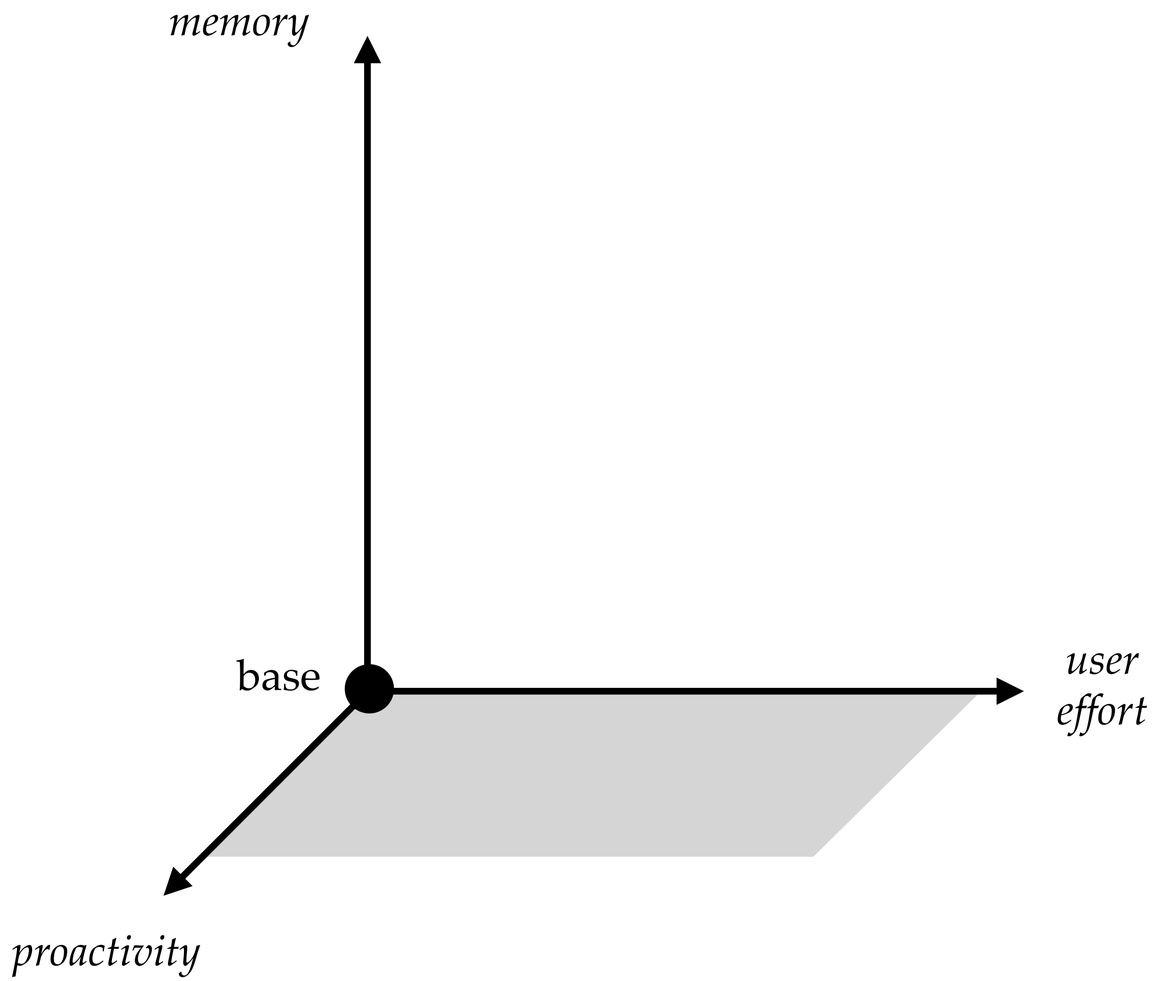
Agents that learn from help they seek

- *Proactivity:* Are agents able to identify when they don't know something and ask for help?
- *Assistance:* Can agents be helped easily by users?
  - Natural language feedback
  - Demonstrations
- *Memory:* Are agents able to represent and use knowledge of past interactions?

Agents that learn from help they seek



Agents that learn from help they seek



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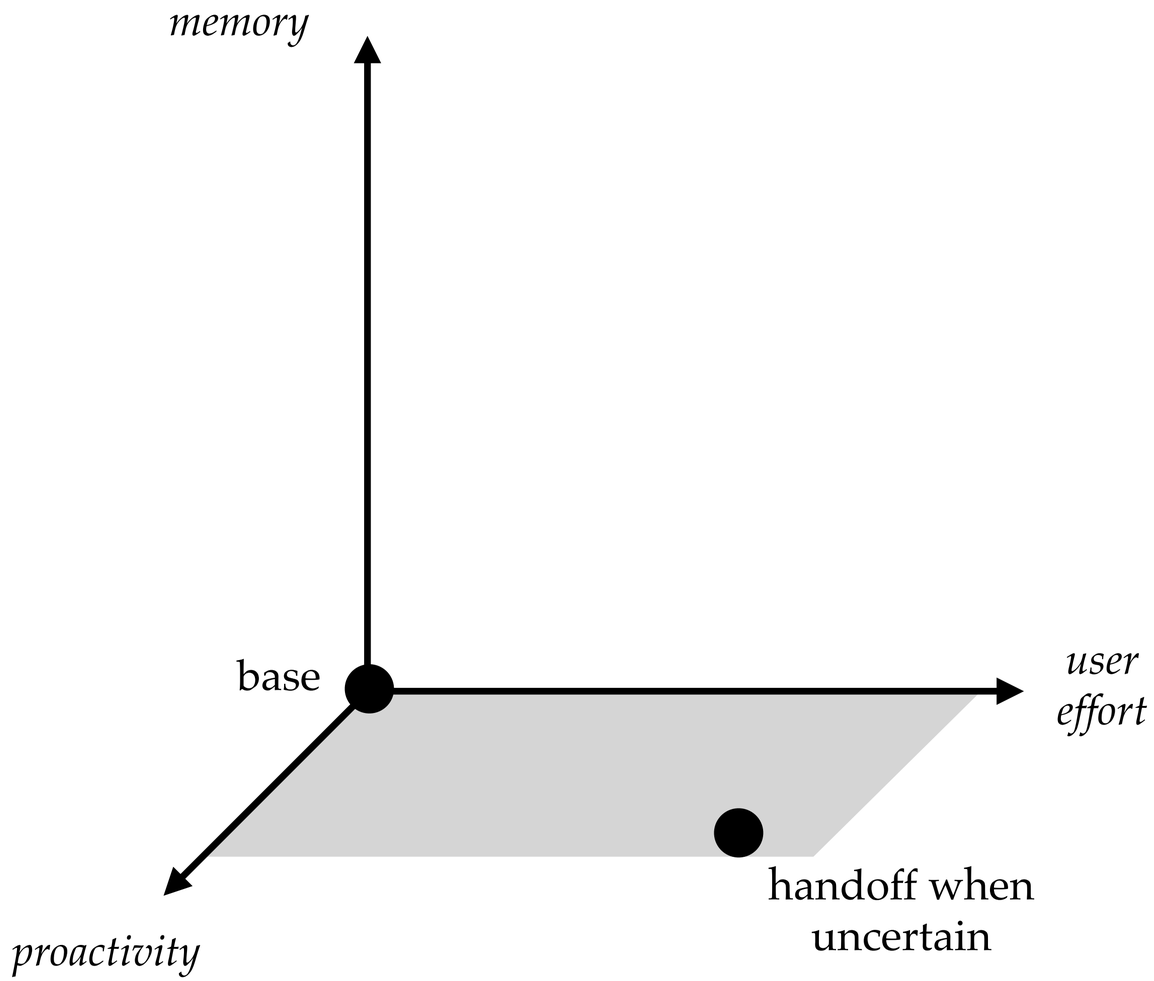

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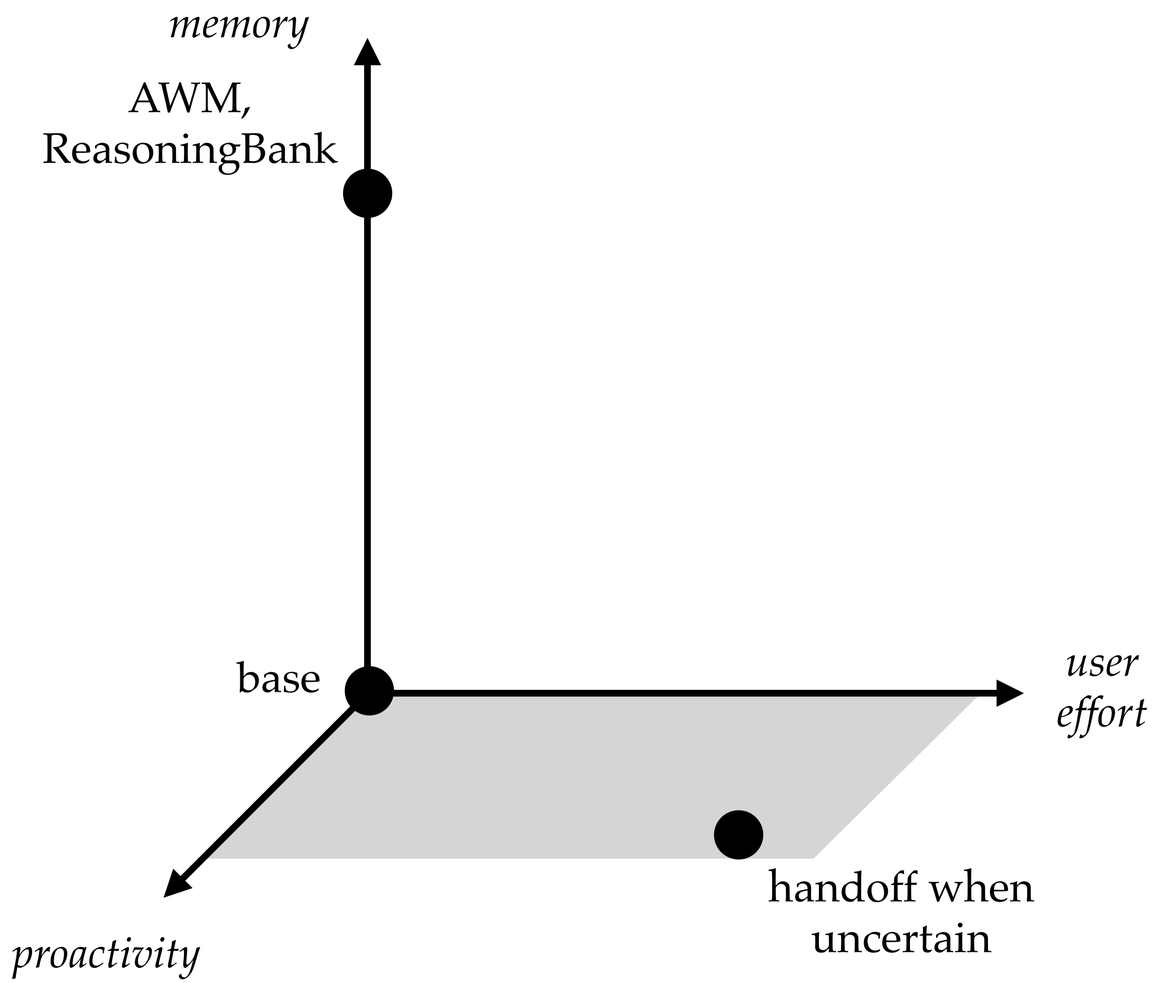
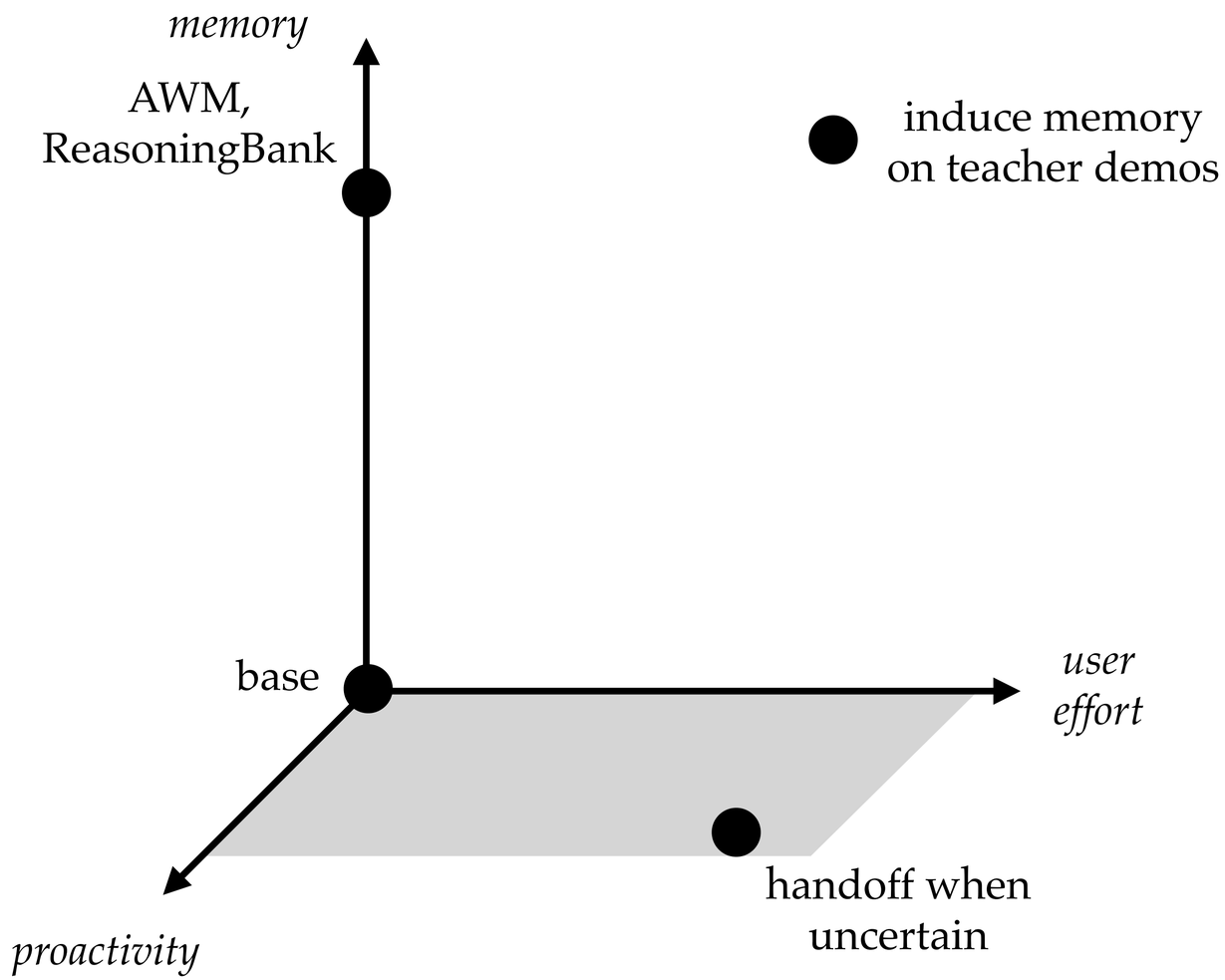
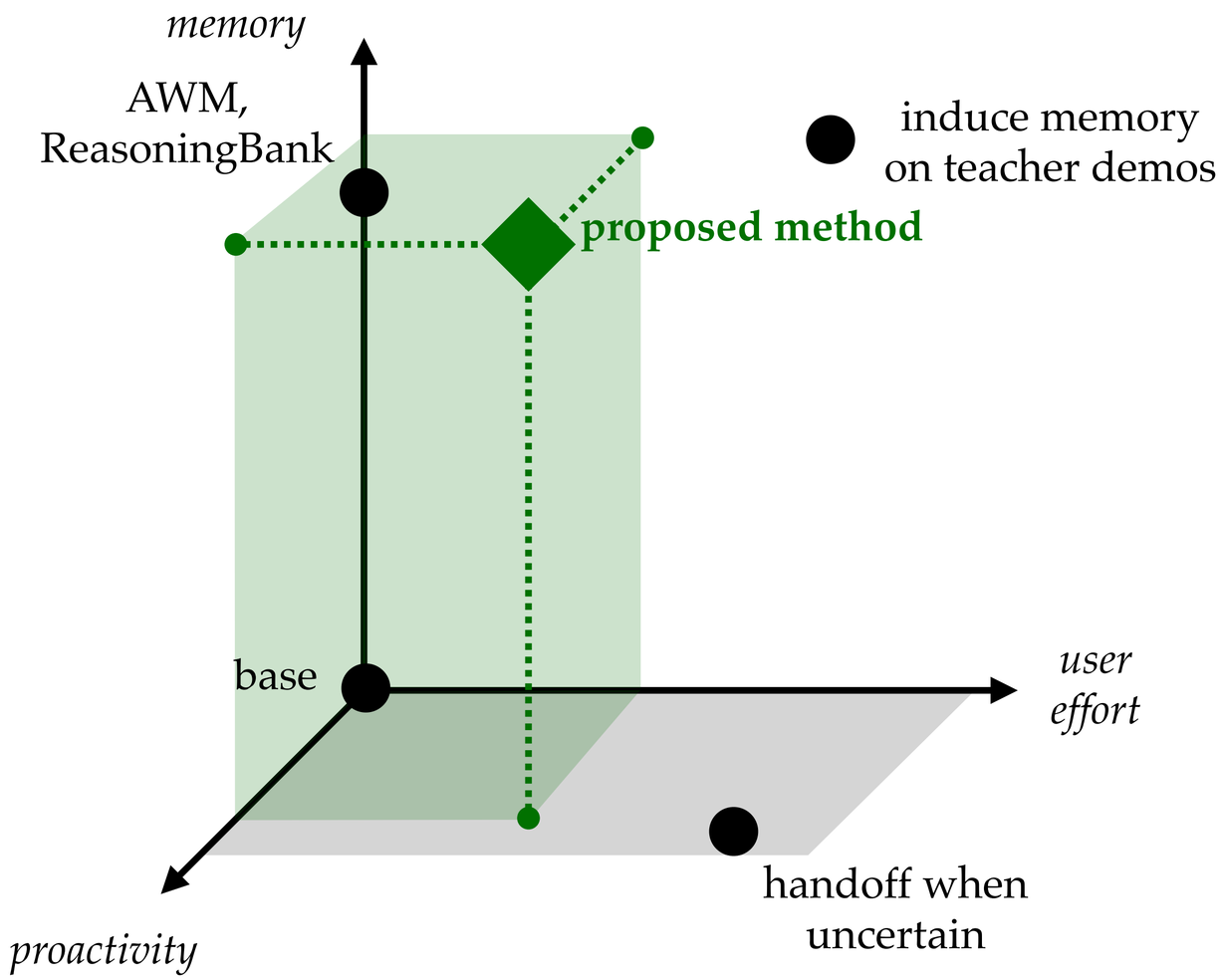


Agents that learn from help they seek



Agents that learn from help they seek



Web agents

- Agents perform complex tasks requiring multiple steps
- Shared structure presents opportunities to apply knowledge from prior interactions
  - WebArena, WorkArena(++)
- Agents can hand control to the user and observe actions, and chat

CowPilot: A Framework for Autonomous and Human-Agent Collaborative Web Navigation

Faria Huq, Zora Zhiruo Wang, Frank F. Xu, Tianyue Ou, Shuyan Zhou, Jeffrey P. Bigham*, Graham Neubig*
Carnegie Mellon University
Published in NAACL 2025 Demo
*Indicates Equal Supervision

Modeling Distinct Human Interaction in Web Agents

Faria Huq[C,†] **Zora Zhiruo Wang**[C,†] **Zhanqiu Guo**[C,‡] **Venu Arvind Arangarajan**[C,‡]
Tianyue Ou[C] **Frank Xu**[C] **Shuyan Zhou**[D] **Graham Neubig**[C] **Jeffrey P. Bigham**[C]
[C]Carnegie Mellon University [D]Duke University
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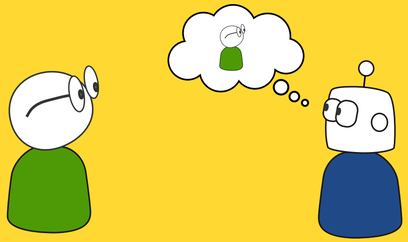
Proposed experiments

- Rollout sequences of tasks solved by a pair of simulated user and agent
  - Agents build memory within task sequence
- Calculate a reward based on
  - number of tasks completed (higher is better)
  - user effort (lower is better)
- Train on high-reward rollouts to teach the agent how to selectively ask for help in context
- Evaluate agents on *success rate* and *number of user actions*

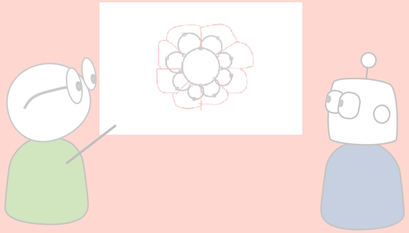
Part I: Reasoning about user intent

*[Vaduguru, Fried, Pu, ICLR 2024; **proposed work**]*



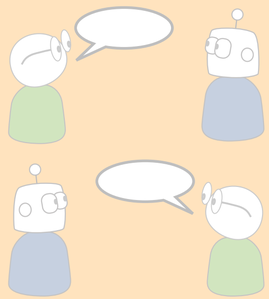
Part II: Reasoning about shared observation

[McCarthy, Vaduguru*,* et al.*, Findings of EMNLP, 2025]*



Part III: Reasoning about common ground over time

[Vaduguru, Hua, Fried, Artzi, ACL 2026; proposed work]



Learning in simulated interactions

- Training in multi-turn interactions with people is challenging
- Training in simulation is shown to be a promising approach
  - Have a user simulator play the role of a user

Learning to Learn from Language Feedback with Social Meta-Learning

Jonathan Cook[1],
[1]Google DeepMind

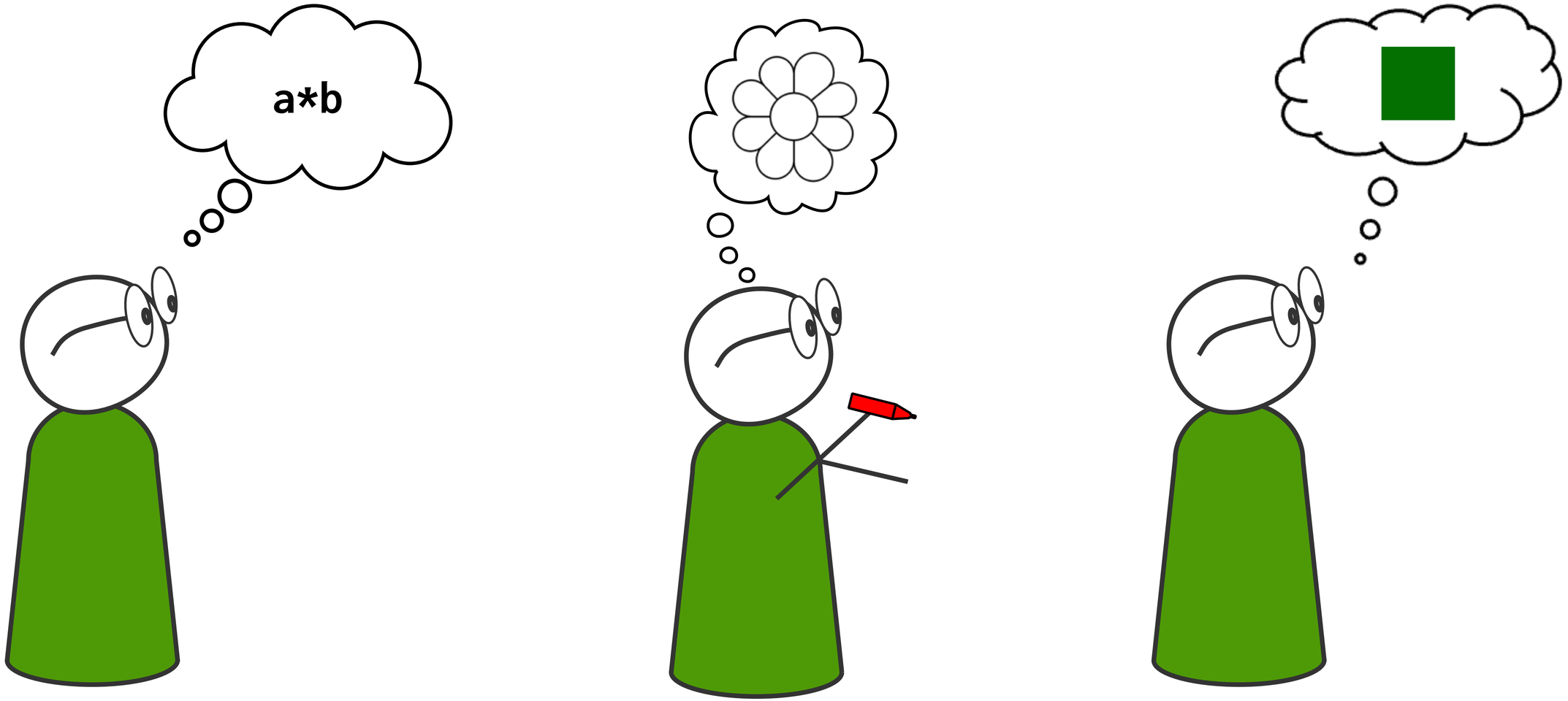
Training Proactive and Personalized LLM Agents

Weiwei Sun[1] Xuhui Zhou[1] Weihua Du[1] Xingyao Wang[2]
Sean Welleck[1] Graham Neubig[1,2] Maarten Sap[1] Yiming Yang[1]
[1]Carnegie Mellon University [2]OpenHands

Collaborative Reasoner: Self-Improving Social Agents with Synthetic Conversations

Ansong Ni* Ruta Desai* Yang Li Xinjie Lei Dong Wang Jiemin Zhang Jane Yu
Ramya Raghavendra Gargi Ghosh Daniel Li Asli Celikyilmaz
Meta FAIR

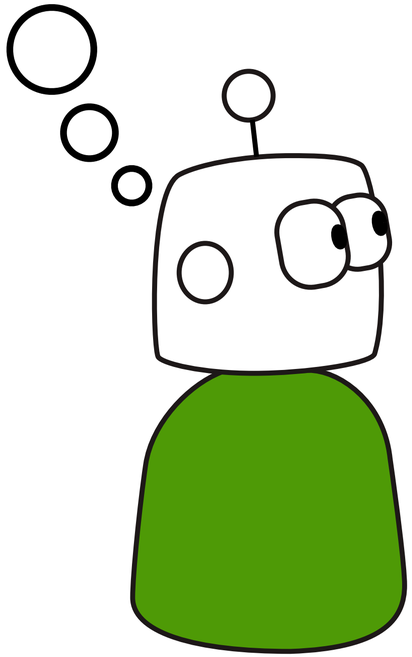
Information asymmetries in simulated interactions

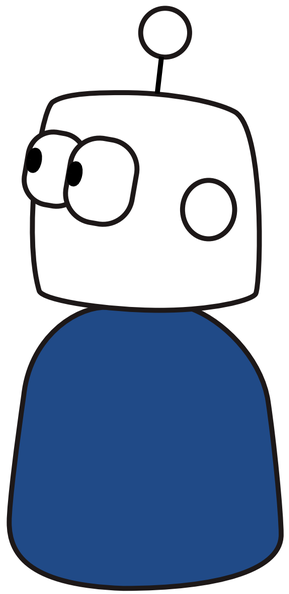


Information asymmetries in simulated interactions

Information asymmetries in simulated interactions

The method **`Header.fromstring`** "...creates an HDU header from a byte string containing the entire header data.". *[…]* A solution is to change **`Header.fromstring`** to *[…]*.



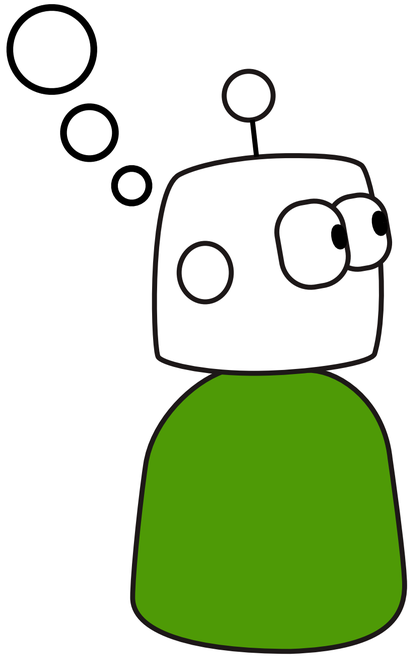


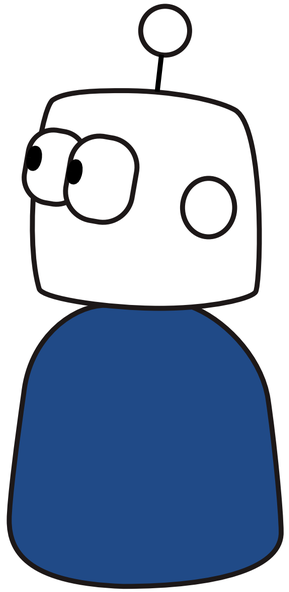
[Vijayvargiya, et al., 2026]

Information asymmetries in simulated interactions

The method **`Header.fromstring`** "...creates an HDU header from a byte string containing the entire header data.". *[…]* A solution is to change **`Header.fromstring`** to *[…]*.

The issue involves a **method** that should create a header from a byte string, *[…]*. A proposal is to modify the method to *[…]*





[Vijayvargiya, et al., 2026]

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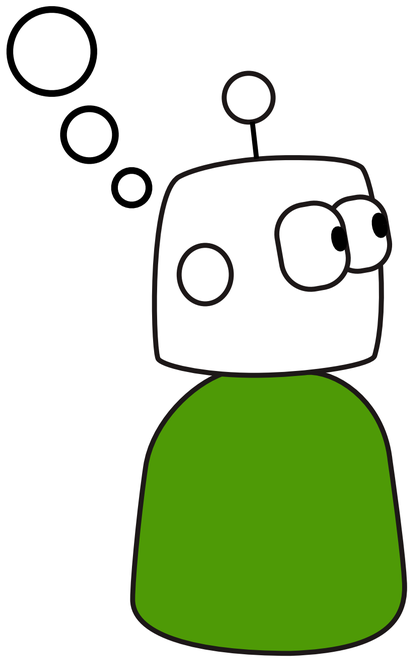
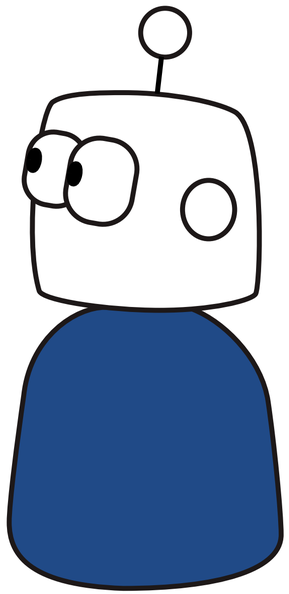


Could you clarify which method you are facing problems with and its location? […]



[Vijayvargiya, et al., 2026]

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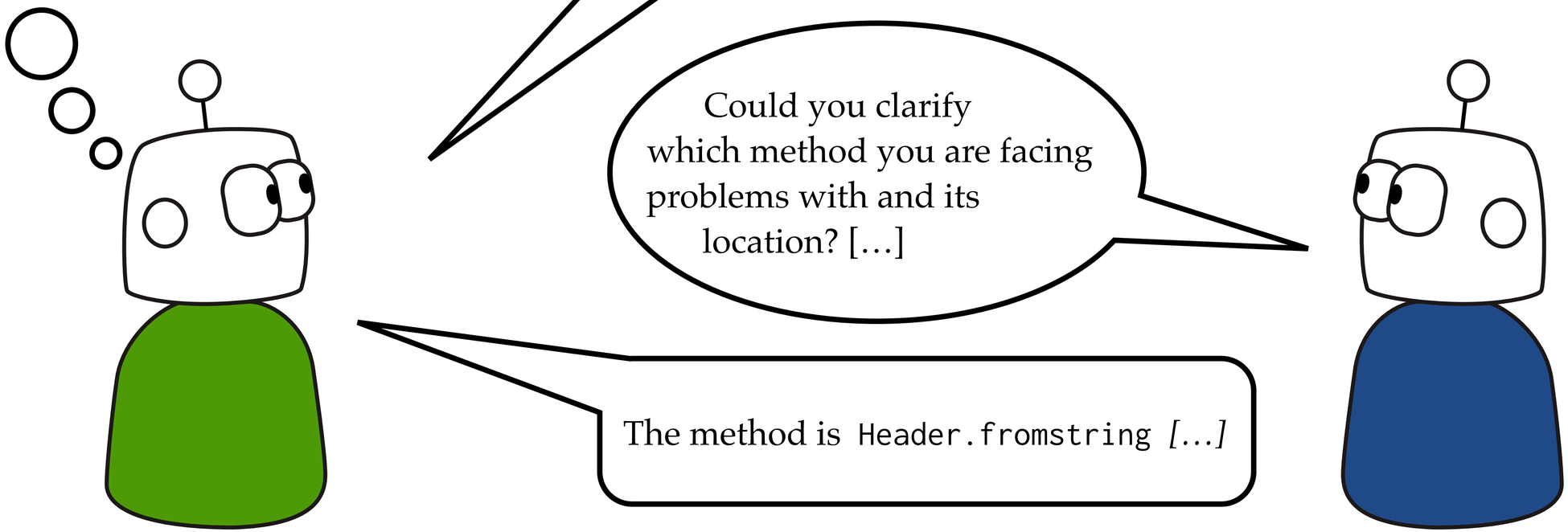
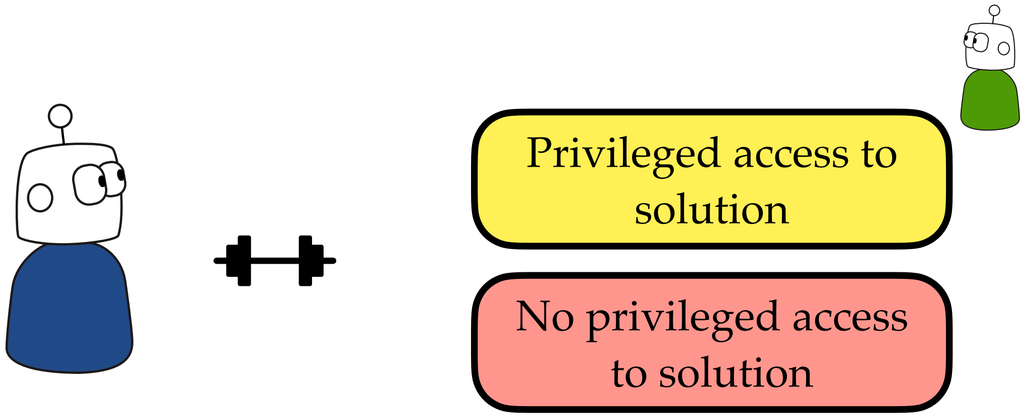


[Vijayvargiya, et al., 2026]

Proposed experiment

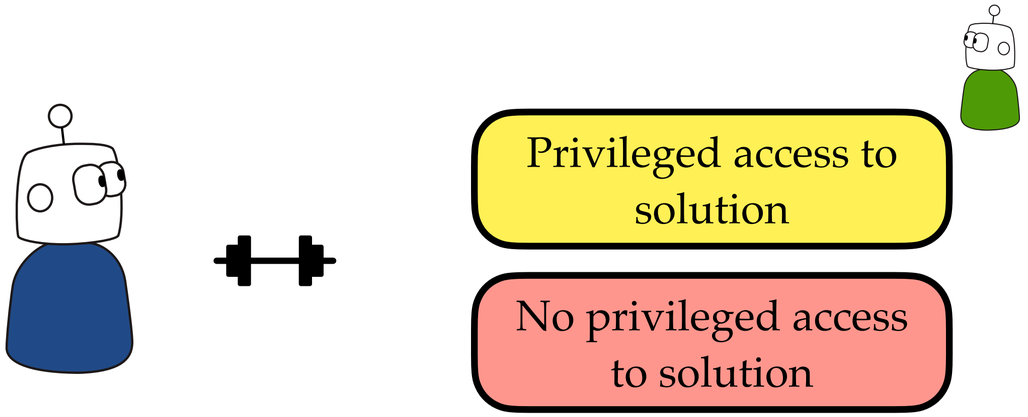
Proposed experiment

- Controlled comparison of training with and without access to solution



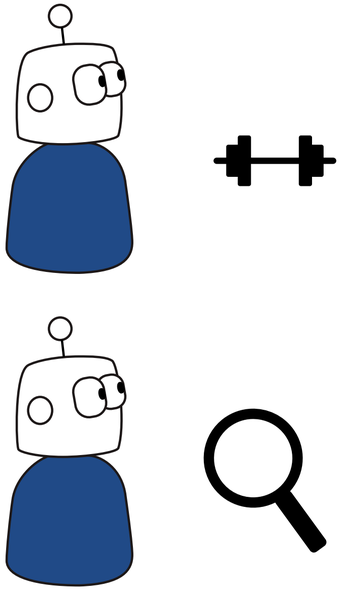
Proposed experiment

- Controlled comparison of training with and without access to solution
  - OmniMATH



Proposed experiment

- Controlled comparison of training with and without access to solution
  - OmniMATH
- Evaluate generalisation to

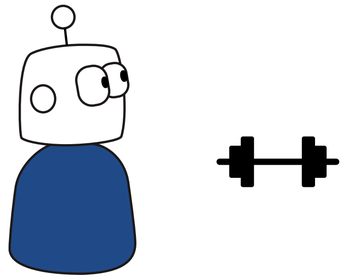


Privileged access to solution

No privileged access to solution

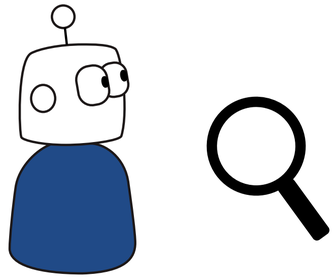
Proposed experiment

- Controlled comparison of training with and without access to solution
  - OmniMATH
- Evaluate generalisation to
  - Models with varying access to solution



Privileged access to solution

No privileged access to solution

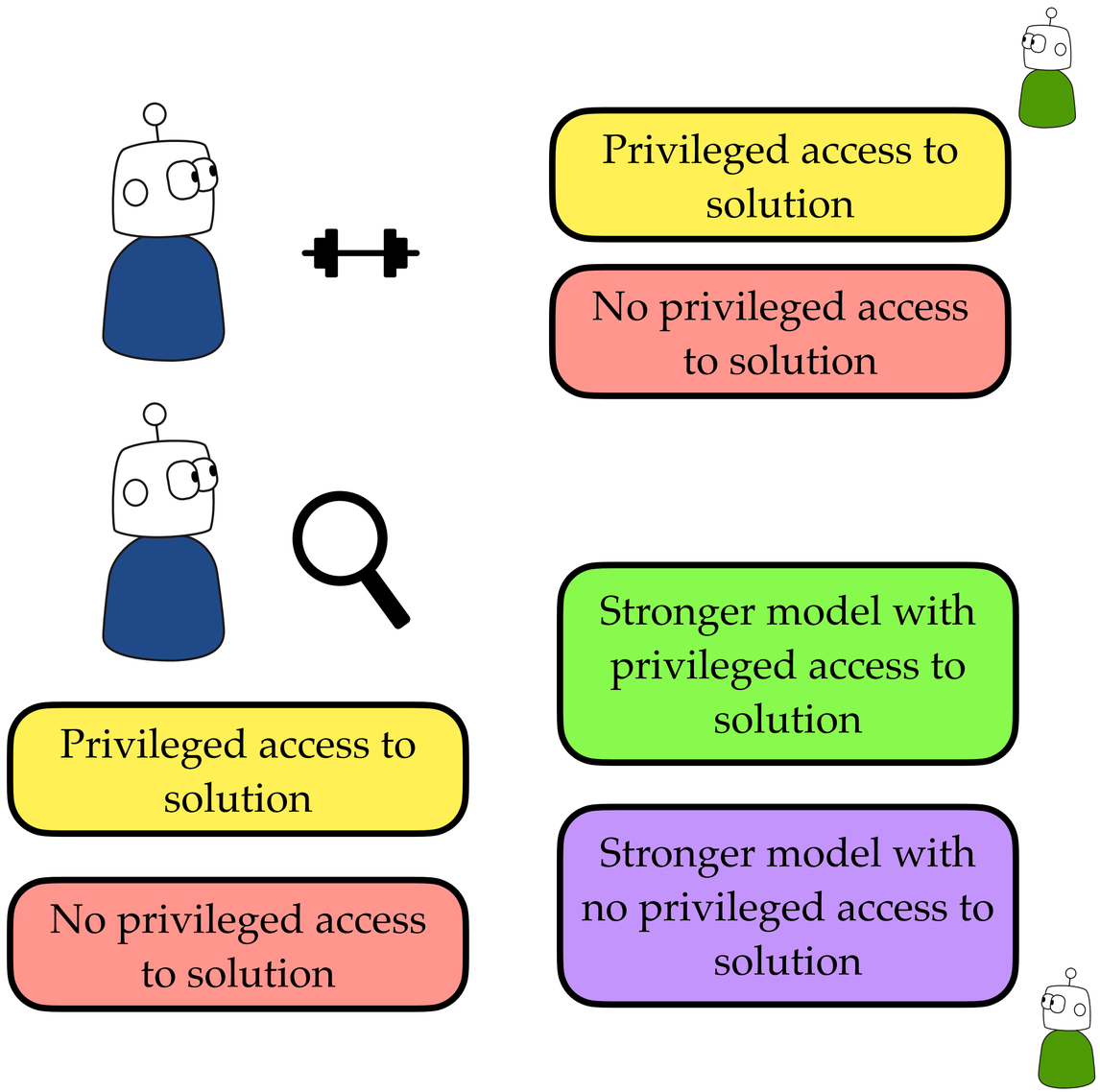


Privileged access to solution

No privileged access to solution

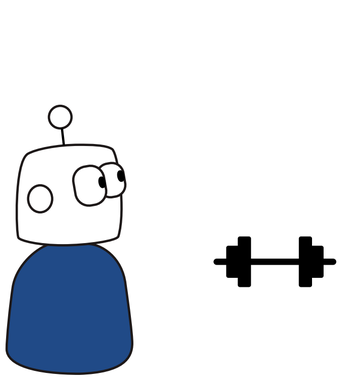
Proposed experiment

- Controlled comparison of training with and without access to solution
  - OmniMATH
- Evaluate generalisation to
  - Models with varying access to solution
  - Stronger and weaker models



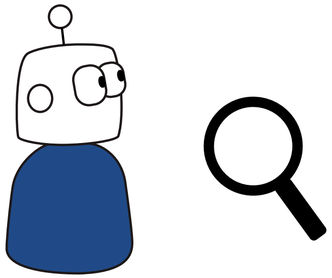
Proposed experiment

- Controlled comparison of training with and without access to solution
  - OmniMATH
- Evaluate generalisation to
  - Models with varying access to solution
  - Stronger and weaker models
- Compare rate of successful solutions



Privileged access to solution

No privileged access to solution



Stronger model with privileged access to solution

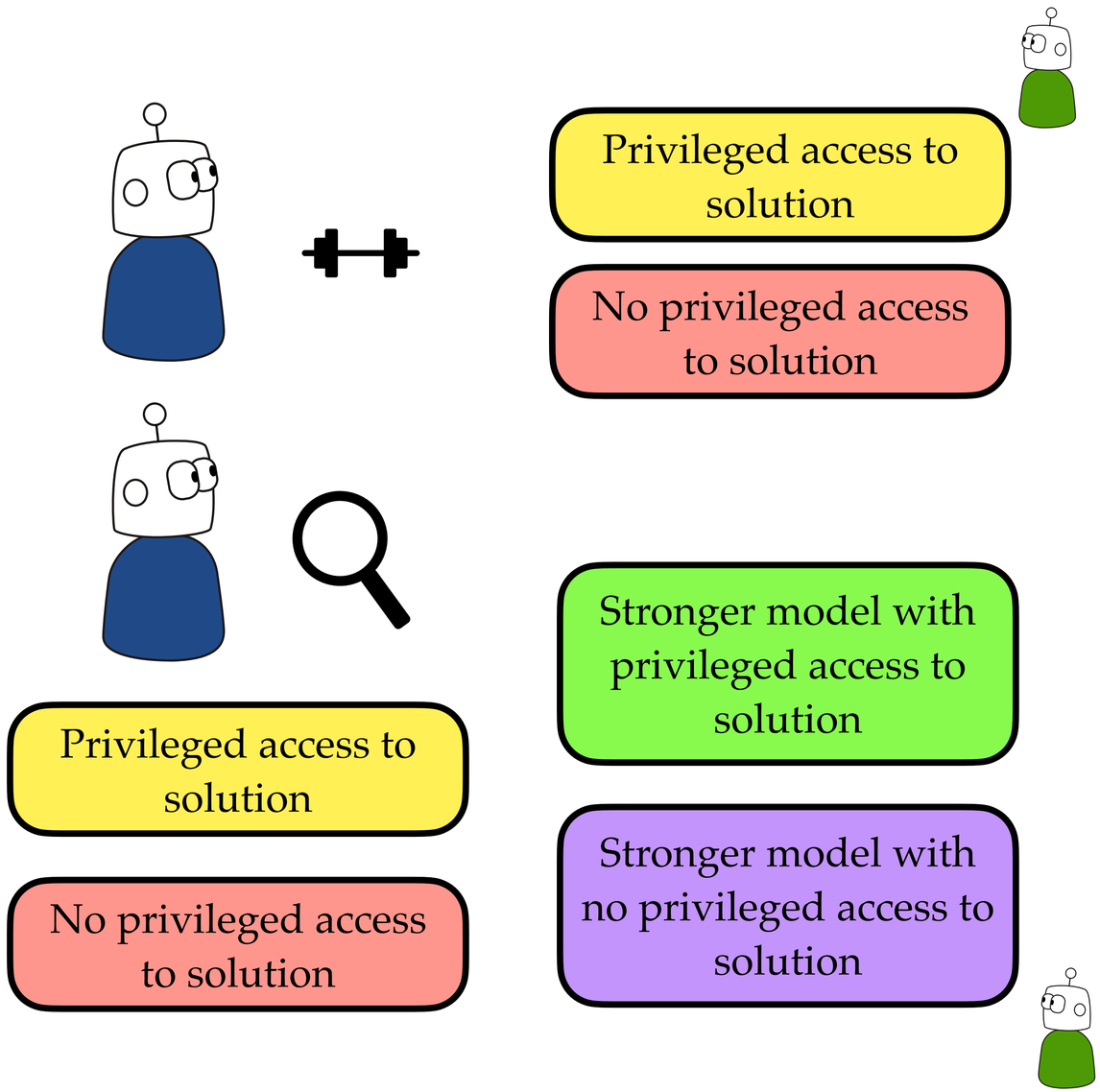
Privileged access to solution

Stronger model with no privileged access to solution

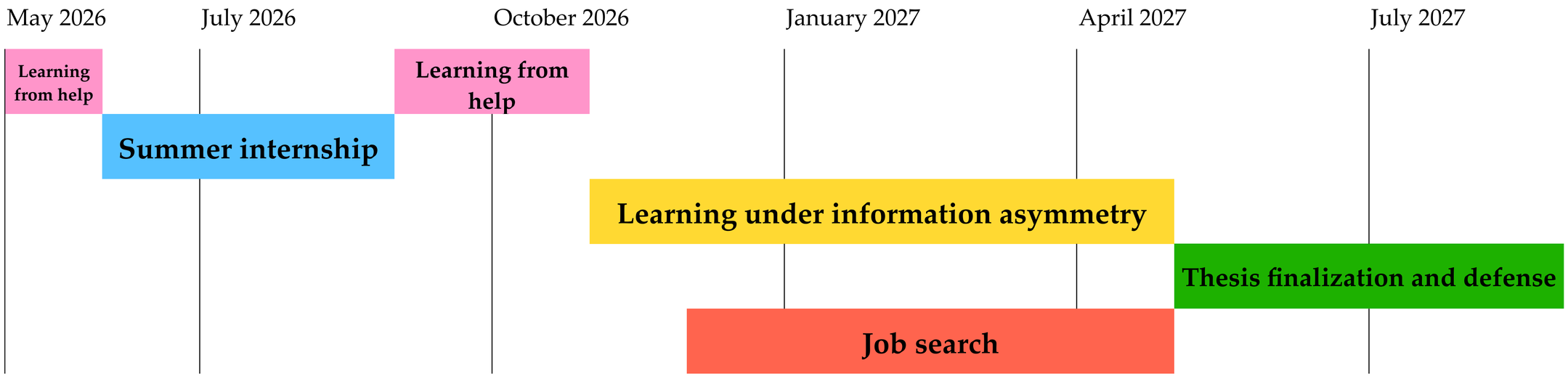
No privileged access to solution

Proposed experiment

- Controlled comparison of training with and without access to solution
  - OmniMATH
- Evaluate generalisation to
  - Models with varying access to solution
  - Stronger and weaker models
- Compare rate of successful solutions
- Characterize behaviors of trained models
  - Turn-taking, clarification, etc.



Timeline


May 2026
July 2026
October 2026
January 2027
April 2027
July 2027
Learning from help
Learning from help
Summer internship
Learning under information asymmetry
Thesis finalization and defense
Job search

Questions and feedback